

Environmental and Nutritional Diseases

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Many diseases are caused or influenced by environmental factors. Broadly defined, the term *environment* encompasses the various settings in which human beings are born, live, learn, work, and play. In each of these settings, the air people breathe, the food and water they consume, and the toxic agents they are exposed to are major determinants of health. The environmental factors that influence our health pertain to individual behavior (“personal environment”) and include tobacco use, alcohol ingestion, nonprescription drug consumption, and diet, and the external (ambient and workplace) environment. In general, in high-resource countries personal behavior has a larger effect on health than the ambient environment, but new

threats related to climate change (described later) may change this equation.

The term *environmental disease* refers to conditions caused by exposure to chemical or physical agents in the ambient, workplace, or personal environment, including diseases of nutritional origin. The impact of environmental disease on health is vast. The International Labour Organization estimates that 340 million occupational accidents occur globally each year, leading to approximately 2.3 million deaths (more than are caused by road accidents and wars combined). In the United States in 2021, there were 2.6 million workplace injuries and illnesses. Disease related to

malnutrition is even more pervasive. In 2021, it was estimated that 828 million people were malnourished—one in every nine persons worldwide. Children are disproportionately affected by undernutrition, which accounts for approximately 50% of childhood mortality worldwide. Estimating the burden of disease in the general population caused by nonoccupational exposures to toxic agents is complicated by the diversity of agents and difficulties in determining the extent and duration of exposures. But whatever the precise numbers, environmental diseases are major causes of disability and suffering and constitute a heavy financial burden, particularly in developing countries.

In this chapter, we first consider the urgent challenge posed by health disparities, and the emerging problem of the health effects of climate change. We then discuss the mechanisms of toxicity of chemical and physical agents and address specific environmental disorders, including those of nutritional origin.

HEALTH DISPARITIES

Health disparities are inequalities in the incidence, prevalence, severity, and burden of diseases, injury, and mortality across groups within a population (Fig. 9.1). Health disparities are generated by multiple factors including race, ethnicity, education level, employment and income status, food security, housing, health care access and health insurance status, neighborhood conditions and infrastructure, and partnership/marital status. The impact of these factors is influenced by personal beliefs and behaviors such as sexism and racism, as well as economic, political, and health care policies. These factors often interact to contribute to health disparities.

For instance, individuals from marginalized groups often encounter barriers in accessing high-quality health care, reside in neglected neighborhoods lacking adequate infrastructure, and experience discrimination, all of which contribute to poor health outcomes.

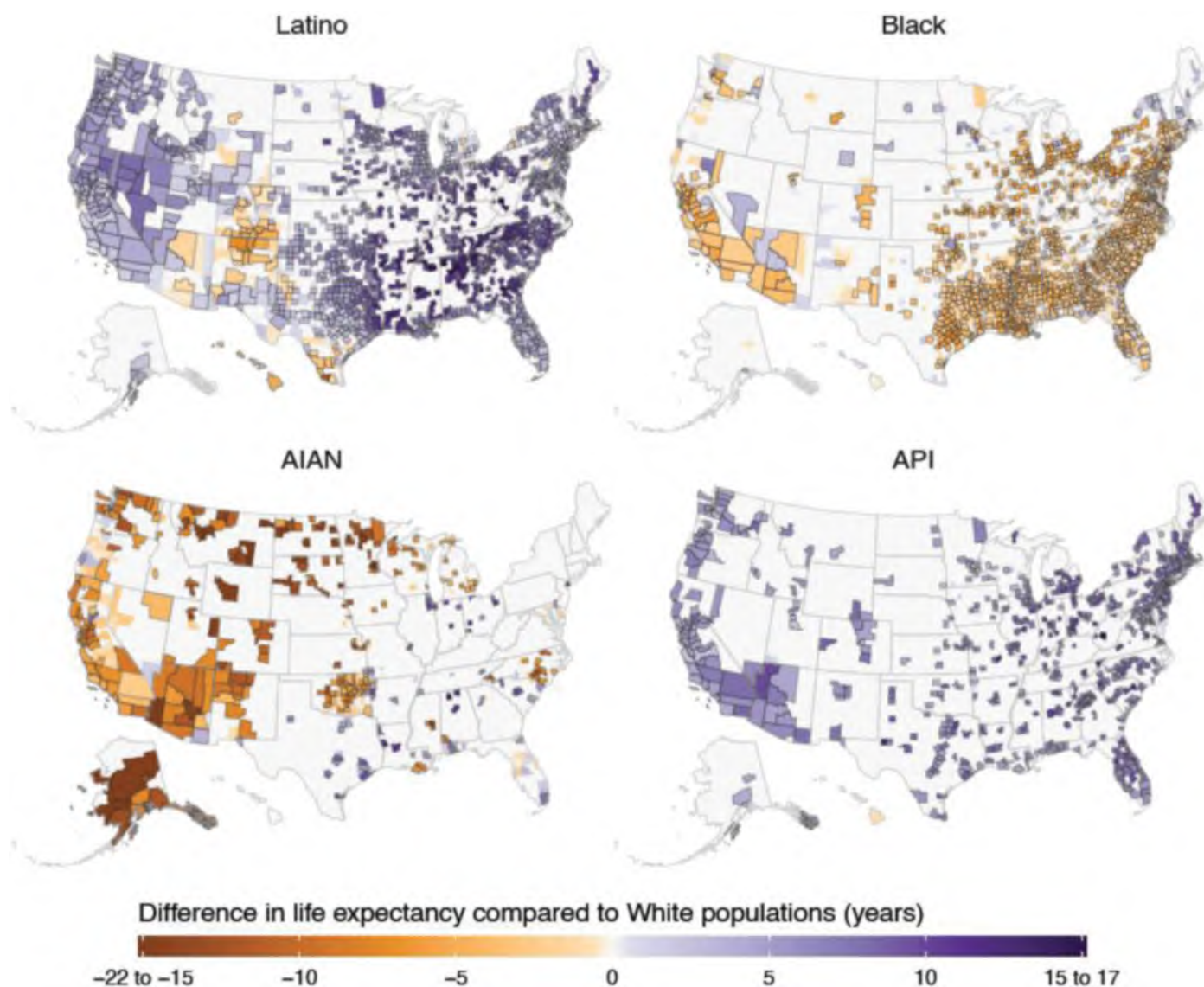


Fig. 9.1 Differences in life expectancy among various racial/ethnic groups compared to the “White” population in counties across the United States in the year 2019. The color scale is truncated at –15 and 15 years. *AIAN*, Non-Hispanic American Indian or Alaska native; *API*, non-Latino and non-Hispanic Asian or Pacific Islander; *Latino*, Latino or Hispanic. (From Dwyer-Lindgren K, et al: Life expectancy by county, race, and ethnicity in the USA, 2000–19: a systematic analysis of health disparities. *Lancet*. 2022;400[10345]:25–38.)

Looming large in any discussion of health disparities is the concept of “race.” Genetic studies conducted as early as the 1960s demonstrated that groups previously considered as separate races such as Africans, Asians, and Europeans could not be differentiated based on genetic polymorphisms, raising questions about the biologic basis for different races. These questions have now been answered by population genetic studies measuring the extent of genetic variation within groups compared to the variation between them and assessing whether any group can be considered an evolutionarily distinct population. These studies have consistently shown greater variation within populations (e.g., African, East Asian, European) than between them. Moreover, due to substantial gene flow between populations, human genetic variation is continuous, making it impossible to sort different populations into genetically discrete groups. Based on these findings, population geneticists and biological anthropologists have concluded that *biologically distinct races do not exist in modern humans*, refuting the notion that race is a genetic risk factor in the occurrence of certain ailments.

However, throughout history societies have used arbitrary criteria such as physical appearance and geographic origin to categorize and differentiate populations into “races,” typically for purposes of exerting social dominance of one race over others. Social subordination of people belonging to certain races has many negative health effects. For example, when compared to Americans of European descent, African Americans are more likely to live close to major roadways, rail lines, ports, and industrial facilities, resulting in increased exposure to air pollution and an increased burden of asthma. Cities with majority African American populations, such as Flint, Michigan, and Jackson, Mississippi, have experienced major water crises that have compromised access to safe drinking water and exposed residents to hazardous levels of environmental contaminants. Largely because of inadequate access to health care, African Americans have an approximately three-fold higher maternal mortality rate, a two-fold higher infant mortality rate, and a two-fold higher COVID-19 mortality rate. Conditions such as type 2 diabetes, hypertension, and obesity are also more prevalent in the African American population. Reduced access to care extends beyond traditional health care settings as well. In instances of cardiac arrests occurring outside of hospitals in public locations, African American and Hispanic individuals are less likely than individuals of European descent to receive potentially lifesaving cardiopulmonary resuscitation from bystanders. Even certain medical tests have led to disparities, such as those measuring kidney function due to race-based adjustments in the algorithms for determining the estimated glomerular filtration rate (eGFR). Removal of race-based corrections would increase the percentage of African American patients diagnosed with chronic kidney disease, leading to earlier diagnosis and timely access to specialty care. Similar race-based corrections have been components of algorithms for clinical decision-making across fields such as cardiology, endocrinology, obstetrics, oncology, pulmonology, urology, surgery, and transplantation; many are now undergoing review due to their potential to perpetuate and amplify race-based health inequities.

Ethnicity, which encompasses multiple characteristics such as culture, country of origin, language, and religion, can also influence health outcomes. In the United States, ethnicity is usually categorized as Hispanic or non-Hispanic. However, it is important to understand that the term “Hispanic” in the US census and many research studies refers to Americans of Spanish origin or descent, whereas the term “Latin American” includes populations from Central and South America, including non-Spanish speaking groups

such as Brazilians, with descent encompassing African, Amerindian, and European ancestry.

Historically, social definitions of race have been based on factors such as skin pigmentation (e.g., White, Black), references to geographic origin (e.g., African American, Asian American), and outdated anthropological terms like “Caucasian.” In this textbook, we will use terms related to geographic origin such as European American and African American when discussing socially defined race and will mention the association of certain genetic disorders (e.g., cystic fibrosis) with ancestry from particular regions, such as (in the case of cystic fibrosis) Northern Europe. However, it is important to remember that because of gene flow between populations, these associations are not absolute. For example, cystic fibrosis affects approximately 1 in 2500 infants of Northern European descent, but also affects 1 in 17,000 African American and 1 in 31,000 Asian American infants. Similarly, sickle cell disease is most common in individuals of West African descent, but the mutation that causes the disease is found at increased frequency in regions throughout the “malaria belt,” including equatorial Africa, Southern Europe, the Middle East, and parts of Asia, due to its protective effect against falciparum malaria.

Thus, while many racial and ethnic associations with disease exist, one must be careful not to allow these associations to bias one’s approach to individual patients.

HEALTH EFFECTS OF CLIMATE CHANGE

Climate change stands to become the preeminent global cause of environmental disease in the twenty-first century and beyond.

Global temperature measurements show that the earth has warmed significantly since the early twentieth century and especially since the mid-1960s. Record-breaking global temperatures have become more frequent, and the 10 warmest years on record occurred from 2014–23, with 2023 being the warmest year on record since global temperature monitoring began in 1850. The average temperature across land and ocean surfaces in 2023 was 1.18°C above the twentieth century average.

The rising land and ocean temperatures have led to a large number of effects that include changes in storm frequency, drought, and flood, as well as large-scale ice losses in Greenland, Antarctica, and the vast majority of the other glaciated regions on earth, as well as dramatic thinning or disappearance of Arctic Ocean ice. The melting of land-based glacial ice and the thermal expansion of the warming oceans has produced approximately 13 to 20 cm of global average sea level rise since 1900, and the sea level currently is rising at a global average rate of 3.5 ± 0.4 mm per year.

The major culprit is the rising atmospheric level of greenhouse gases, particularly carbon dioxide (CO₂) released through the burning of fossil fuels, ozone (an important air pollutant, discussed later), and methane. These gases, along with water vapor, produce the so-called greenhouse effect by absorbing energy radiated from the earth’s surface that otherwise would be lost into space. The annual average level of atmospheric CO₂ (about 419.3 ppm) in 2023 was higher than at any point in approximately 650,000 years and, without changes in human behavior, is expected to increase to above 700 ppm by the end of this century—levels not experienced for tens of millions of years and more than double the preindustrial level of 280 ppm. This increase stems not only from increased CO₂ production but also from deforestation and the attendant decrease in carbon fixation by plants. Depending on the computer model used, increased levels of greenhouse gases are projected to cause the global temperature to rise by ~3.5°C by the year 2100, a rate of change unseen in the last 10,000 years.

The health consequences of climate change will depend on its extent and rapidity, the severity of the ensuing consequences, and humankind's ability to mitigate the damaging effects (Fig. 9.2). Based on current trends, the World Health Organization (WHO) estimates that approximately 250,000 deaths will occur annually between 2030 and 2050 due to climate change. This estimate stems from the expectation that climate change will increase the incidence and severity of many diseases, including:

- *Cardiovascular, cerebrovascular, and respiratory diseases*, all of which will be exacerbated by heat waves and air pollution, related in part to wildfires.
- *Gastroenteritis, cholera, and other foodborne and waterborne infectious diseases*, caused by contamination due to floods and disruption of clean water supplies and sewage treatment after heavy rains and other environmental disasters.
- *Vector-borne infectious diseases* such as malaria and dengue fever resulting from changes in vector number and geographic distribution related to increased temperatures, crop failures, and more extreme weather variation (e.g., more frequent and severe El Niño events).
- *Malnutrition*, caused by changes in local climate that disrupt crop production. Such changes are anticipated to be most severe in tropical locations, in which average temperatures may already be near or above crop tolerance levels; it is estimated that by 2080, agricultural productivity may decline by 10% to 25% in some low-resource countries due to climate change.

Beyond these disease-specific effects, it is estimated that the melting of glacial ice, particularly in Greenland and other parts of the Northern Hemisphere, combined with the thermal expansion of warming oceans, will raise sea levels by 2 to 6 feet by 2100. Approximately 10% of the world's population—roughly 600 million people—live in low-lying areas that are at risk for flooding even if the rise in ocean levels is at the low end of these estimates. For example, a rise in sea level by 1.5 feet will submerge 70% of the land mass of the Maldives islands, while a 3-foot rise will completely inundate the islands, a change in sea level that is projected to occur by 2100. The resulting displacement of people will disrupt lives and commerce, creating conditions ripe for political unrest, war, and poverty—the “vectors” of malnutrition, sickness, and death.

Worldwide recognition of the potentially catastrophic effects of climate change led in late 2015 to a historic meeting of 196 countries in Paris, France, at which the participating countries agreed to the following objective: To hold the increase in the global average temperature to well below 2°C above preindustrial levels and to pursue efforts to limit the temperature increase to 1.5°C above preindustrial levels, recognizing that this would significantly reduce the risks and impacts of climate change. However, the major producers of greenhouse gases, particularly the United States and China, have not made the difficult changes needed to curb greenhouse emissions, and it seems likely that dire predictions related to climate change will be realized in the coming years.

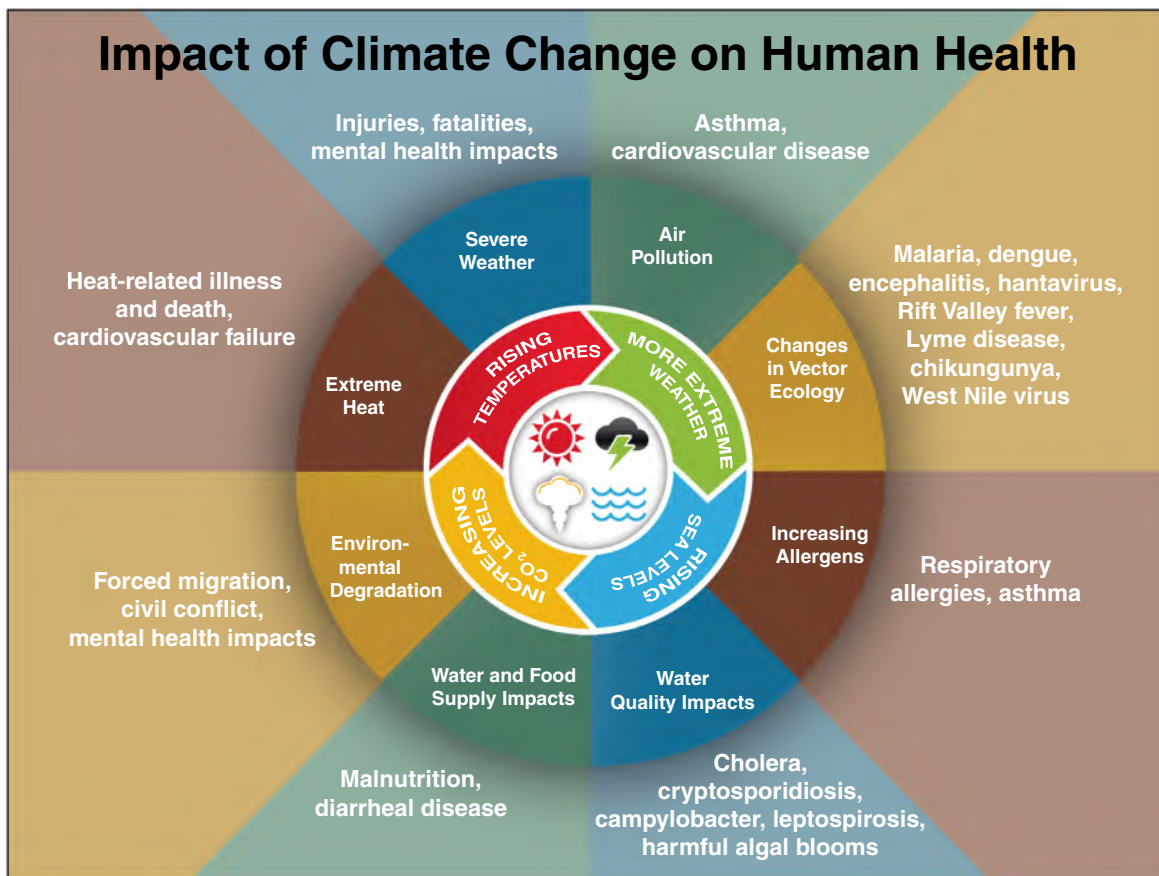


Fig. 9.2 Climate change and impact on health outcomes. This figure illustrates the most immediate impacts of climate change (increasing CO₂ levels, rising temperatures, more extreme weather, rising sea levels), their effects on environmental factors, and downstream effects on health outcomes. (From CDC: <https://www.cdcfoundation.org/climate-health/resources/>.)

TOXICITY OF CHEMICAL AND PHYSICAL AGENTS

Toxicology, defined as the science of poisons, studies the distribution, effects, and mechanisms of action of toxic agents. More broadly, it also includes the study of the effects of physical agents such as radiation and heat. Of the approximately 100,000 chemicals in commercial use in the United States, only a small proportion has been tested experimentally for health effects. Several agencies in the United States set permissible levels of exposure to known environmental hazards (e.g., the maximum level of carbon monoxide [CO] in air that is noninjurious or the tolerable levels of radiation that are “safe”). Factors such as the complex interaction between various pollutants and the age, genetic predisposition, and different tissue sensitivities of exposed persons create wide variations in individual sensitivity to toxic agents, limiting the value of establishing “safe levels” for entire populations. Nevertheless, such cutoffs are useful for comparative studies of the effects of agents between specific populations and for estimating risk of disease in heavily exposed individuals.

We now consider some basic principles relevant to the effects of toxic chemicals and drugs.

- The *definition of a poison* is not straightforward. It is basically a quantitative concept that depends on dosage. The quote from Paracelsus in the 16th century that “all substances are poisons; the right dosage differentiates a poison from a remedy” is still valid today, given the number of pharmaceutical drugs with potentially harmful effects.
- **Xenobiotics** are exogenous chemicals in the air, water, food, and soil that are absorbed into the body through inhalation, ingestion, and skin contact (Fig. 9.3).
- Chemicals may be excreted in urine or feces; eliminated in expired air; or accumulate in bone, fat, brain, or other tissues.
- Chemicals may act at the site of entry or at other sites following transport through the blood.
- *Most solvents and drugs are lipophilic*, which facilitates their transport in the blood by lipoproteins and their penetration through the plasma membrane into cells.
- Most solvents, drugs, and xenobiotics are metabolized to form inactive water-soluble products (detoxification) or are activated to form toxic metabolites. The reactions that metabolize xenobiotics into nontoxic products or that activate xenobiotics to generate toxic compounds (Fig. 9.4) occur in two phases. In *phase I reactions*, chemicals undergo hydrolysis, oxidation, or reduction. Products of phase I reactions are often metabolized into water-soluble compounds through *phase II reactions*, which include glucuronidation, sulfation, methylation, and conjugation with glutathione (GSH). Water-soluble compounds are readily excreted.
- The most important catalyst of phase I reactions is the cytochrome P-450 enzyme system. Cytochrome P-450 enzymes (CYPs) are a large family of heme-containing enzymes, each with preferred substrate specificities. These enzymes are primarily expressed in hepatocytes, in which they localize to the endoplasmic reticulum, but are also found in skin, lungs, gastrointestinal mucosa, and other organs. **The P-450 system catalyzes reactions that either detoxify xenobiotics or, less commonly, convert xenobiotics into active compounds that cause cellular injury.** Both types of reactions may produce, as a by-product, reactive oxygen species (ROS), which can cause cellular damage (Chapter 2). Examples of metabolic activation of chemicals through CYPs are the production of the toxic trichloromethyl free radical from carbon tetrachloride in the liver and the generation of a DNA-binding metabolite from benzo[a]pyrene, a carcinogen present in cigarette smoke, in the lung. CYPs participate in the metabolism of alcohol (discussed later) and many common therapeutic drugs, such as acetaminophen, barbiturates, warfarin, and anticonvulsants.

There is great variation in the activity of CYPs among individuals. The variation may be a consequence of genetic polymorphisms in specific

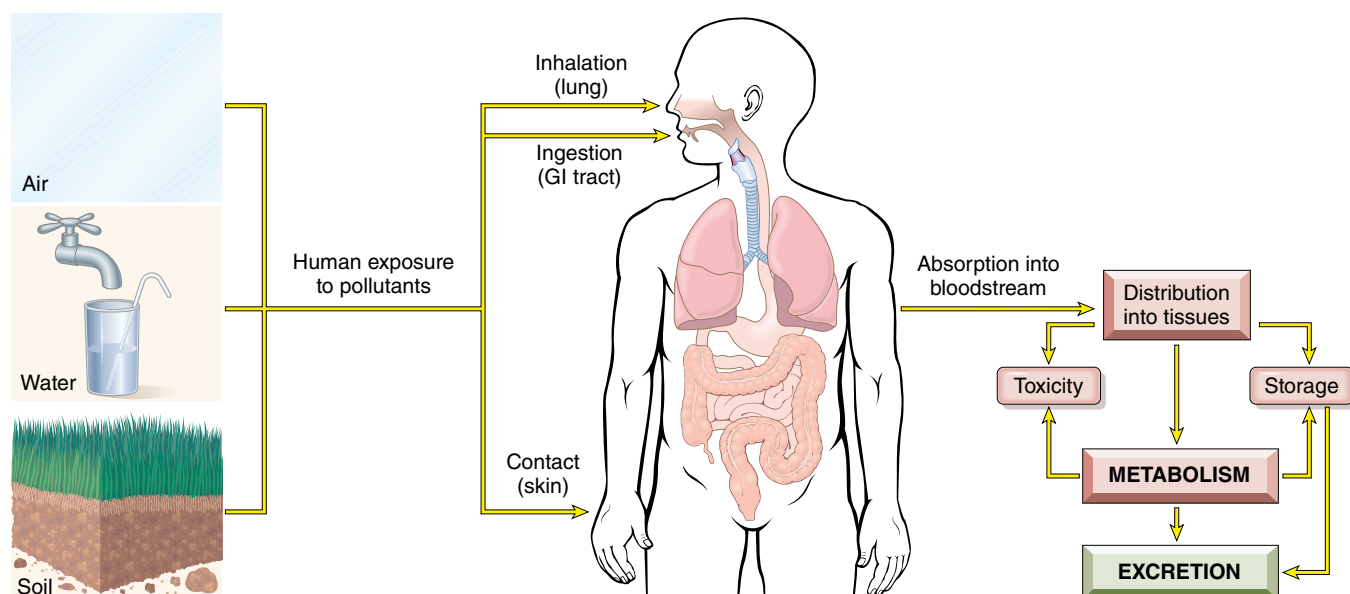


Fig. 9.3 Human exposure to pollutants. Pollutants present in the air, water, and soil can be absorbed through the lungs, gastrointestinal (GI) tract, and skin. Once inside the body, these pollutants either act at the site of absorption or are transported via the bloodstream to various organs, where they can be stored or metabolized. Xenobiotics, such as environmental chemicals and toxins, may be metabolized to water-soluble compounds (detoxification) that are excreted or, in some cases, may be metabolized into toxic compounds (activation).

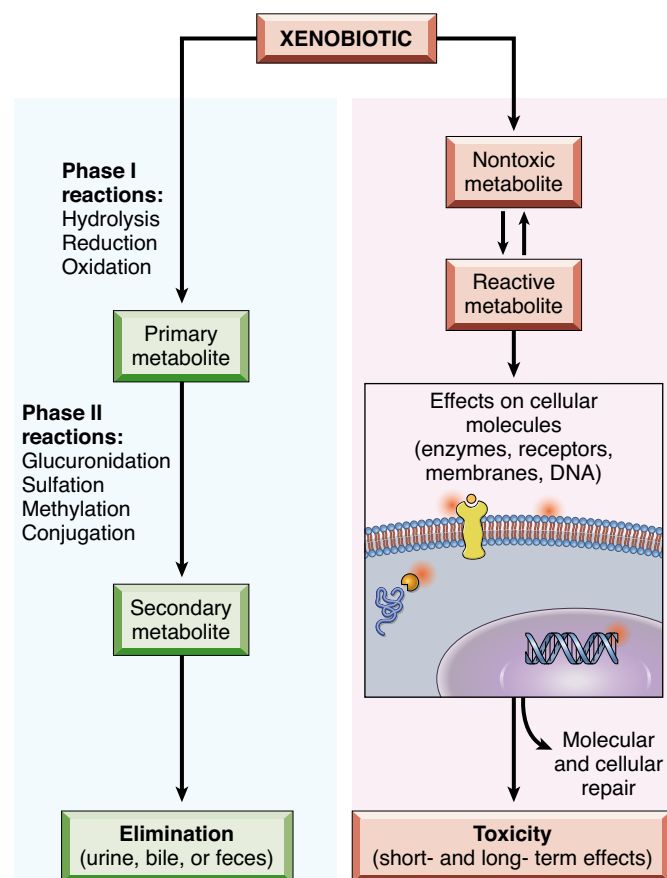


Fig. 9.4 Xenobiotic metabolism. (A) In some instances, xenobiotics are metabolized within the body to nontoxic metabolites that are eliminated (detoxification, *left panel*). Alternatively, xenobiotic metabolism may give rise to reactive metabolites that are harmful to cellular components (*right panel*). If cellular repair mechanisms are not effective, both short-term and long-term toxicity occurs. (Based on Hodgson E. *A Textbook of Modern Toxicology*. 3rd ed. Wiley; 2004.)

CYPs, but more commonly it is due to exposure to drugs or chemicals that either induce or diminish CYP activity. Known CYP inducers include environmental chemicals, drugs, smoking, alcohol, and hormones. By contrast, fasting or starvation can decrease CYP activity. Inducers of CYP act by binding to specific nuclear receptors, which then heterodimerize with the *retinoid X receptor* (RXR) to form a transcriptional activation complex that associates with promoter elements located in the 5'-flanking regions of CYP genes. Nuclear receptors participating in CYP induction include the aryl hydrocarbon receptor, the peroxisome proliferator-activated receptors (PPARs), and two orphan nuclear receptors, constitutive androstane receptor (CAR) and pregnane X receptor (PXR).

This brief overview of the general mechanisms of toxicity provides the background for the discussion of environmental diseases presented in this chapter.

ENVIRONMENTAL POLLUTION

Air Pollution

Air pollution is a significant cause of morbidity and mortality worldwide, particularly among at-risk individuals with preexisting pulmonary or cardiac disease. Air is precious to life but can also carry many potential causes of disease. Airborne microorganisms have long been significant causes of morbidity and mortality, especially in low-resource

countries. More widespread are airborne chemicals and particulate pollutants, especially in developed nations. Here we consider these hazards in outdoor and indoor air.

Outdoor Air Pollution

The ambient air in industrialized nations is contaminated with an unsavory mixture of gaseous and particulate pollutants, more heavily in cities and in proximity to heavy industry. In the United States, the Environmental Protection Agency (EPA) sets and enforces National Ambient Air Quality Standards (NAAQS) to limit air pollution. These standards regulate six pollutants: sulfur dioxide (SO₂), carbon monoxide (CO), ozone (O₃), nitrogen dioxide (NO₂), lead (Pb), and particulate matter (PM), which is categorized by size, with PM₁₀ indicating particles less than 10 μm in diameter and PM_{2.5} referring to particles less than 2.5 μm in diameter. Collectively, these contribute to the formation of *smog* (smoke and fog) that sometimes blankets large cities such as Beijing, Los Angeles, Houston, Cairo, New Delhi, Mexico City, and São Paulo. While air pollution may seem like a modern issue, concerns about its impact on health stretch back centuries. John Evelyn wrote in 1661 that inhabitants of London suffered from “Catharrs, Phthisicks and Consumptions” (bronchitis, pneumonia, and tuberculosis) and breathed “nothing but an impure and thick mist, accompanied by a fuliginous and filthy vapour, which renders them obnoxious to a thousand inconveniences, corrupting the lungs, and disordering the entire habit of their bodies.” The first environmental control law, proclaimed by Edward I in 1306, was straightforward in its simplicity: “whoever should be found guilty of burning coal shall suffer the loss of his head.” What has changed in modern times is the nature and sources of air pollutants and the types of regulations that control their emission.

Although the lungs bear the brunt of the adverse consequences, air pollutants can affect many other organ systems, possibly by inducing systemic inflammation. Here we will make some general comments on smoking and pollutant-caused lung diseases, which are covered in more detail in Chapter 15. Major health effects of outdoor pollutants are summarized in Table 9.1. Ozone, sulfur dioxide, particulates, and CO are discussed next.

Ozone (O₃) is produced by interaction of ultraviolet (UV) radiation and oxygen (O₂) in the stratosphere and accumulates in the so-called ozone layer 10 to 30 miles above the earth's surface. This layer protects life on earth by absorbing the most dangerous UV radiation emitted by the sun. However, in 1985, scientists discovered that ozone was nearly completely depleted over Antarctica and was thinned elsewhere, a dire effect stemming from the widespread use of chlorofluorocarbon (CFCs) gases in air conditioners and refrigerators and as aerosol propellants. When released into the atmosphere, CFCs rise into the stratosphere and trigger chemical reactions that destroy ozone molecules. Due to prevailing stratospheric air currents, the resulting depletion is most profound in polar regions, particularly over Antarctica during the winter months. Recognition of the problem led to the Montreal Protocol in 1987, international agreements that call for a complete phase-out of chlorofluorocarbon use by 2020 in high-resource countries and by 2040 in low-resource countries. Decreased use of chlorofluorocarbons over the past 30 years has reduced the size of the yearly ozone “hole” over Antarctica, indicating that this global environmental challenge is being met successfully.

In contrast to the “good” ozone in the stratosphere, ozone that accumulates in the lower atmosphere (ground-level ozone) is one of the most pernicious air pollutants. Ground-level ozone is a gas formed by the reaction of nitrogen oxides and volatile organic compounds in the presence of sunlight. These chemicals are released by industrial emissions and motor vehicle exhaust. Ozone toxicity is in large part mediated by the production of free radicals, which injure epithelial cells along the respiratory tract and type I alveolar cells and cause the release of inflammatory mediators. Healthy individuals exposed to ozone experience upper respiratory tract inflammation and mild symptoms (decreased

TABLE 9.1 Health Effects of Outdoor Air Pollutants

Pollutant	Populations at Risk	Effects
Ozone	Healthy adults and children	Decreased lung function Increased airway reactivity Lung inflammation
	Athletes, outdoor workers	Decreased exercise capacity
	Asthmatics	Increased hospitalizations
Nitrogen dioxide	Healthy adults	Increased airway reactivity
	Asthmatics	Decreased lung function
	Children	Increased respiratory infections
Sulfur dioxide	Healthy adults	Increased respiratory symptoms
	Individuals with chronic lung disease	Increased mortality
	Asthmatics	Increased hospitalization Decreased lung function
Acid aerosols	Healthy adults	Altered mucociliary clearance
	Children	Increased respiratory infections
	Asthmatics	Decreased lung function Increased hospitalizations
Particulates	Children	Increased respiratory infections
	Individuals with chronic lung or heart disease	Decreased lung function
	Asthmatics	Excess mortality Increased attacks

From Bascom R, Bromberg PA, Costa DL, et al. Health effects of outdoor air pollution. *Am J Respir Crit Care Med*. 1996;153:477-498.

lung function and chest discomfort), but exposure is much more dangerous for people with asthma or chronic obstructive lung disease.

To make matters worse, ozone often combines with other agents such as sulfur dioxide and particulates to create a veritable “witches’ brew” of pollutants. Sulfur dioxide is produced by power plants burning coal and oil, from copper smelting, and as a by-product of paper mills. Released into the air, it may be converted into sulfuric acid and sulfuric trioxide, which cause a burning sensation in the nose and throat, difficulty in breathing, and asthma attacks in susceptible individuals.

Particulate matter (known as “soot”) is a particularly important cause of morbidity and mortality related to pulmonary inflammation and secondary cardiovascular effects. Based on studies of large cities in the United States, it is estimated that there is a 0.5% increase in overall daily mortality for every 10 mg/m³ increase in 10-μm particles (PM₁₀) in outdoor air, mainly due to exacerbations of pulmonary and cardiac disease. In addition, recent experimental work suggests that inhaled particulates are potent promoters of lung cancer. Particulates are emitted by coal- and oil-fired power plants, by industrial processes burning these fuels, and by diesel exhaust. Although the particles have not been well characterized chemically or physically, fine or ultrafine particles less than 10 μm in diameter are the most harmful. They are readily inhaled into the alveoli, where they are phagocytosed by macrophages and neutrophils, which respond by releasing inflammatory mediators. In contrast, particles that are greater than 10 μm in diameter are of lesser consequence because they are generally removed in the nose or trapped by the mucociliary epithelium of the airways. The US EPA recently proposed to revise the

primary annual PM_{2.5} standard down from its current level of 12.0 μg/m³ to a range of 9.0 to 10.0 μg/m³, and other groups have urged even stricter goals.

CO is a systemic asphyxiant that is an important cause of accidental and suicidal death. CO is a nonirritating, colorless, tasteless, odorless gas that is produced during any process that results in the incomplete oxidation of hydrocarbons. From the standpoint of human health, the most important environmental source of CO is the burning of carbonaceous materials, as occurs in automotive engines, furnaces, and cigarettes. CO is short-lived in the atmosphere, being rapidly oxidized to CO₂; thus, elevated levels in ambient air are transient and occur only in close proximity to sources of CO. Chronic poisoning may occur in individuals working in environments such as tunnels, underground garages, and highway toll booths with high exposures to automobile fumes. Of greater concern is acute toxicity. In a small, closed garage, the average running car can produce sufficient CO to induce coma or death within 5 minutes, and CO concentrations can also rapidly rise to toxic levels with improper use of gasoline-powered electricity generators (e.g., during power outages) or following mine fires. CO kills in part by inducing central nervous system (CNS) depression, which appears so insidiously that victims are often unaware of their plight. Hemoglobin has a 200-fold greater affinity for CO than for oxygen, and the resultant carboxyhemoglobin cannot carry O₂. Systemic hypoxia develops when the hemoglobin is 20% to 30% saturated with CO; unconsciousness and death are likely with 60% to 70% saturation.

MORPHOLOGY

Chronic poisoning by CO develops because carboxyhemoglobin, once formed, is very stable. Even with low-level, but persistent, exposure to CO, carboxyhemoglobin may rise to life-threatening levels in the blood. The slowly developing hypoxia can insidiously evoke widespread ischemic changes in the CNS; these are particularly marked in the basal ganglia and lenticular nuclei. With cessation of exposure to CO, the patient usually recovers, but there may be permanent neurologic sequelae, such as impairment of memory, vision, hearing, and speech. The diagnosis is made by measuring carboxyhemoglobin levels in the blood.

Acute poisoning by CO is generally a consequence of accidental exposure or suicide attempts. In light-skinned individuals, **acute poisoning is marked by a characteristic generalized cherry-red color of the skin and mucous membranes**, which results from high levels of carboxyhemoglobin. This effect of CO on coloration may result in a failure to recognize the oxygen-starved state of the victim (and is used by the meat industry in the United States to keep meat appearing fresh—caveat emptor!). If death occurs rapidly, morphologic changes may not be present; with longer survival, the brain may be slightly edematous, with punctate hemorrhages and hypoxia-induced neuronal changes. The morphologic changes are not specific and stem from systemic hypoxia.

Indoor Air Pollution

As homes are increasingly designed to exclude the outside environment, the potential for pollution of the indoor air increases. The most common pollutant is tobacco smoke (discussed later), but additional offenders are CO, nitrogen dioxide (both already mentioned as outdoor pollutants), and asbestos (Chapter 15). Volatile substances containing polycyclic aromatic hydrocarbons generated by cooking oils and coal burning are important indoor pollutants in low-resource parts of the world, particularly parts of Asia. Only a few comments about other agents are made here.

- *Smoke from burning organic materials*, containing various oxides of nitrogen and carbon particulates, is an irritant that predisposes exposed

persons to lung infections and may contain carcinogenic polycyclic hydrocarbons. It is estimated that one-third of the world's people burn carbon-containing material such as wood, dung, charcoal, kerosene, lignite, and peat in their homes for cooking, heating, and light.

- *Bioaerosols* carry a range of agents, including microbes capable of causing infectious diseases such as Legionnaires' disease, viral pneumonia, and the common cold, to less threatening but nonetheless distressing allergens derived from pet dander, dust mites, fungi, and molds that are variously responsible for rhinitis, eye irritation, and asthma.
- *Radon*, a radioactive gas derived from uranium widely present in soil and in homes, can cause lung cancer in uranium miners. It is also suspected that low-level chronic exposures in the home increase lung cancer risk, particularly in those who smoke tobacco. Radon is the number one cause of lung cancer among nonsmokers, according to EPA estimates. Overall, radon is the second leading cause of lung cancer. Radon is responsible for about 21,000 lung cancer deaths every year. About 2900 of these deaths occur among people who have never smoked.
- *Formaldehyde* is used in the manufacture of building materials (e.g., cabinetry, furniture, adhesives) and may accumulate in the air in poorly ventilated housing. At concentrations of 0.1 ppm or higher, it causes breathing difficulties and a burning sensation in the eyes and throat and can trigger asthma attacks. Formaldehyde is classified as a carcinogen for humans and animals.
- *Sick building syndrome* is a condition that develops in office workers who spend their days mostly indoors. It is characterized by headaches and respiratory problems. Its underlying cause remains elusive; it may be a consequence of exposure to one or more indoor pollutants, possibly due to poor ventilation.

KEY CONCEPTS

Environmental Diseases and Environmental Pollution

- Environmental diseases are health conditions caused by exposure to harmful chemical or physical agents in the ambient, workplace, and personal environments.
- Xenobiotics are foreign chemicals that enter the body through inhalation, ingestion, and skin contact and can either be eliminated or accumulate in fat, bone, brain, and other tissues.
- Xenobiotics can be converted into nontoxic products or activated to generate toxic compounds through a two-phase reaction process that involves the cytochrome P-450 system.
- The most common and important air pollutants are ozone (which in combination with oxides and particulate matter forms smog), sulfur dioxide, acid aerosols, and particles less than 10 μm in diameter.
- CO poisoning is an important cause of death from accidents and suicide; it binds hemoglobin with high affinity, leading to systemic asphyxiation associated with CNS depression.
- A variety of pollutants including smoke, bioaerosols, radon, and formaldehyde may accumulate in indoor air and cause disease.

Metals as Environmental Pollutants

Lead, mercury, arsenic, and cadmium are the heavy metals most associated with harmful effects in humans.

Lead

Lead is a readily absorbed metal that binds to sulfhydryl groups in proteins and interferes with calcium metabolism, effects that cause hematologic, skeletal, neurologic, gastrointestinal, and renal toxicities. Lead exposure may occur through contaminated air, food, and water. For most of the twentieth century the major sources of lead in the environment were lead-containing house paints and gasoline.

Lead remains a significant health hazard, especially for children, despite regulations limiting its use in residential paints and the ban on lead in gasoline introduced in the United States in 1996. There are many sources of lead in the environment such as from mining, foundries, batteries, and spray painting that constitute occupational hazards. Flaking lead paint in older houses and soil contamination pose major hazards to youngsters. Blood levels of lead in children living in older homes containing lead-based paint or lead-contaminated dust often exceed 3.5 $\mu\text{g}/\text{dL}$, the level at which the Centers for Disease Control and Prevention (CDC) recommends that measures be taken to limit further exposure. A dramatic case of lead contamination of drinking water occurred in Flint, Michigan, in 2014 to 2016. The Flint water crisis occurred when the source of water supply to the city was changed from Lake Huron to the Flint River. Because water from the Flint River had a higher chloride concentration than the lake waters, it leached lead from century-old lead pipes. This caused an increase in lead levels in tap water above the acceptable limit of 15 parts per billion (ppb) in about 25% of the homes and in some cases as high as 13,200 ppb. As a result, 6000 to 12,000 residents, including many children, developed very high lead levels in their blood. Ingested lead is particularly harmful to children because they absorb more than 50% of lead from food (as compared to approximately 15% in adults) and have a more permeable blood-brain barrier, factors that increase their risk of brain damage. The main clinical features of lead poisoning in children and adults are shown in Figs. 9.5 and 9.6.

Most absorbed lead (80% to 85%) is taken up into newly deposited tooth and bone, where it competes with calcium, binds phosphates, and has a half-life of 20 to 30 years. About 5% to 10% of the absorbed

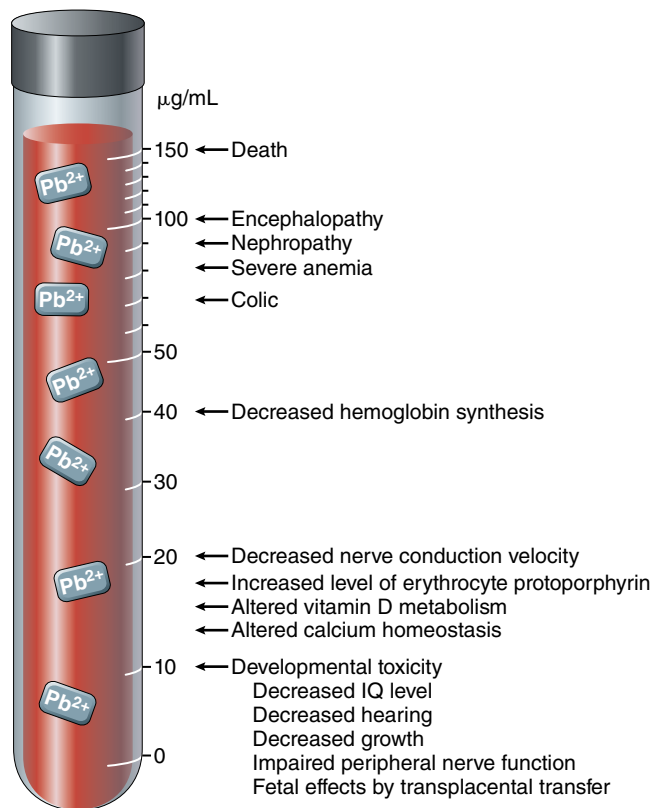


Fig. 9.5 Effects of lead poisoning in children related to blood levels. Pb^{2+} , Lead. (Modified from Bellinger DC, Bellinger AM. Childhood lead poisoning: the tortuous path from science to policy. *J Clin Invest*. 2006;116:853-857.)

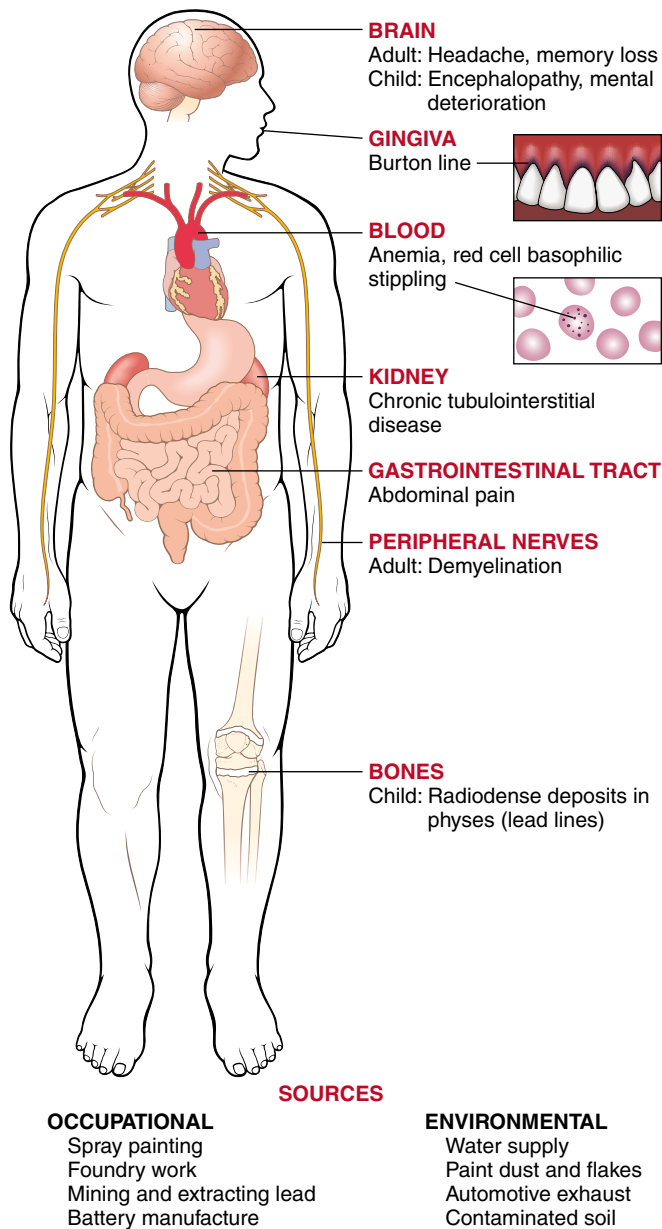


Fig. 9.6 Pathologic features of lead poisoning in adults.

lead remains in the blood, and the remainder is distributed throughout soft tissues. Excess lead is toxic to nervous tissues in adults and children; peripheral neuropathies predominate in adults, whereas central effects are more common in children. The effects of chronic lead exposure in children may be subtle, producing mild dysfunction, including reduced intelligence quotient (IQ), learning disabilities, and delayed psychomotor development. At higher doses, however, the results can be devastating, taking the form of blindness, psychoses, seizures, coma, and even death. Lead poisoning is more dangerous in children than in adults. Lead-induced peripheral neuropathies in adults such as those leading to wristdrop and footdrop generally remit with chelation therapy and cessation of exposure, but both peripheral and central nervous system abnormalities in children are usually irreversible.

Excess lead also affects many other tissues. It interferes with the normal remodeling of calcified cartilage and primary bone trabeculae in the epiphyses in children, causing increased bone density detected on x-rays as radiodense *lead lines* (Fig. 9.7A). Lead lines of a different sort may

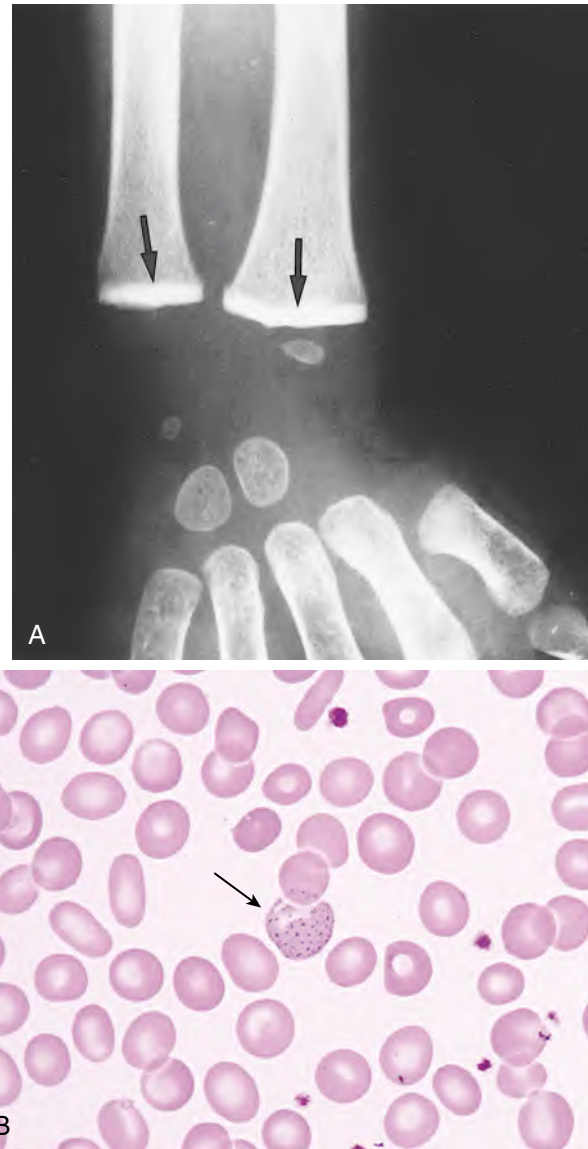


Fig. 9.7 Clinical findings in lead poisoning. (A) Impaired remodeling of calcified cartilage in the epiphyses (arrows) of the wrist has caused a marked increase in their radiodensity, making them as radiopaque as cortical bone. (B) Basophilic stippling. Coarse stippling in red cells (arrow) is common in lead poisoning as well as other anemias caused by impaired hemoglobin synthesis. (A, Courtesy Dr. GW Dietz, Department of Radiology, University of Texas Southwestern Medical School, Dallas. B, From McPherson RA, Pincus MR. *Henry's Clinical Diagnosis and Management by Laboratory Methods*. 22nd ed. Elsevier; 2011:526.)

occur in the gums, where lead stimulates hyperpigmentation; darkening of the tongue may also occur. Lead inhibits the healing of fractures by increasing chondrogenesis and delaying cartilage mineralization. Excretion of lead occurs by way of the kidneys, and acute exposure to large amounts of lead may cause damage to proximal tubules.

Lead has a high affinity for sulfhydryl groups and interferes with two enzymes involved in heme synthesis: delta-aminolevulinic acid dehydratase and ferrochelatase. Decreased activity of these enzymes impairs iron incorporation into heme, leading to *microcytosis* (small red cells) and anemia. Lead also inhibits sodium- and potassium-dependent adenosine triphosphatases (ATPases) in cell membranes,

an effect that may increase the fragility of red cells, causing hemolysis. The diagnosis of lead poisoning requires constant vigilance. It may be suspected based on neurologic changes in children or unexplained microcytic anemia with basophilic stippling in red cells in adults and children. Elevated blood lead and red cell free protoporphyrin levels (greater than 50 $\mu\text{g/dL}$) or, alternatively, zinc protoporphyrin levels, are required for definitive diagnosis. In milder cases of lead exposure, anemia may be the only abnormality.

MORPHOLOGY

The major anatomic targets of lead toxicity are the bone marrow and blood, nervous system, gastrointestinal tract, and kidneys (see Fig. 9.6).

Blood and marrow changes occur in days to weeks and are characteristic. In the peripheral blood the defect in hemoglobin synthesis appears as a **microcytic, hypochromic anemia** that is often accompanied by mild **hemolysis**. Even more distinctive is **punctate basophilic stippling of red cells** (Fig. 9.7B).

Brain damage is prone to occur in children. In young children, sensory, motor, intellectual, and psychologic deficits have been described. Lead toxicity in a pregnant female may impair brain development in the fetus. The anatomic changes underlying the subtle functional deficits are ill-defined, but there is concern that these defects may be permanent. At the more severe end of the spectrum there is marked brain edema, demyelination of the cerebral and cerebellar white matter, and necrosis of cortical neurons accompanied by diffuse astrocytic proliferation. In adults the CNS is less often affected, but frequently a **peripheral demyelinating neuropathy** appears, typically involving the motor nerves of the most frequently used muscles. Thus, the extensor muscles of the wrist and fingers are often the first to be affected (causing wristdrop), followed by paralysis of the peroneal muscles (causing footdrop).

The **gastrointestinal tract** is also a major source of clinical manifestations. Lead colic is characterized by severe, poorly localized abdominal pain.

Kidneys may develop proximal tubular damage associated with intranuclear inclusions consisting of protein aggregates. Chronic renal damage leads eventually to interstitial fibrosis and renal failure. Decreases in uric acid excretion can lead to gout (saturnine gout).

Mercury

Like lead, mercury binds strongly to sulfhydryl groups in certain proteins with high affinity, leading to damage in the CNS and the kidney. Mercury has had many uses throughout history, for example, as a pigment in cave paintings, a cosmetic, a remedy for syphilis, and a component of diuretics. Alchemists tried (without much success) to produce gold from mercury. Poisoning from inhalation of mercury vapors has long been recognized and is associated with tremor, gingivitis, and bizarre behavior, such as that displayed by the Mad Hatter in *Alice in Wonderland*. There are three forms of mercury: metallic mercury (also referred to as elemental mercury), inorganic mercury compounds (mostly mercuric chloride), and organic mercury (mostly methyl mercury). Today, the main sources of exposure to mercury are contaminated fish (methyl mercury) and mercury vapors released from metallic mercury in dental amalgams, a possible occupational hazard for dental workers. In some areas of the world, mercury used in gold mining has contaminated rivers and streams.

Inorganic mercury from the natural degassing of the earth's crust or from industrial contamination is converted to organic compounds such as methyl mercury by bacteria. Methyl mercury enters the food chain, and in carnivorous fish such as swordfish, shark, and bluefish, mercury levels may be 1 million times higher than in the surrounding water. Almost 90% of ingested methyl mercury is absorbed in the gastrointestinal tract. The

consumption of fish contaminated by the release of methyl mercury from industrial sources in Minamata Bay and the Agano River in Japan caused widespread morbidity and mortality. Acute exposure through consumption of bread made from grain treated with a methyl mercury-based fungicide in Iraq in 1971 resulted in hundreds of deaths and thousands of hospitalizations. The medical disorders associated with the Minamata episode became known as *Minamata disease* and include cerebral palsy, deafness, blindness, intellectual disability, and major CNS defects in children exposed in utero. The lipid solubility of methyl mercury and metallic mercury facilitates their accumulation in the brain, disturbing neuromotor, cognitive, and behavioral functions. Intracellular GSH, by acting as a sulfhydryl donor, is the main protective mechanism against mercury-induced CNS and kidney damage.

Mercury continues to be released into the environment by power plants and other industrial sources, and there are serious concerns about the effects of chronic low-level exposure to methyl mercury in the food supply. To protect against potential fetal brain damage, the CDC has recommended that pregnant females avoid consumption of fish known to contain high levels of mercury. Ingested mercury can injure the gut and cause ulcerations and bloody diarrhea. In the kidneys, mercury can cause acute tubular necrosis and renal failure. Chronic exposure can cause nephrotic syndrome.

Arsenic

Arsenic salts interfere with several aspects of cellular metabolism, leading to toxicities that are most prominent in the gastrointestinal tract, nervous system, skin, and heart. Arsenic was the poison of choice in Renaissance Italy, with members of the Borgia and Medici families being highly skilled practitioners of the art of its use. Because of its favored use as an instrument of assassination among royal families, arsenic has been called "the poison of kings and the king of poisons." Deliberate poisoning is rare today, but unintentional exposure to arsenic is an important health problem in many areas of the world. Arsenic is found naturally in soils and water and is used in products such as wood preservers and herbicides and other agricultural products. It may be released into the environment from mines and smelting industries. Arsenic is present in Chinese and Indian herbal medicine, and high concentrations of inorganic arsenic are present in groundwater in countries such as Bangladesh, Chile, and China. It is estimated that 40 million people in Bangladesh drink water contaminated with arsenic, constituting one of the greatest environmental cancer risks yet uncovered.

The most toxic forms of arsenic are the trivalent compounds arsenic trioxide, sodium arsenite, and arsenic trichloride. If ingested in large quantities, arsenic causes acute gastrointestinal, cardiovascular, and CNS toxicities that are often fatal. These effects may be attributed in part to interference with mitochondrial oxidative phosphorylation since trivalent arsenic can replace the phosphates in ATP. However, arsenic also has pleiotropic effects on the activity of several other enzymes and ion channels, and these too may contribute to certain toxicities.

- *Neurologic effects* usually occur 2 to 8 weeks after exposure and consist of a sensorimotor neuropathy that causes paresthesias, numbness, and pain.
- *Cardiovascular effects* include hypertension and prolonged Q-Tc interval with ventricular arrhythmias.
- *Skin changes* consisting of numerous hypopigmented macules on a background of hyperpigmentation, producing a raindrop-like appearance, and hyperkeratotic papules on the palms and soles, which occur with chronic exposure.
- *Increased risk for the development of cancers* is the most serious consequence of chronic exposure, particularly of the lung, bladder, and skin. Arsenic-induced skin tumors differ from those induced by sunlight; they are often multiple and usually appear on the palms

and soles. The mechanisms of arsenic carcinogenesis in skin and lung have not been elucidated but may involve defects in nucleotide excision repair mechanisms that protect against DNA damage.

Cadmium

Cadmium is preferentially toxic to the kidneys and the lungs through uncertain mechanisms that may involve increased production of ROS. In contrast to the other metals discussed in this section, cadmium toxicity is a relatively modern problem. It is an occupational and environmental pollutant generated by mining, electroplating, and production of nickel-cadmium batteries, which are usually disposed of as household waste. Cadmium can contaminate the soil and plants directly or through fertilizers and irrigation water. Food is the most important source of cadmium exposure for the general population. Its toxic effects require its uptake into cells via transporters such as ZIP8, which normally serves as a transporter for zinc.

The principal toxic effects of excess cadmium are a form of obstructive lung disease caused by necrosis of alveolar epithelial cells and renal tubular damage that may progress to end-stage renal disease. Cadmium exposure can also cause skeletal abnormalities associated with calcium loss. Cadmium-containing water used to irrigate rice fields in Japan caused a disease in postmenopausal females known as *itai-itai* (“ouch-ouch”), a combination of osteoporosis and osteomalacia associated with renal disease. Cadmium exposure is also associated with an elevated risk of lung cancer, which has been demonstrated in workers exposed occupationally and in populations living near zinc smelters. Cadmium is not directly genotoxic and most likely produces DNA damage through the generation of ROS (Chapter 2).

KEY CONCEPTS

Toxic Effects of Heavy Metals

- Lead, mercury, arsenic, and cadmium are the heavy metals most associated with toxic effects in humans.
- Children absorb more ingested lead than adults; the main source of exposure for children is lead-containing paint in older housing and lead-containing drinking water.
- Excess lead causes CNS defects in children and peripheral neuropathy in adults. It also interferes with the remodeling of cartilage and causes anemia by interfering with hemoglobin synthesis.
- The major source of exposure to mercury is contaminated fish. The developing brain is highly sensitive to methyl mercury, which accumulates in the CNS.
- Exposure of the fetus to high levels of mercury in utero may lead to Minamata disease, characterized by cerebral palsy, deafness, and blindness.
- Arsenic is naturally found in soil and water and is a component of some wood preservatives and herbicides. Excess arsenic interferes with mitochondrial oxidative phosphorylation and the function of a variety of proteins. It causes toxic effects in the gastrointestinal tract, CNS, and cardiovascular system; long-term exposure causes skin lesions and carcinomas.
- Cadmium from nickel-cadmium batteries and chemical fertilizers can contaminate soil. Excess cadmium causes obstructive lung disease and kidney damage.

OCCUPATIONAL HEALTH RISKS: INDUSTRIAL AND AGRICULTURAL EXPOSURES

More than 10 million occupational injuries occur annually in the United States, and approximately 65,000 people die because of occupational injuries and illnesses. About 10% of these deaths occur due to falls and other traumatic injuries, while the remainder are secondary to

toxic exposures that lead to cancer, lung disease, and other potentially fatal conditions. Industrial exposures to toxic agents are as varied as the industries themselves. They range from merely annoying irritations of respiratory airways by formaldehyde or ammonia fumes to lung cancers arising from exposure to asbestos, arsenic, or uranium. Human diseases associated with occupational exposures are listed in Table 9.2. In addition to toxic metals (already discussed), other important agents that contribute to environmental diseases include:

- **Organic solvents** are widely used in huge quantities worldwide. Some, such as *chloroform* and *carbon tetrachloride*, are found in degreasing and dry-cleaning agents and paint removers. Acute exposure to high levels of these agents can cause dizziness and confusion, CNS depression, and even coma. Lower levels may cause liver and kidney toxicity. Occupational exposure to *benzene* and *1,3-butadiene* increases the risk of leukemia. Benzene is oxidized to an epoxide through hepatic CYP2E1, a component of the P-450 enzyme system already mentioned. The epoxide and other metabolites disrupt progenitor cell differentiation in the bone marrow and may lead to marrow aplasia and acute myeloid leukemia.
- **Polycyclic hydrocarbons** are released during the combustion of coal and gas, particularly at the high temperatures used in steel foundries, and are also present in tar and soot. (Pott identified soot as the cause of scrotal cancers in chimney sweeps in 1775, as mentioned in Chapter 7). Polycyclic hydrocarbons are among the most potent carcinogens, and industrial exposures have been implicated in the causation of lung and bladder cancer.
- **Organochlorines** (and halogenated organic compounds in general) are synthetic products that resist degradation and are lipophilic. Organochlorines used as pesticides include *dichlorodiphenyltrichloroethane* (DDT) and its metabolites and agents such as lindane, aldrin, and dieldrin. Nonpesticide organochlorines include *polychlorinated biphenyls* (PCBs) and *dioxin* (2,3,7,8-tetrachlorodibenzo-*p*-dioxin [TCDD]). DDT was banned in the United States in 1973. Acute DDT poisoning in humans causes neurologic toxicity. Most organochlorines are endocrine disruptors and have antiestrogenic or antiandrogenic activity in laboratory animals, but long-term health effects in humans have not been firmly established.

Dioxins and PCBs can cause skin disorders such as folliculitis and acneiform dermatosis known as chloracne, which consists of acne, cyst formation, hyperpigmentation, and hyperkeratosis, generally around the face and behind the ears. It can be accompanied by abnormalities in the liver and CNS. Because PCBs induce the P-450 enzyme system, workers exposed to these substances may show altered drug metabolism. Environmental disasters in Japan and China in the late 1960s caused by the consumption of rice oil contaminated by PCBs poisoned about 2000 people in each episode. The primary manifestations of the disease (yushō in Japan, yu-cheng in China) were chloracne and hyperpigmentation of the skin and nails.

- **Bisphenol A (BPA)** is used in the synthesis of polycarbonate food and water containers and of epoxy resins that line almost all food bottles and cans; as a result, exposure to BPA is virtually ubiquitous in humans. BPA has long been known as a potential endocrine disruptor. Several large retrospective studies have linked elevated urinary BPA levels to heart disease in adult populations. In addition, infants who drink from BPA-containing containers may be particularly susceptible to the endocrine effects of BPA. In 2010, Canada was the first country to list BPA as a toxic substance, and the largest makers of baby bottles and “sippy” cups have stopped using BPA in the manufacturing process. The extent of the human health risks associated with BPA remains uncertain.

TABLE 9.2 Human Diseases Associated With Occupational Exposures

Organ/System	Effect	Toxicant
Cardiovascular system	Heart disease	Carbon monoxide, lead, solvents, cobalt, cadmium
Respiratory system	Nasal cancer	Isopropyl alcohol, wood dust
	Lung cancer	Radon, asbestos, silica, bis(chloromethyl) ether, nickel, arsenic, chromium, mustard gas, uranium
	Chronic obstructive pulmonary disease	Grain dust, coal dust, cadmium
	Hypersensitivity	Beryllium, isocyanates
	Irritation	Ammonia, sulfur oxides, formaldehyde
	Fibrosis	Silica, asbestos, cobalt
Nervous system	Peripheral neuropathies	Solvents, acrylamide, methyl chloride, mercury, lead, arsenic, DDT
	Ataxic gait	Chlordane, toluene, acrylamide, mercury
	CNS depression	Alcohols, ketones, aldehydes, solvents
	Cataracts	Ultraviolet radiation
Urinary system	Renal toxicity	Mercury, lead, glycol ethers, solvents
	Bladder cancer	Naphthylamines, 4-aminobiphenyl, benzidine, rubber products
Reproductive system	Male infertility	Lead, phthalate plasticizers, cadmium
	Female infertility	Lead, mercury
	Stillbirth	Lead, mercury
	Teratogenesis	Mercury, polychlorinated biphenyls
Hematopoietic system	Leukemia	Benzene
Skin	Folliculitis and acneiform dermatosis	Polychlorinated biphenyls, dioxins, herbicides
	Cancer	Ultraviolet radiation
Gastrointestinal tract	Liver angiosarcoma	Vinyl chloride

CNS, Central nervous system; DDT, dichlorodiphenyltrichloroethane.

Data from Leigh JP, Markowitz SB, Fahs M, Shin C, Landrigan PJ. Occupational injury and illness in the United States. Estimates of costs, morbidity, and mortality. *Arch Intern Med.* 1997;157:1557-1568; Mitchell FL. Hazardous waste. In: Rom WN, ed. *Environmental and Occupational Medicine*. 2nd ed. Little, Brown; 1992:1275; and Levi PE. Classes of toxic chemicals. In: Hodgson E, Levi PE, eds. *A Textbook of Modern Toxicology*. Appleton & Lange; 1997:229.

- Perfluoroalkyl and polyfluoroalkyl substances (known as PFAS) were only recently recognized as important environmental exposures. PFAS have a carbon backbone with variable numbers of linked fluorine atoms and are widely used to create “nonstick” coatings in plastics and cookware and in fire suppressing foams. They are chemically inert and accumulate in the environment and the body, earning them the nickname “forever chemicals.” Epidemiologic and animal studies have linked PFAS exposure to endocrine abnormalities, developmental delays in children, immune dysfunction, and increased risk of certain cancers.
- *Vinyl chloride*, used in the synthesis of polyvinyl resins, can cause angiosarcoma of the liver, a rare type of liver tumor.
- Inhalation of *mineral dusts* causes chronic, nonneoplastic lung diseases called *pneumoconioses*. This group of disorders includes diseases induced by organic and inorganic particulates as well as chemical fume- and vapor-induced nonneoplastic lung diseases. The most common pneumoconioses are caused by exposures to coal dust (in mining of hard coal), silica (in sandblasting and stone cutting), asbestos (in mining, fabrication, and insulation work), and beryllium (in mining and fabrication). Exposure to these agents nearly always occurs in the workplace. The increased risk of cancer because of asbestos exposure, however, extends to family members of asbestos workers and to other persons exposed outside the workplace. Pneumoconioses and their pathogenesis are discussed in Chapter 15.

EFFECTS OF TOBACCO

Smoking is the most readily preventable cause of death in humans. The main culprit is cigarette smoking, but smokeless tobacco (e.g.,

snuff, chewing tobacco) is also harmful to health and an important cause of oral cancer. The use of tobacco products not only creates personal risks, but passive tobacco inhalation from the environment (*second-hand smoke*) can cause lung cancer in nonsmokers. More than 20 million US residents have died of smoking-related diseases since the 1964 Surgeon General’s report on the adverse effects of smoking. Of these, almost 2.5 million died from inhalation of second-hand smoke. More than 10 times as many people in the United States have died because of cigarette smoking than have died in all the wars fought by the United States in its entire history. Annually, tobacco is responsible for more than 400,000 deaths in the United States, one-third of these due to lung cancer. Indeed, tobacco is the leading exogenous cause of human cancers, including 90% of lung cancers.

Worldwide, in 2023, there were approximately 1.3 billion smokers who consumed 5 trillion cigarette-equivalents of tobacco. Two-thirds of smokers live in 10 countries, led by China, which accounts for about 30%, and India with about 10%, followed by Indonesia, the United States, Russia, Bangladesh, Japan, Turkey, Vietnam, and the Philippines.

Between 1990 and 2020, the worldwide prevalence of smoking decreased 28% in males and 38% in females. The largest decreases in smoking prevalence were observed in Brazil (73%), Norway (54%), and Senegal (51%), while decreases of over 45% were seen in Iceland, Denmark, Canada, Australia, Colombia, and Costa Rica. These drops are attributable to tobacco control policies such as the 17 Articles of the WHO Framework Convention on Tobacco Control (FCTC), which is one of the most widely adopted United Nations Treaties.

Other trends, however, are concerning. The world’s most populous country, China, has become the world’s largest producer and consumer

TABLE 9.3 Effects of Selected Tobacco Smoke Constituents

Substance	Effect
Tar	Carcinogenesis
Polycyclic aromatic hydrocarbons	Carcinogenesis
Nicotine	Ganglionic stimulation and depression; tumor promotion
Phenol	Tumor promotion; mucosal irritation
Benzo[a]pyrene	Carcinogenesis
Carbon monoxide	Impaired oxygen transport and utilization
Formaldehyde	Toxicity to cilia; mucosal irritation
Nitrogen oxides	Toxicity to cilia; mucosal irritation
Nitrosamine	Carcinogenesis

of cigarettes. China has approximately 340 million smokers who account about one-third of the world's tobacco consumption. It is estimated that more than 2.4 million people in China and 1 million people in India die each year of smoking-related diseases. Overall, according to the World Health Organization, cigarette smoking causes more than 8 million deaths annually (7 million directly from tobacco use and 1.3 million from second-hand smoke), mostly from cardiovascular disease, various types of cancers, and chronic respiratory problems.

The harm caused by smoking increases with the amount a person smokes. This is often measured in *pack-years*, which multiply the average number of packs smoked daily by the total number of smoking years. The cumulative effect is striking. While about 75% of nonsmokers reach age 70, only about 50% of smokers live that long. The good news is that cessation of smoking improves overall health and lifespan. Within 5 years of quitting, the risk of death from heart disease drops dramatically. Lung cancer mortality also decreases approximately 20% within 5 years, but the excess risk of lung cancer persists for several decades after smoking cessation.

The number of potentially noxious chemicals in tobacco smoke is extraordinary. Tobacco smoke contains a complex mixture of 7000 chemicals, more than 60 of which have been identified as carcinogens. Table 9.3 provides only a partial list and includes various types of injuries produced by these agents.

Nicotine, an alkaloid present in tobacco leaves, is strongly addictive. Nicotine binds to nicotinic acetylcholine receptors in the brain and stimulates the release of catecholamines from sympathetic neurons. This activity is responsible for the acute effects of smoking, such as increased heart rate and blood pressure and elevated cardiac contractility and output. Recent studies indicate that in addition to being addictive, nicotine has other untoward effects. Exposure to nicotine during pregnancy can harm fetal brain development and can also increase the risk of preterm birth and stillbirth.

Smoking and Lung Cancer

Cigarette smoke has a direct irritant effect on the tracheobronchial mucosa, leading to increased mucus production (bronchitis), and contains multiple chemicals (Table 9.4) that act as potent carcinogens involved in the development of lung cancer in humans. CYPs and phase II enzymes metabolize these carcinogens to intermediates with increased water solubility, facilitating their excretion. However, some intermediates produced by CYPs are electrophilic and form DNA adducts that are repaired by error-prone mechanisms, leading to potentially oncogenic mutations (Chapter 7). Most tellingly, sequencing of the genomes of lung cancers that occur in smokers has revealed the presence of thousands of mutations of a type that is produced by carcinogens in tobacco

TABLE 9.4 Suspected Organ-Specific Carcinogens in Tobacco Smoke

Organ	Carcinogen
Lung, larynx	Polycyclic aromatic hydrocarbons 4-(Methylnitrosamino)-1-(3-pyridyl)-1-butanone (NNK) Polonium 210
Esophagus	NNN
Pancreas	NNK
Bladder	4-Aminobiphenyl, 2-naphthylamine
Oral cavity (smoking)	Polycyclic aromatic hydrocarbons, NNK, NNN
Oral cavity (snuff)	NNK, NNN, polonium 210

NNK, Nicotine-derived nitrosamine ketone; NNN, N'-nitrosonornicotine. Data from Szczesny LB, Holbrook JH. Cigarette smoking. In: Rom WH, ed. *Environmental and Occupational Medicine*. 2nd ed. Little, Brown; 1992:1211.

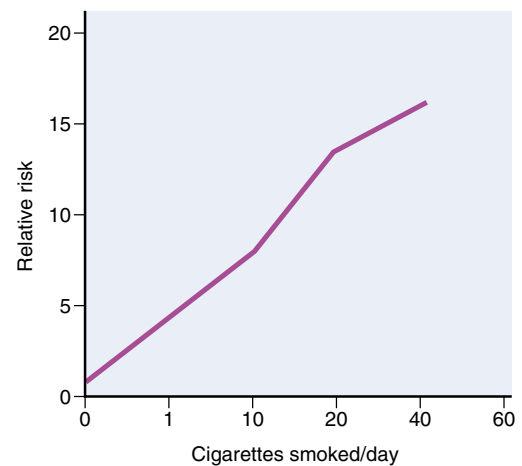


Fig. 9.8 Effects of cigarettes smoked per day on lung cancer risk. (Modified from Stewart BW, Kleihues P, eds. *World Cancer Report*. IARC Press; 2003.)

smoke in experimental models. The risk of developing lung cancer is strongly related to the number of pack-years of cigarettes smoked per day (Fig. 9.8). Moreover, smoking increases the risk of other carcinogenic influences. Witness the 10-fold higher incidence of lung carcinomas in asbestos workers and uranium miners who smoke over those who do not smoke and the interaction between tobacco consumption and alcohol in the development of oral and laryngeal cancers (Fig. 9.9).

Smoking and Other Diseases

In addition to lung cancer, smoking is linked to many other malignant and nonmalignant disorders that affect numerous organ systems (Fig. 9.10):

- *Cancers of the esophagus, pancreas, bladder, kidney, cervix, liver, colon, and hemopoietic cells.*
- *Chronic obstructive pulmonary disease (COPD).* Agents in smoke recruit leukocytes to the lung and increase local elastase production leading to COPD and its major manifestations, emphysema and chronic bronchitis, conditions that are discussed in Chapter 15. Smoking also exacerbates asthma and increases the risk for pulmonary tuberculosis.
- *Atherosclerosis and its major complications, myocardial infarction and stroke.* The causal mechanisms probably relate to several factors including increased platelet aggregation, decreased myocardial oxygen supply (because of significant lung disease coupled with the hypoxia related to the CO content of cigarette smoke) accompanied

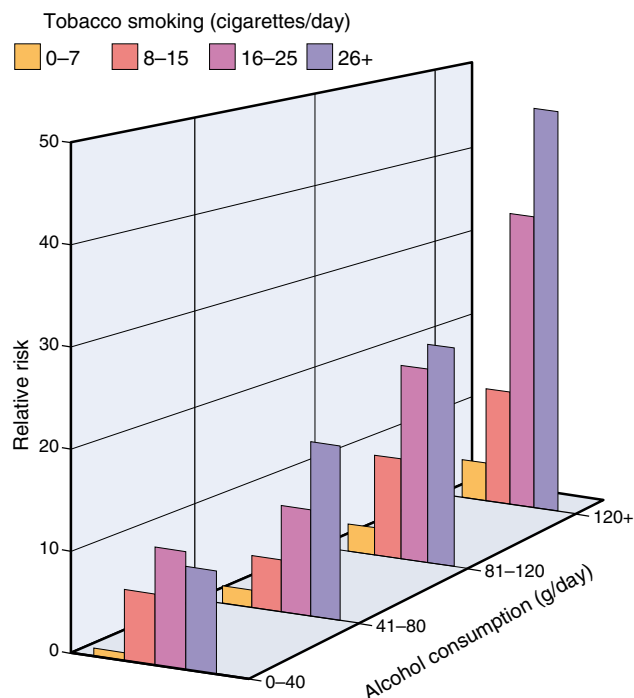


Fig. 9.9 Multiplicative increase in the risk of laryngeal cancer due to the interaction between cigarette smoking and alcohol consumption. (Modified from Stewart BW, Kleihues P, eds. *World Cancer Report*. IARC Press; 2003.)

by increased myocardial oxygen demand and a decreased threshold for ventricular fibrillation. Smoking has a multiplicative effect on the incidence of myocardial infarction when combined with hypertension and hypercholesterolemia.

- **Smoking harms the developing fetus.** Maternal smoking increases the risk of spontaneous abortions and preterm births and results in *intrauterine growth retardation* (Chapter 10). Birth weights of infants born to mothers who stopped smoking before pregnancy are, however, normal.
- **Smoking also increases the risk of several other disorders, including type 2 diabetes, rheumatoid arthritis, age-related macular degeneration, ectopic pregnancy, and erectile dysfunction.**

Exposure to environmental tobacco smoke (*passive smoke inhalation*) carries many of the same health risks as active smoking. It is estimated that the relative risk of lung cancer in nonsmokers exposed to environmental smoke is about 1.3 times higher than that of nonsmokers who are not exposed to smoke. In the United States, the CDC estimates that over 7000 lung cancer deaths in nonsmokers can be attributed each year to environmental tobacco smoke. Even more striking is the increased risk of coronary atherosclerosis and fatal myocardial infarction. Studies report that every year 30,000 to 60,000 cardiac deaths in the United States are associated with exposure to passive smoke. Children living in a household with an adult who smokes have an increased frequency of respiratory illnesses such as asthma. Passive smoke inhalation in nonsmokers can be estimated by measuring the blood levels of cotinine, a metabolite of nicotine. In the United States, median cotinine levels in nonsmokers have decreased by more than 70% since around 2000 because of the adoption of nonsmoking policies in public places. However, passive exposure to tobacco smoke in the home remains a major public health concern, particularly for children. Clearly, the transient pleasure of smoking comes with a heavy long-term price.

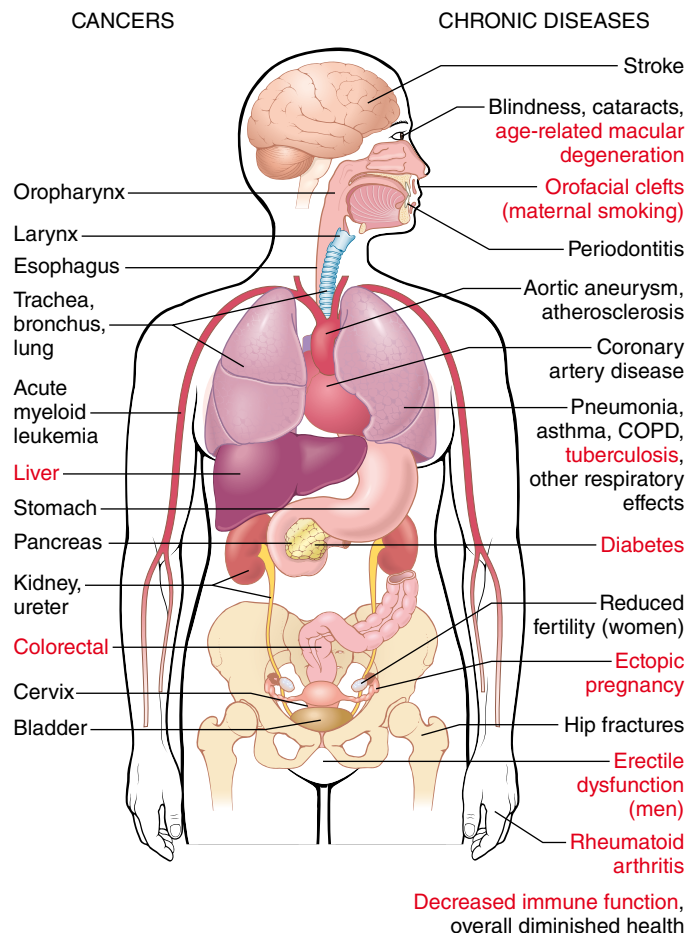


Fig. 9.10 Health consequences causally linked to smoking. Items in red are new diseases added in the 2016 report from the Surgeon General. *COPD*, Chronic obstructive pulmonary disease. (From US Department of Health and Human Services. *The Health Consequences of Smoking—50 Years of Progress: A Report of the Surgeon General*; 2016.)

Vaping involves using electronic cigarettes (e-cigarettes) or other similar devices that deliver vaporized nicotine and flavorings. These devices serve as a smoking-cessation tool for long-term tobacco smokers but have also gained popularity as recreational products, alarmingly among adolescents and children. While e-cigarettes are seen as a less harmful alternative to cigarette smoking due to the reduced exposure to harmful chemicals, vaping still comes with substantial health risks, including nicotine addiction, respiratory symptoms, and lung damage from the inhalation of chemicals.

In 2019, health officials identified a new lung condition linked to vaping. This condition, EVALI (e-cigarette or vaping product use–associated lung injury), affected thousands of people and caused dozens of deaths. Examination of lung tissue revealed lipid-laden pulmonary alveolar macrophages and vacuolated pneumocytes, suggesting a chemical-induced pneumonitis. Many affected patients vaped mixtures of nicotine, cannabinoids, and vitamin E acetate, which is suspected of causing lung damage.

EFFECTS OF ALCOHOL

The safety of ethanol consumption depends on various factors, including the amount and frequency of consumption. Excessive amounts of alcohol cause serious physical and psychologic damage. Moderate amounts—defined as one to two drinks daily—have generally not been

KEY CONCEPTS

Health Effects of Tobacco

- Smoking is the most prevalent preventable cause of human death.
- Tobacco smoke contains more than 7000 compounds. Among these are nicotine, which is responsible for tobacco addiction, and potent carcinogens—mainly, polycyclic aromatic hydrocarbons, nitrosamines, and aromatic amines. Nicotine also has other adverse effects, particularly on fetal development, and is associated with preterm birth and stillbirth.
- Approximately 90% of lung cancers occur in smokers. Smoking is also associated with an increased risk of cancers of the oral cavity, larynx, esophagus, stomach, bladder, and kidney, some forms of leukemia, as well as liver and colorectal cancer. Cessation of smoking reduces the risk of lung cancer.
- Smokeless tobacco use is an important cause of oral cancers. Tobacco consumption interacts with alcohol in multiplying the risk of oral, laryngeal, and esophageal cancer and increases the risk of lung cancers from occupational exposures to asbestos, uranium, and other agents.
- Tobacco use is an important risk factor for development of atherosclerosis and myocardial infarction, peripheral vascular disease, and cerebrovascular disease. In the lungs, in addition to cancer, it predisposes to emphysema, chronic bronchitis, and chronic obstructive pulmonary disease.
- Maternal smoking increases the risk of spontaneous abortion, premature birth, and intrauterine growth retardation.

considered detrimental; however, increasingly newer studies suggest that even small amounts of alcohol are not completely safe and are linked to the potential for serious health risks. In this section, we describe alcohol metabolism and the major health consequences associated with alcohol abuse.

Despite all the attention given to illicit drugs such as cocaine and opiates, excessive alcohol use is a far more widespread hazard and claims many more lives. According to the 2021 National Survey on Drug Use and Health, 52% of adults (over 18 years of age) and 7% of youths (ages 12 to 17) in the United States reported using alcohol in the previous month. Even more worrisome is the fact that 28.6 million adults (11.3% of this age group) and nearly 900,000 youths (3.4% of this age group) have alcohol use disorder (AUD). AUD is a chronic relapsing condition characterized by an impaired ability to stop or control alcohol use despite adverse social, occupational, or health consequences. It is estimated that alcohol consumption is responsible for the deaths of more than 140,000 people (~97,000 males, and 43,000 females) annually in the US. In 2022, alcohol-impaired driving resulted in over 13,000 deaths (about one-third of overall driving fatalities). Among all cirrhosis-related deaths, over 50% are alcohol related. Worldwide, alcohol accounts for approximately 3 million deaths every year, representing 5.3% of all deaths.

After consumption, ethanol is absorbed unaltered in the stomach and small intestine and then distributes throughout the body in direct proportion to the blood level. Less than 10% is excreted unchanged in the urine, sweat, and breath. The amount exhaled is proportional to the blood level and forms the basis for the breath test used by law enforcement agencies. An individual with blood alcohol concentration of 80 mg/dL (0.08%) or more is considered legally impaired and can face penalties for driving under the influence or driving while intoxicated in most states, but lower limits are often set for individuals under the age of 21 and operators of commercial vehicles. For an average individual, a blood alcohol concentration of 0.08% is reached after consumption of three standard drinks, about three (12 ounce) bottles of beer, 15 oz of wine, or 4 to 5 oz of 80-proof distilled spirits. Drowsiness occurs at 200 mg/dL, stupor at 300 mg/dL, and coma, with possible respiratory arrest, at higher levels. The rate of metabolism affects the blood alcohol

level. Individuals with alcohol use disorder develop tolerance to alcohol. They metabolize alcohol at a higher rate than normal and hence show lower peak levels of alcohol than average for the same amount of alcohol consumed. Most of the alcohol in the blood is metabolized to acetaldehyde in the liver by three enzyme systems: alcohol dehydrogenase, cytochrome P-450 isoenzymes, and catalase (Fig. 9.11). Of these, the main enzyme involved in alcohol metabolism is alcohol dehydrogenase, located in the cytosol of hepatocytes. At high blood alcohol levels, however, the microsomal ethanol-oxidizing system also plays an important role. This system involves cytochrome P-450 enzymes, particularly the CYP2E1 isoform, located in the smooth endoplasmic reticulum. Induction of P-450 enzymes by alcohol explains the increased susceptibility of heavy drinkers to other compounds metabolized by the same enzyme system, which include drugs (acetaminophen, cocaine), anesthetics, carcinogens, and industrial solvents. Of note, however, when alcohol is present in the blood at high concentrations, it competes with other CYP2E1 substrates and may delay the catabolism of other drugs, thereby potentiating their effects. Catalase is of minor importance, being responsible for only about 5% of alcohol metabolism. Acetaldehyde produced by these systems is in turn converted by acetaldehyde dehydrogenase to acetate, which is used in the mitochondrial respiratory chain or in lipid synthesis.

Several toxic effects result from ethanol metabolism. Listed here are only the most important of these.

- *Acetaldehyde, the direct product of alcohol oxidation, has many toxic effects and is responsible for some of the acute effects of alcohol.* The efficiency of alcohol metabolism varies among populations, depending on the expression levels of alcohol dehydrogenase and aldehyde dehydrogenase as well as the presence of genetic variants that alter enzyme activity. About 50% of Asians have very low aldehyde dehydrogenase activity due to the substitution of lysine for glutamine at residue 487 (the normal allele is termed ALDH2*1, and the inactive variant is designated as ALDH2*2). The ALDH2*2 protein has dominant-negative activity, such that even one copy of the ALDH2*2 allele reduces enzyme activity significantly. Individuals homozygous for the ALDH2*2 allele are completely unable to oxidize acetaldehyde and cannot tolerate alcohol, experiencing nausea, flushing, tachycardia, and hyperventilation after its ingestion.
- *Alcohol oxidation by alcohol dehydrogenase causes the reduction of nicotinamide adenine dinucleotide (NAD) to NADH, with a consequent decrease in NAD and increase in NADH.* NAD is required for fatty acid oxidation in the liver and for the conversion of lactate into pyruvate. Its deficiency is a main cause of the accumulation of fat in the liver of alcoholics. The increase in the NADH/NAD ratio in alcoholics also causes lactic acidosis.
- *ROS generation.* Metabolism of ethanol in the liver by CYP2E1 produces ROS, which causes lipid peroxidation of hepatocyte membranes. Alcohol also provokes the release of endotoxin (lipopolysaccharide) from gram-negative bacteria in the intestinal flora, which stimulates the production of tumor necrosis factor (TNF) and other cytokines from macrophages and Kupffer cells, leading to hepatic injury.

The adverse effects of ethanol can be classified as acute or chronic. **Acutely, alcohol exerts its effects mainly on the CNS, but it may induce hepatic and gastric changes that are reversible if alcohol consumption is discontinued.** Even with moderate intake of alcohol, multiple fat droplets accumulate in the cytoplasm of hepatocytes (fatty change or hepatic steatosis). The gastric changes are acute gastritis and ulceration. In the CNS, alcohol is a depressant, first affecting subcortical structures (probably the high brainstem reticular formation) that modulate cerebral cortical activity. Consequently, there is stimulation and disordered cortical, motor, and intellectual behavior. At

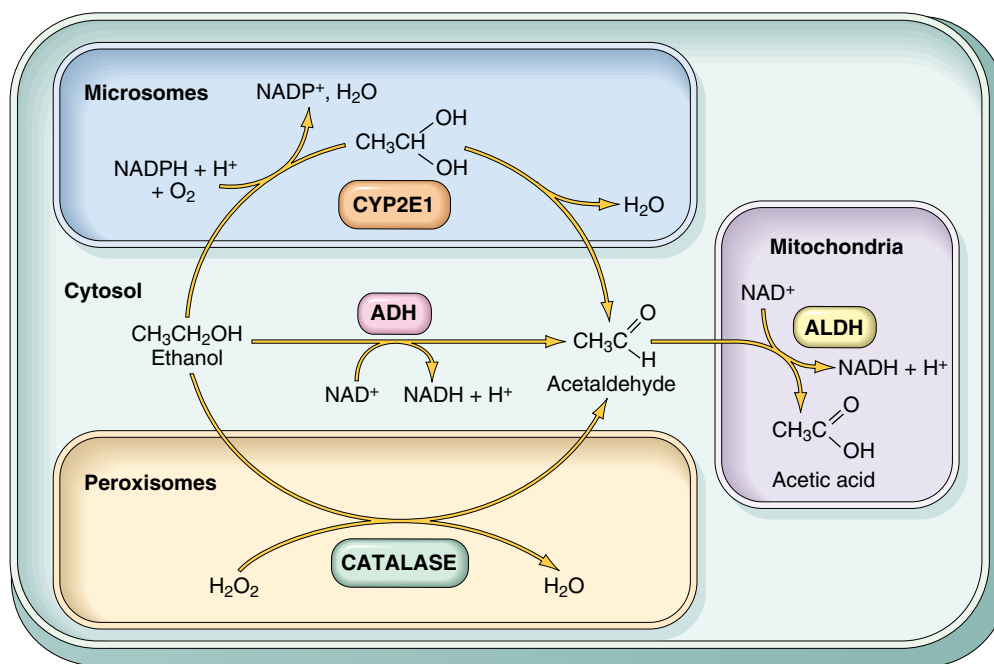


Fig. 9.11 Ethanol metabolism. Ethanol is oxidized to acetaldehyde via three different routes, eventually leading to the production of acetic acid. Oxidation by alcohol dehydrogenase (ADH) takes place in the cytosol, whereas the cytochrome P-450 system and its CYP2E1 isoform are located in the endoplasmic reticulum (microsomes), and catalase is located in peroxisomes. Oxidation of ethanol by ADH is the most important route, while catalase is responsible for only approximately 5% of ethanol metabolism. Oxidation of ethanol by CYPs may generate reactive oxygen species (not shown). Subsequent oxidation of acetaldehyde by aldehyde dehydrogenase (ALDH) occurs in mitochondria. (From Parkinson A, Ogilvie BW, Buckley DB, et al. Biotransformation of xenobiotics. In Klaassen CD, ed. *Casarett and Doull's Toxicology: The Basic Science of Poisons*. 6th ed. McGraw-Hill; 2001:133.)

progressively higher blood levels, cortical neurons and then lower medullary centers are depressed, including those that regulate respiration. Respiratory arrest may follow.

Excessive chronic alcohol use affects not only the liver and stomach, but virtually all other organs and tissues as well. People with alcohol use disorder experience significant morbidity and have a shortened lifespan, related principally to damage to the liver, gastrointestinal tract, CNS, cardiovascular system, and pancreas.

- The *liver* is the main site of chronic injury. In addition to fatty change mentioned above, chronic alcohol use disorder causes alcoholic hepatitis and cirrhosis, as described in Chapter 18. Cirrhosis is associated with portal hypertension and an increased risk for the development of hepatocellular carcinoma.
- In the *gastrointestinal tract*, chronic alcohol use disorder can cause massive bleeding from gastritis, gastric ulcer, or esophageal varices (associated with cirrhosis), which may be fatal.
- *Neurologic effects.* Thiamine (vitamin B_1) deficiency is common in those with alcohol use disorder; the principal lesions resulting from this deficiency are peripheral neuropathies and Wernicke-Korsakoff syndrome (see Table 9.9 later and Chapter 28); cerebral atrophy, cerebellar degeneration with loss of neurons in the anterior vermis that are responsible for coordination, and optic neuropathy may also occur.
- *Cardiovascular effects.* Alcohol has diverse effects on the cardiovascular system. Injury to the myocardium may produce dilated congestive cardiomyopathy (alcoholic cardiomyopathy, discussed in Chapter 12). Chronic alcoholism is also associated with an increased incidence of hypertension, and heavy alcohol consumption, with attendant liver injury, results in decreased levels of high-density

lipoprotein (HDL), increasing the likelihood of coronary heart disease.

- *Pancreatitis.* Excessive alcohol intake increases the risk of acute and chronic pancreatitis (Chapter 19).
- *Fetus.* The use of ethanol during pregnancy can cause fetal alcohol syndrome, which is marked by microcephaly, growth retardation, and facial abnormalities in the newborn and reduction in mental functions as the child grows older. It is difficult to establish the minimal amount of alcohol consumption that can cause fetal alcohol syndrome, but consumption during the first trimester of pregnancy is particularly harmful.
- *Carcinogenesis.* Chronic alcohol consumption is associated with an increased incidence of cancer of the oral cavity, esophagus, and liver. Even light or moderate drinking increases the risk of cancer. In females, low to moderate intake (12 oz beer or 5 oz of wine) incurs a slightly higher risk of breast cancer. Acetaldehyde is considered the main agent associated with alcohol-induced laryngeal and esophageal cancer, and acetaldehyde-DNA adducts have been detected in some tumors from these tissues. Individuals with one copy of the $\text{ALDH}2^*$ 2 aldehyde dehydrogenase allele who drink are at a higher risk for developing cancer of the esophagus. As mentioned earlier, alcohol and cigarette smoke synergize in the causation of various cancers.
- *Malnutrition.* Ethanol is a substantial source of energy (empty calories). Severe chronic alcohol use disorder may lead to malnutrition and nutritional deficiencies, particularly of the B vitamins.

Some studies have shown an association between moderate amounts of alcohol intake (about 20 to 30 g per day, corresponding to approximately 250 mL of wine) and a reduced risk of death from heart

disease. Possible mechanisms include increasing HDL cholesterol levels, inhibition of platelet aggregation, providing antioxidants, and lowering of fibrinogen levels. However, it is difficult to establish cause and effect from these studies, and the benefits may be largely attributable to confounding factors such as income, lifestyle, and the eating habits of individuals who consume moderate amounts of alcohol daily. Other approaches for increasing HDL cholesterol levels such as exercise and increasing the intake of antioxidant rich foods may provide similar benefits.

KEY CONCEPTS

Alcohol—Metabolism and Health Effects

- Acutely, excessive alcohol consumption causes drowsiness at blood levels of approximately 200 mg/dL. Stupor and coma develop at higher levels.
- Alcohol is oxidized to acetaldehyde in the liver by alcohol dehydrogenase, by the cytochrome P-450 system, and by catalase, which is of minor importance. Diminished ability to metabolize acetaldehyde is associated with acute toxicity and an increased risk of certain cancers.
- Alcohol oxidation by alcohol dehydrogenase depletes NAD, leading to accumulation of fat in the liver and metabolic acidosis.
- The main effects of chronic alcohol use disorder are fatty liver, alcoholic hepatitis, and cirrhosis, which leads to portal hypertension and increases the risk for development of hepatocellular carcinoma.
- Chronic alcohol use disorder can cause bleeding from gastritis and gastric ulcers, peripheral neuropathy associated with thiamine deficiency, alcoholic cardiomyopathy, and acute and chronic pancreatitis.
- Chronic alcohol use disorder is a major risk factor for cancers of the oral cavity, larynx, and esophagus. The risk is greatly increased by concurrent smoking or use of smokeless tobacco.

INJURY BY THERAPEUTIC DRUGS AND DRUGS OF ABUSE

Injury by Therapeutic Drugs (Adverse Drug Reactions)

Adverse drug reactions (ADRs) refer to undesired effects that occur when people take medications at the prescribed dose. These reactions are very common. While most ADRs are not life-threatening, some can be serious, leading to over 100,000 deaths annually. ADRs are also common in outpatient settings and nursing homes. One notable example is discoloration of the skin caused by the accumulation of an oxidized metabolite of the antibiotic minocycline (Fig. 9.12). More frequently observed are drug reactions that arise from direct drug actions or immune-mediated hypersensitivity reactions. Drug-induced hypersensitivity reactions often present as skin rashes, although they can sometimes mimic autoimmune disorders such as systemic lupus erythematosus (Chapter 6) or manifest as conditions such as hemolytic anemia or immune thrombocytopenia (Chapter 13).

People over 65 are more likely to experience ADRs. Table 9.5 lists common pathologic findings associated with adverse drug reactions, along with the drugs most frequently involved. Many of the drugs that produce adverse reactions, such as antineoplastic agents, are highly potent, and the adverse reactions are considered acceptable risks of treatment. In this section, adverse reactions to commonly used drugs are examined, beginning with the undesirable effects of anticoagulants, menopausal hormone therapy (MHT), and oral contraceptives (OCs) and then discussing the effects of acetaminophen and aspirin.

Anticoagulants

Anticoagulants administered to individuals to prevent thrombosis and embolism increase the risk of uncontrolled bleeding (particularly

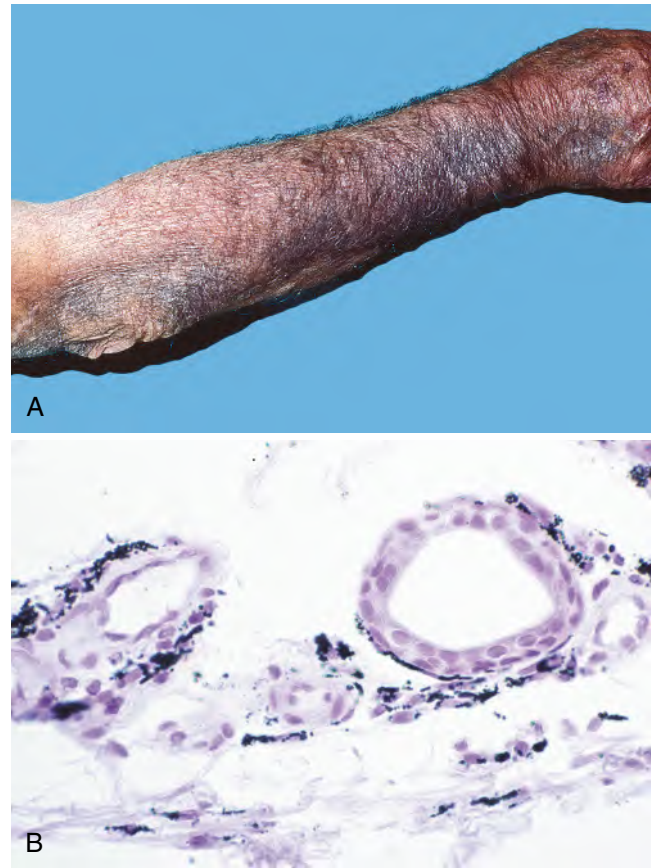


Fig. 9.12 Adverse drug reaction. Skin pigmentation caused by minocycline, a long-acting tetracycline derivative. (A) Diffuse blue-gray pigmentation of the forearm. (B) Deposition of drug metabolite/iron/melanin pigment particles in the dermis. (Courtesy Dr. Zsolt Argenyi, Department of Pathology, University of Washington, Seattle.)

after trauma), which may be severe or fatal. There are two main classes of oral anticoagulants, the vitamin K antagonist warfarin, and direct inhibitors of the coagulation cascade such as the thrombin inhibitor dabigatran and the Factor Xa inhibitors apixaban and rivaroxaban. While warfarin is inexpensive, many drugs and foods rich in vitamin K interfere with its metabolism or abrogate its function, so its effects must be monitored with regular blood sampling, and maintaining anticoagulation in a safe therapeutic range is often difficult. Direct oral anticoagulants (commonly referred to as DOACs) avoid these food and drug interactions (allowing them to be given in fixed doses), do not require routine monitoring, have efficacy and safety profiles that are superior to warfarin, and result in lower mortality. Thus, outside of a few specific settings (e.g., patients with atrial fibrillation and valve abnormalities, or with advanced kidney and liver disease), DOACs have emerged as the preferred drugs in a large majority of patients in need of chronic anticoagulation. While the cost of DOACs is much higher than that of warfarin, recent studies suggest they are nonetheless cost effective due to lower rates of bleeding and stroke. Moreover, the safety profile of DOACs has been improved by the development of DOAC inhibitors that rapidly reverse anticoagulation.

Menopausal Hormone Therapy

As females approach menopause, their estrogen levels naturally decline. This can cause a variety of symptoms, including vasomotor (hot flashes and night sweats) and genitourinary symptoms (vaginal dryness, urinary urgency and frequency, and recurrent urinary-tract infections). Reduced estrogen is also linked to an increased risk of osteoporosis, cardiovascular

TABLE 9.5 Common Adverse Drug Reactions and Their Agents

Reaction	Major Offenders
Bone Marrow and Blood Cells^a	
Granulocytopenia, aplastic anemia, pancytopenia	Antineoplastic agents, chloramphenicol
Hemolytic anemia, thrombocytopenia	Penicillin, methyldopa, quinidine, heparin
Cutaneous	
Urticaria, macules, papules, vesicles, petechiae, exfoliative dermatitis, fixed drug eruptions, abnormal pigmentation	Antineoplastic agents, sulfonamides, hydantoins, some antibiotics, and many other agents
Cardiac	
Arrhythmias	Theophylline, hydantoins, digoxin
Cardiomyopathy	Doxorubicin, daunorubicin
Renal	
Glomerulonephritis	Penicillamine
Acute tubular necrosis	Aminoglycoside antibiotics, cyclosporin, amphotericin B
Tubulointerstitial disease with papillary necrosis	Phenacetin, salicylates
Pulmonary	
Asthma	Salicylates
Acute pneumonitis	Nitrofurantoin
Interstitial fibrosis	Busulfan, nitrofurantoin, bleomycin
Hepatic	
Fatty change	Tetracycline
Diffuse hepatocellular damage	Halothane, isoniazid, acetaminophen
Cholestasis	Chlorpromazine, estrogens, contraceptive agents
Systemic	
Anaphylaxis	Penicillin
Drug-induced lupus erythematosus	Hydralazine, procainamide
Bleeding	Warfarin, dabigatran
Central Nervous System	
Tinnitus and dizziness	Salicylates
Acute dystonic reactions and parkinsonian syndrome	Phenothiazine antipsychotics
Respiratory depression	Sedatives

^aAffected in almost half of all drug-related deaths.

disease, and cognitive decline in aging females. Menopausal hormone therapy (MHT) is used to manage these symptoms. The most common type of MHT (previously referred to as hormone replacement therapy) consists of combinations of estrogens and progesterone or a synthetic progestin. Because of the risk of uterine cancer, therapy with estrogen alone is used only in females who have had a hysterectomy. Initially used to mitigate hot flashes and other symptoms of menopause, early clinical studies suggested that MHT use in postmenopausal females could prevent or slow the progression of osteoporosis (Chapter 26) and reduce the likelihood of myocardial infarction and cognitive decline. However, in 2002,

initial publication of findings by the Women's Health Initiative surprised the medical community by reporting that a large prospective placebo-controlled trial failed to find support for some of the presumed beneficial effects of the therapy. This study involved approximately 17,000 females who were taking a combination of estrogen (conjugated equine estrogens) and a synthetic progestin (medroxyprogesterone acetate). Although MHT did reduce the number of fractures in females on treatment, researchers also reported that after 5 years of treatment, combination MHT increased the risk of breast cancer (Chapter 23), stroke, and venous thromboembolism (VTE) and had no effect on the incidence of coronary heart disease. The shock waves produced by these findings led to a drastic decrease in the use of MHT, from 22% to 4.7% in 2010, which was accompanied by an apparent drop in the incidence of newly diagnosed breast cancers.

However, during the past few years there has been a reappraisal of the risks and benefits of MHT. These newer analyses showed that **MHT effects depend on the type of hormone therapy regimen used (combination estrogen-progestin versus estrogen alone); the age and risk factor status of the female at the start of treatment; the duration of the treatment; and the hormone dose, formulation, and route of administration.** The current risk/benefit consensus can be summarized as follows:

- *Combination estrogen-progestin increases the risk of breast cancer* after a median time of 5 to 6 years but there is no increase in the risk for ovarian cancer.
- *MHT may have a protective effect on the development of atherosclerosis and coronary disease when started in females younger than age 60 years*, but there is increased risk of coronary heart disease among females who started MHT after the age of 60 years. These data support the notion of a “timing hypothesis”—that there may be a critical therapeutic window for MHT effects on the cardiovascular system. Protective effects in younger females depend in part on the response of estrogen receptors in healthy vascular endothelium. Despite the potential cardiovascular benefits in younger females, MHT is not recommended for prevention of cardiovascular disease.
- *MHT increases the risk of stroke and VTE including deep vein thrombosis and pulmonary embolism in females who start therapy after the age of 60 years.* The increase in VTE is more pronounced during the first 2 years of treatment and in females who have other risk factors, such as immobilization and hypercoagulable states caused by prothrombin or factor V Leiden mutations (Chapter 4). The risks of VTE and stroke appear to be lower with transdermal than oral estrogen, possibly because this mode of administration avoids first pass hepatic metabolism of estrogen, which can result in increased levels of triglycerides. Even in younger females, transdermal administration may be preferred in those with obesity and high triglyceride levels. As can be appreciated from these associations, the assessment of risks and benefits when considering the use of MHT in females is complex. The current consensus is that oral agents have an important role in the management of menopausal symptoms in early menopause, particularly in females less than 60 years old without contraindications such as obesity and elevated risk of thrombosis who are within 10 years of menopause. Oral agents should be tapered at 3 to 5 years or followed by transdermal therapy if symptoms persist. MHT is generally not recommended for the prevention of cardiovascular disease and dementia but may be appropriate for prevention of osteoporosis in certain females.

Oral Contraceptives

Worldwide, millions of females use hormonal contraception. Oral contraceptives (OCs) typically contain a synthetic estradiol and a variable amount of a progestin, but some preparations contain only progestins. They act by inhibiting ovulation or preventing implantation. Currently

prescribed OCs contain much smaller amounts of estrogens (as little as 20 µg of ethinyl estradiol) than earlier formulations and are associated with fewer side effects. Opill (norgestrel), which contains only the progestin norgestrel, is now available in the United States as the first OC that does not require a prescription (i.e., “over-the-counter”), and transdermal and implantable OC formulations have also become available. Hence the results of epidemiologic studies must be interpreted in the context of the OC, its dosage, and its delivery system. Nevertheless, there is good evidence to support the following conclusions.

- **Breast carcinoma.** The prevailing view is that OCs do not increase breast cancer risk.
- **Endometrial cancer and ovarian cancers.** OCs have a protective effect against these tumors that may last for decades following cessation of OC use.
- **Cervical cancer.** OCs may increase risk of cervical carcinomas in females infected with human papillomavirus.
- **Thromboembolism.** Most studies indicate that OCs, including low-dose (less than 50 µg of estrogen) preparations, are associated with a two-fold to four-fold increased risk of venous thrombosis and pulmonary thromboembolism due to a hypercoagulable state induced by elevated hepatic synthesis of coagulation factors. This risk may be even higher with third-generation OCs that contain synthetic progestins, particularly in females who are carriers of the factor V Leiden mutation. To put this complication into context, however, the risk of thromboembolism associated with OC use is two to six times lower than the risk of thromboembolism associated with pregnancy.
- **Cardiovascular disease.** There is considerable uncertainty about the risk of atherosclerosis and myocardial infarction in users of OCs. It seems that OCs do not increase the risk of coronary artery disease in females younger than 30 years or in older females who are non-smokers, but the risk approximately doubles in females older than 35 years who smoke.
- **Hepatic adenoma.** There is a well-defined association between the use of OCs and this rare benign hepatic tumor, especially in older females who have used OCs for prolonged periods.

Ultimately, the pros and cons of OCs must be viewed in the context of their wide applicability and acceptance as a form of contraception.

Acetaminophen

Acetaminophen is the most used analgesic in the United States. It is present in more than 300 products, alone or in combination with other agents. In the United States, it is the cause of about 50% of cases of acute liver failure, 30% of which are fatal. Intentional overdose (attempted suicide) is the most common cause of acetaminophen toxicity in Great Britain, while unintentional overdose is the most frequent cause in the United States, representing almost 50% of the total intoxication cases.

At therapeutic doses, about 95% of acetaminophen undergoes detoxification in the liver by phase II enzymes and is excreted in the urine as glucuronide or sulfate conjugates (Fig. 9.13). About 5% or less is metabolized through the activity of CYPs (primarily CYP2E) to NAPQI (*N*-acetyl-*p*-benzoquinoneimine), a highly reactive metabolite. NAPQI is normally inactivated by conjugation to GSH, but when acetaminophen is taken in large doses, unconjugated NAPQI accumulates and causes hepatocellular injury, leading to centrilobular necrosis that may progress to liver failure. The injury produced by NAPQI involves two mechanisms: (1) covalent binding to hepatic proteins, which causes damage to cellular membranes and mitochondrial dysfunction, and (2) depletion of GSH, making hepatocytes more susceptible to ROS-induced injury. Because alcohol induces CYP2E in the liver, toxicity can occur at lower doses in heavy drinkers.

The window between the usual dose (0.5 g) and the toxic dose (15 to 25 g) is large, and the drug is ordinarily very safe. Toxicity begins with nausea, vomiting, diarrhea, and sometimes shock, followed in a few days by evidence of jaundice. Overdose of acetaminophen can be treated in its early stages (within 12 hours) by administration of *N*-acetylcysteine, which restores GSH levels. In serious overdose, liver failure ensues, starting with centrilobular necrosis that may extend to entire lobules; in such circumstances liver transplantation is the only hope for survival. Some patients also show evidence of concurrent renal damage.

Aspirin (Acetylsalicylic Acid)

Knowledge of the medical benefits of salicylates dates back more than 3500 years to the Sumerians and Egyptians, who used salicin-containing extracts of the bark of willow trees for relief of pain, fever, and inflammation. In 1897, chemists at Bayer synthesized

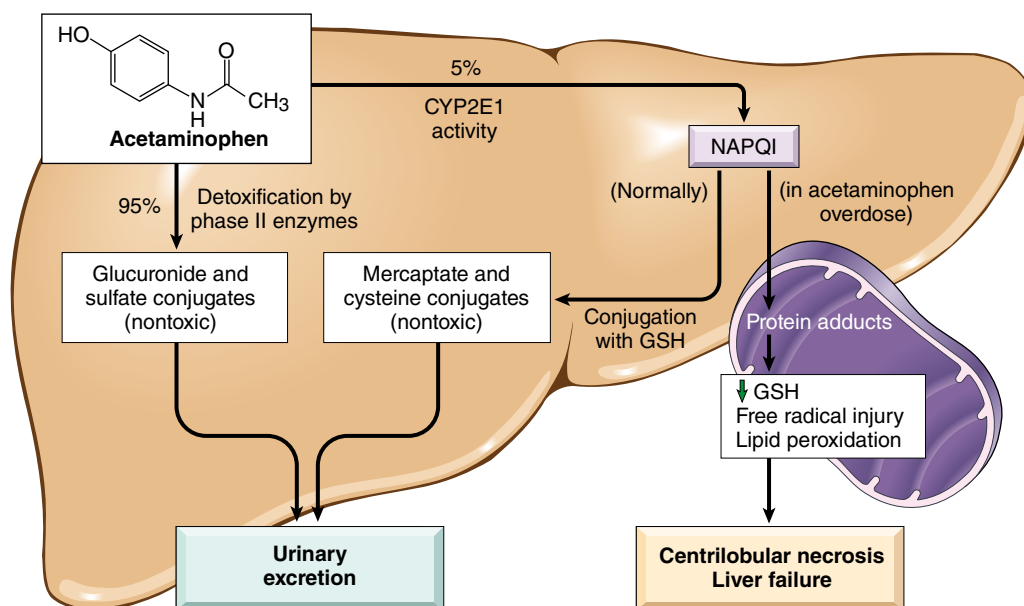


Fig. 9.13 Acetaminophen metabolism and toxicity. GSH, Glutathione; NAPQI, *N*-acetyl-*p*-benzoquinoneimine.

acetylsalicylic acid and in 1899 it was branded aspirin, comprising the word acetyl and *Spiraea ulmaria*, the tree from which the salicylic acid was extracted. Aspirin overdose may result from accidental ingestion by young children; in adults, overdose is frequently suicidal.

Acute salicylate toxicity produces several pathophysiologic alterations. Initially, stimulation of the respiratory center in the medulla increases the rate and depth of respiration, leading to “blowing off” of CO₂ and respiratory alkalosis. However, salicylate toxicity subsequently uncouples oxidative phosphorylation and inhibits the Krebs cycle, changes that lead to hyperpyrexia, increased dependence on glycolysis, accumulation of pyruvate and lactate, and metabolic acidosis. Metabolic acidosis increases levels of nonionized forms of salicylate in the blood that diffuse into the brain and produce effects ranging from nausea to coma. Ingestion of 2 to 4 g by children or 10 to 30 g by adults may be fatal, but survival has been reported after ingestion of doses five times larger.

Chronic aspirin toxicity (salicylism) develops in persons who take 3 g or more daily for long periods of time for treatment of chronic pain or inflammatory conditions. It is manifested by headaches, dizziness, ringing in the ears (tinnitus), hearing impairment, mental confusion, drowsiness, nausea, vomiting, and diarrhea. The CNS changes may progress to convulsions and coma. The morphologic consequences of chronic salicylism are varied. Most often there is an acute erosive gastritis (Chapter 17), which may produce gastrointestinal bleeding and lead to gastric ulceration. A bleeding tendency may appear concurrently with chronic toxicity because aspirin acetylates platelet cyclooxygenase and irreversibly blocks the production of thromboxane A₂, an activator of platelet aggregation. Petechial hemorrhages may appear in the skin and internal viscera, and bleeding from gastric ulcerations may be exacerbated by the clotting defect. With the recognition of gastric ulceration and bleeding as an important complication of ingestion of large doses of aspirin, chronic toxicity is now quite uncommon.

Injury by Nontherapeutic Agents (Drug Use Disorder)

Drug use disorder is a pervasive public health problem defined by repeated or chronic use of mind-altering substances beyond therapeutic or social norms that may lead to drug addiction and overdose. According to the 2023 United Nations Office on Drugs and Crime report, it is estimated that about 296 million people worldwide

(roughly 5.8% of the global population aged 15 to 64 years) used nonprescription drugs at least once during 2021. In the United States in 2021 it was estimated that 61.2 million people used nonprescription drugs, including 40% of young adults between 18 and 25 years of age, and that 46.3 million people (16.5% of the population) aged 12 or older met the criteria for drug use disorder. The appalling toll of drug use has risen sharply recently. In 2023, approximately 107,000 people died in the United States of drug overdose, with roughly 70% of these deaths attributable to synthetic opioids (primarily fentanyl), a four-fold increase over the past decade.

Commonly used nontherapeutic agents are listed in Table 9.6. Considered here are cocaine, opioids, amphetamines, and marijuana, among others.

Cocaine

According to the United Nations World Drug Report 2023, there were 21.6 million users of cocaine in 2021 worldwide. Of these, 11.4 million users were in the Americas, 5.2 million in Europe, 2.3 million in Africa, and 2.1 million in Asia.

Cocaine is extracted from the leaves of the coca plant and is usually prepared as a water-soluble powder, cocaine hydrochloride. Cocaine sold on the street is liberally diluted with talcum powder, lactose, or other look-alikes. It can be snorted or dissolved in water and injected subcutaneously or intravenously. Crystallization of the pure alkaloid yields nuggets of crack, so called because of the cracking or popping sound it makes when heated to produce vapors that are inhaled. The pharmacologic actions of cocaine and crack are identical, but crack is far more potent.

Cocaine produces intense euphoria and neurologic stimulation, making it one of the most addictive drugs known. Experimental animals will press a lever more than 1000 times and forgo food and drink to obtain it. In the cocaine user, although physical dependence generally does not occur, the psychologic withdrawal is profound and can be extremely difficult to treat. Intense cravings are particularly severe in the first several months after abstinence and can recur for years. **Cocaine facilitates neurotransmission in the CNS, where it blocks the reuptake of dopamine; similarly, in adrenergic peripheral nerves, it blocks the reuptake of epinephrine and norepinephrine while stimulating the presynaptic release of norepinephrine.** The euphoria is due to its enhancement of brain dopamine activity, especially in the so-called mesolimbic dopamine reward pathway.

TABLE 9.6 Commonly Used Nontherapeutic Agents

Class	Molecular Target	Example
Opioid narcotics	Mu opioid receptor (agonist)	Heroin, hydromorphone Oxycodone Methadone Fentanyl
Sedative-hypnotics	GABA _A receptor (agonist)	Barbiturates Ethanol Benzodiazepines
Psychomotor stimulants	Dopamine transporter (antagonist) Serotonin receptors (toxicity)	Cocaine Amphetamines 3,4-Methylenedioxymethamphetamine (MDMA, Ecstasy)
Phencyclidine-like drugs	NMDA glutamate receptor channel (antagonist)	Phencyclidine (PCP, angel dust) Ketamine
Cannabinoids	CB1 cannabinoid receptors (agonist)	Marijuana Hashish
Hallucinogens	Serotonin 5-HT ₂ receptors (agonist)	Lysergic acid diethylamide (LSD) Mescaline Psilocybin

GABA, γ -Aminobutyric acid; 5-HT₂, 5-hydroxytryptamine; NMDA, N-methyl D-aspartate.

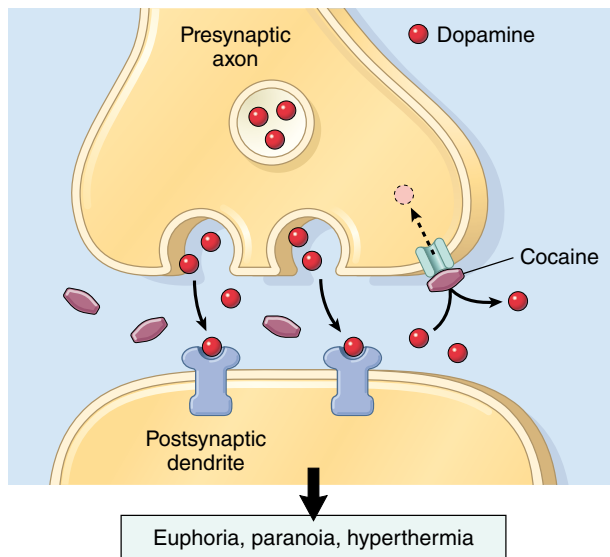
The acute and chronic effects of cocaine on various organ systems are:

- **Cardiovascular effects.** The most serious cardiac effects of cocaine relate to its sympathomimetic activity resulting from blockade of reuptake of epinephrine and norepinephrine at adrenergic nerve endings (Fig. 9.14). The net effect is the accumulation of these two neurotransmitters in synapses, resulting in excess stimulation, manifested by tachycardia, hypertension, and peripheral vasoconstriction. Cocaine may also induce myocardial ischemia by causing coronary artery vasoconstriction and by enhancing platelet aggregation and thrombus formation. These effects of cocaine increase myocardial oxygen demand and decrease coronary blood flow, setting the stage for myocardial ischemia and possible myocardial infarction. Cocaine can also precipitate lethal arrhythmias by enhanced sympathetic activity as well as by disrupting normal ion

(K^+ , Ca^{2+} , Na^+) transport in the myocardium. These toxic effects are not necessarily dose related, and a fatal event may occur in a first-time user with what is a typical mood-altering dose. They may also be potentiated by cigarette smoking, which increases the likelihood of cocaine-induced coronary vasospasm.

- **CNS.** The most common acute effects on the CNS are hyperpyrexia (thought to be caused by aberrations of the dopaminergic pathways that control body temperature) and seizures.
- **Effects on pregnancy.** In pregnant females, cocaine may cause acute decreases in blood flow to the placenta, resulting in fetal hypoxia and spontaneous abortion. Neurologic development may be impaired in the fetus of a pregnant female who is a chronic drug user.
- **Other effects.** Chronic cocaine use may cause (1) perforation of the nasal septum in snorters, (2) decreased lung diffusing capacity in those who inhale the smoke, and (3) dilated cardiomyopathy.

CENTRAL NERVOUS SYSTEM SYNAPSE



SYMPATHETIC NEURON–TARGET CELL INTERFACE

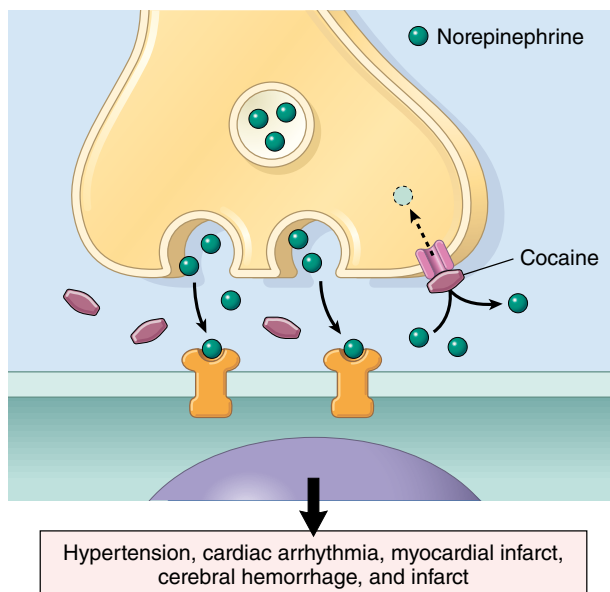


Fig. 9.14 Effect of cocaine on neurotransmission. The drug inhibits reuptake of the neurotransmitters dopamine and norepinephrine in the central and peripheral nervous systems.

Opioids and Opiates

According to the 2023 United Nations Office on Drugs and Crime report, in 2021 there were 31.5 million users of opiates (naturally occurring chemicals derived from poppy plants, such as morphine) and 60.4 million users of opioids (derivatives or synthetic variants of opiates) worldwide. Opioid drugs that are frequently used in non-therapeutic settings include prescription drugs such as oxycodone, hydrocodone, tramadol, and methadone.

Problems associated with opioid use have been markedly exacerbated by the increased illicit production of the highly potent, cheap synthetic opioid fentanyl, which is often mixed with or substituted for other drugs. Fentanyl is a safe and effective sedative and analgesic when used appropriately but is often fatal when used outside of medical supervision, particularly when used unknowingly by individuals who have not developed tolerance to opioids. The usual cause of death is profound respiratory suppression, a predictable effect of opioids that can be rapidly reversed if antagonists such as naloxone (Narcan) are administered in a timely fashion. With a death toll of over 80,000 people in 2023 in the United States, the problem of opioid use has assumed epidemic proportions and shows no sign of abating.

Heroin is a street drug derived from the poppy plant that is closely related to morphine. As sold on the street, it is cut (diluted) with an agent (often talc or quinine); thus, the size of the dose is not only variable but also usually unknown to the buyer. Heroin, along with any contaminating substances, is usually self-administered intravenously or subcutaneously. The effects on the CNS are varied and include euphoria, hallucinations, somnolence, and sedation. Heroin has a wide range of other adverse physical effects related to (1) the pharmacologic action of the agent, (2) reactions to the cutting agents or contaminants, (3) hypersensitivity reactions to the drug or its adulterants (quinine itself has neurologic, renal, and auditory toxicity), and (4) diseases contracted incident to the use of contaminated needles. Some of the most important adverse effects are:

- **Sudden death.** Sudden death, usually related to overdose, is an ever-present risk because drug purity is generally unknown (ranging from 2% to 90%). The yearly mortality among heroin users in the United States is estimated to be between 1% and 3%. The addition of very potent synthetic opioids such as fentanyl to heroin, a common practice among dealers, greatly increases the chances of fatal overdose. Sudden death can also occur if heroin is taken after tolerance for the drug, built up over time, is lost (as during a period of incarceration). The mechanisms of death include respiratory depression, arrhythmia and cardiac arrest, and severe pulmonary edema.
- **Pulmonary injury.** Pulmonary complications include moderate to severe edema, septic embolism from endocarditis, lung abscess, opportunistic infections, and foreign-body granulomas from talc and other adulterants. Although granulomas occur principally in

the lung, they are sometimes found in the mononuclear phagocyte system, particularly in the spleen, liver, and lymph nodes that drain the upper extremities. Examination under polarized light often highlights trapped talc crystals, sometimes enclosed within foreign-body giant cells.

- **Infections.** Infectious complications are common, particularly of the skin and subcutaneous tissue, heart valves, liver, and lungs. In a series of addicted patients admitted to the hospital, more than 10% had endocarditis, which often takes a distinctive form involving right-sided heart valves, particularly the tricuspid. Most cases are caused by *Staphylococcus aureus*, but fungi and a multitude of other organisms have also been implicated. Viral hepatitis is the most common infection among addicted persons and is acquired by the sharing of needles. In the United States, this practice has also led to a high incidence of human immunodeficiency virus (HIV) infection in intravenous drug users.
- **Skin.** Cutaneous lesions are probably the most frequent telltale sign of heroin addiction. Acute changes include abscesses, cellulitis, and ulcerations due to subcutaneous injections. Scarring at injection sites, hyperpigmentation over commonly used veins, and thrombosed veins are the usual sequelae of repeated intravenous inoculations.
- **Kidneys.** Kidney disease is a relatively common hazard. The two forms most frequently encountered are amyloidosis (generally secondary to skin infections) and focal and segmental glomerulosclerosis; both induce proteinuria and nephrotic syndrome.

Amphetamines and Related Drugs

Methamphetamine is an addictive drug also known as “speed” or “meth.” It is related to amphetamine but has stronger effects in the CNS. The number of people in the United States who used methamphetamine more than doubled between 2009 and 2021, from 1.2 million to 2.5 million, with nearly 25% of users reporting injecting methamphetamine and 50% meeting diagnostic criteria for methamphetamine use disorder. There are high rates of co-occurring substance use or mental illness among adults using methamphetamine. Methamphetamine acts by releasing dopamine in the brain, which inhibits presynaptic neurotransmission at corticostriatal synapses, slowing glutamate release. Methamphetamine produces a feeling of euphoria, which is followed by a “crash.” Long-term use leads to violent behaviors, confusion, and psychosis marked by paranoia and hallucinations.

Marijuana

It is estimated that worldwide 219 million people used marijuana (or “pot”) in 2021, making it far and away the most widely used illicit drug globally. In 2019, marijuana was also the most used illegal drug (per the federal government) in United States, with 48.2 million people (18% of Americans) using the drug at least once during that year. It is still illegal in most of the world’s countries, but some countries have decriminalized its use and others have specifically legalized medical and personal use. In the United States as of 2023, 38 states allowed the medical use of cannabis products and nearly half had legalized personal use of marijuana. The Food and Drug Administration has approved the use of several cannabis-derived and cannabis-related drug products with a prescription from a licensed health care provider. In 2024, the United States announced plans to change marijuana from a schedule I drug to a schedule III drug, which would place it in the same category as over-the-counter drugs like acetaminophen.

Marijuana is made from the leaves of the *Cannabis sativa* plant, which contain the psychoactive substance Δ^9 -tetrahydrocannabinol (THC). About 5% to 10% of THC is absorbed when it is smoked in a hand-rolled cigarette (“joint”). Marijuana use causes euphoria and a

sense of relaxation. In many people it causes a heightened sensory perception (e.g., brighter colors), laughter, altered perception of time, and increased appetite. Among the beneficial effects of marijuana is its potential use to treat nausea secondary to cancer chemotherapy and as an agent capable of decreasing pain in some chronic conditions that are otherwise difficult to treat. The functional and organic CNS consequences of marijuana smoking have received the most scrutiny. Its use distorts sensory perception and impairs motor coordination, but these acute effects generally clear in 4 to 5 hours. With continued use, these changes may progress to cognitive and psychomotor impairments, such as inability to judge time, speed, and distance, a potential cause of automobile accidents. Marijuana increases the heart rate and sometimes blood pressure, and it may cause angina in a person with coronary artery disease.

The respiratory system is also affected by chronic marijuana smoking; laryngitis, pharyngitis, bronchitis, cough and hoarseness, and asthma-like symptoms have been described, along with mild but significant airway obstruction. Marijuana cigarette smoke contains several carcinogens that are also present in tobacco smoke. Furthermore, smoking a marijuana cigarette, compared with a tobacco cigarette, is associated with a three-fold increase in the amount of tar inhaled and retained in the lungs, presumably because of the larger puff volume, deeper inhalation, and longer breath holding. With the advent of e-cigarettes and vaping, an increasing number of users are inhaling aerosolized THC, a practice that has been associated with severe lung injury in some users, possibly due to the presence of adulterants. Paradoxically, some heavy users develop cannabis hyperemesis syndrome, marked by intractable nausea and vomiting that only remits with cessation of use.

Whether marijuana causes physical dependence and addiction remains controversial. According to the US National Institute on Drug Abuse and the CDC, chronic use of marijuana, especially if the exposure started before age 18 can give rise to marijuana use disorder in up to 30% of users. Dependence in such individuals is manifested by withdrawal symptoms such as moodiness and sleep difficulties that may last up to 2 weeks when not taking the drug. A small subset of those who develop dependence may become addicted; these individuals cannot stop using the drug even though it interferes with many aspects of their life. The exact number of people who become addicted is uncertain since epidemiologic studies have not used uniform criteria to define addiction to marijuana.

Other Drugs

The variety of drugs that have been tried by individuals seeking “new experiences” (highs, lows, “out-of-body experiences”) are legion. These drugs include various stimulants, depressants, analgesics, and hallucinogens (Table 9.6). Among these are 1-(1-phenylcyclohexyl) piperidine (PCP), or phencyclidine, and ketamine (related anesthetic agents); lysergic acid diethylamide (LSD), the most potent hallucinogen known; ecstasy (3,4-methylenedioxymethamphetamine [MDMA]); and “bath salts,” synthetic cathinones that are chemically related to khat, a widely used stimulant in East Africa. Not much is known about the long-term deleterious effects of any of these agents. Acutely, LSD has unpredictable effects on mood, affect, and thought, sometimes leading to bizarre and dangerous behaviors. Chronic use of ecstasy may deplete the CNS of serotonin, potentially leading to sleep disorders, depression, anxiety, and aggressive behavior.

On the other hand, therapeutic uses for psychoactive drugs are emerging. Ketamine derivatives taken in low doses have recently been approved by the FDA for treatment of severe depression, and many clinical trials exploring the use of hallucinogens as treatment for psychiatric conditions such as posttraumatic stress disorder are ongoing. Initial trials have shown some short-term benefit for depression in selected populations.

KEY CONCEPTS

Drug Injury

- Drug injury may be caused by therapeutic drugs (adverse drug reactions) or nontherapeutic agents (drug abuse).
- Antineoplastic agents, anticoagulants, MHT preparations and OCs, acetaminophen, and aspirin are among the therapeutic drugs involved most frequently.
- MHT increases the risk of endometrial and breast cancers and thromboembolism and does not appear to protect against ischemic heart disease in females over 60 years of age. OCs have a protective effect against endometrial and ovarian cancer but increase the risk of thromboembolism and hepatic adenoma.
- Overdose of acetaminophen may cause centrilobular liver necrosis, leading to liver failure. Early treatment with agents that restore GSH levels may limit toxicity.
- Aspirin produces gastric ulceration and blocks the production of thromboxane A_2 in platelets, which may produce bleeding.
- Common drugs leading to drug use disorder include sedative-hypnotics (barbiturates, ethanol), psychomotor stimulants (cocaine, methamphetamine, ecstasy), opioid narcotics (heroin, oxycodone), hallucinogens, and cannabinoids (marijuana). Widespread availability of synthetic opioids, particularly fentanyl, has led to a sharp increase in deaths due to drug overdose.

INJURY BY PHYSICAL AGENTS

Injury induced by physical agents is divided into the following categories: mechanical trauma, thermal injury, electrical injury, and injury produced by ionizing radiation. Each type is considered separately.

Mechanical Trauma

Mechanical forces may inflict a variety of forms of damage. The type of injury depends on the shape of the colliding object, the amount of energy discharged at impact, and the tissues or organs that bear the impact. Bone and head injuries result in unique damage and are discussed elsewhere (Chapters 26 and 28). All soft tissues react similarly to mechanical forces, and the patterns of injury can be divided into abrasions, contusions, lacerations, incised wounds, and puncture wounds. This is just a small sampling of the various forms of trauma encountered by forensic pathologists, who deal with wounds produced by shooting, stabbing, blunt force, traffic accidents, and other causes. In addition to morphologic analyses, forensic pathology now includes molecular methods for identity testing and sophisticated methods to detect the presence of foreign chemicals. Details about the practice of forensic pathology can be found in specialized textbooks.

Thermal Injury

Both excessive heat and excessive cold are important causes of injury. Burns are the most common cause of thermal injury and are discussed first; a brief discussion of hyperthermia and hypothermia follows.

Thermal Burns

In the United States, nearly half a million people require medical treatment for burn injuries each year. Eighty percent of burns are caused by fire or by scalding, the latter being a major cause of injury in children. It is estimated that approximately 3500 persons die each year due to injuries caused by fire and smoke inhalation, mostly in homes. Since the 1970s, marked decreases have been seen in mortality rates and the length of hospitalizations of burn patients. In recent years there were approximately 45,000 burn hospitalizations annually. In specialized burn centers, which treat about half of hospitalized patients, the survival rate is an impressive 95%. This remarkable improvement is due to

a better understanding of the systemic effects of extensive burns, the prevention of wound infection, and the use of treatments that promote the healing of skin surfaces.

The clinical significance of a burn injury depends on the following factors:

- *Depth of the burns*
- *Percentage of body surface involved*
- *Internal injuries* caused by the inhalation of hot and toxic fumes
- *Promptness and efficacy of therapy*, especially fluid and electrolyte management and prevention or control of wound infections

Burns used to be classified as first degree to fourth degree, according to the depth of the injury (first-degree burns being the most superficial), but are now classified as superficial, partial-thickness, and full-thickness.

- *Superficial burns* (formerly known as *first-degree burns*) are confined to the epidermis.
- *Partial-thickness burns* (formerly known as *second-degree burns*) involve injury to the dermis.
- *Full-thickness burns* (formerly known as *third-degree burns*) extend to the subcutaneous tissue. Full-thickness burns may also damage muscle tissue underneath the subcutaneous tissue (these were known formerly as fourth-degree burns).

The three most serious complications after a burn injury are shock, sepsis, and respiratory insufficiency. Particularly in burns of more than 20% of the body surface, there is a rapid (within hours) shift of body fluids into the interstitial compartments throughout the body due to the *systemic inflammatory response syndrome*, leading to shock (Chapter 4). Because of widespread vascular leakiness, generalized edema, including pulmonary edema, can be severe. An important pathophysiologic effect of burns is the development of a hypermetabolic state associated with excess heat loss and an increased need for nutritional support. It is estimated that when more than 40% of the body surface is burned, the resting metabolic rate doubles.

The burn site is ideal for the growth of microorganisms; the serum and debris provide nutrients, and the burn injury compromises blood flow, blocking effective inflammatory responses. As a result, virtually all burns become colonized with bacteria. Infections are defined by the presence of greater than 10^5 bacteria per gram of tissue, and invasive local infection is defined by the presence of greater than 10^5 bacteria per gram in unburned adjacent tissue. The most common offender is the opportunist *Pseudomonas aeruginosa*, but antibiotic-resistant strains of other common hospital-acquired bacteria, such as methicillin-resistant *S. aureus*, and fungi, particularly *Candida* species, may also be involved. Furthermore, systemic inflammatory response syndrome may impair or dysregulate both innate and adaptive immune responses (Chapter 4). Direct bacteremic spread and release of toxic substances such as endotoxin from the local site may also have dire consequences. Pneumonia or septic shock with renal failure and/or acute respiratory distress syndrome (Chapter 15) are the most common serious sequelae.

Systemic organ failure resulting from burn sepsis has greatly diminished during the past 30 years because of the introduction of techniques for early excision and grafting of the burn wound. Removal of the burn wound decreases infection and reduces the need for reconstructive surgery. Grafting is done with split-thickness skin grafts; dermal substitutes, which serve as a bed for cell repopulation, may be used in large full-thickness burns.

Injury to the airways and lungs may develop within 24 to 48 hours after the burn and may result from the direct effect of heat on the mouth, nose, and upper airways or from the inhalation of heated air and noxious gases in the smoke. Water-soluble gases, such as chlorine,

sulfur oxides, and ammonia, may react with water to form acids or alkalis, particularly in the upper airways, producing inflammation and swelling that may lead to partial or complete airway obstruction. Lipid-soluble gases, such as nitrous oxide and products of burning plastics, are more likely to reach deeper airways, producing pneumonitis.

In burn survivors the development of hypertrophic scars, both at the site of the original burn and at donor graft sites, and itching may become long-term, difficult-to-treat problems. Hypertrophic scarring is a common complication of burn injury marked by excessive deposition of collagen in the healing wound bed; its etiology is not well understood.

MORPHOLOGY

Grossly, full-thickness burns are white or charred, dry, and painless (because of destruction of nerve endings), whereas, depending on the depth, partial-thickness burns are pink or mottled with blisters and are painful. Histologically, devitalized tissue reveals coagulative necrosis, while adjacent vital tissue quickly accumulates inflammatory cells and marked exudation.

Hyperthermia

Hyperthermia occurs most frequently in individuals exposed to elevated ambient temperatures, a risk that is exacerbated by climate change. The manifestations of prolonged exposure to high temperatures may take the form of heat cramps, heat exhaustion, or heat stroke. Another uncommon but important cause of hyperthermia is *malignant hypertension*, which is triggered by exposure to certain anesthetics rather than heat.

- *Heat cramps* result from loss of electrolytes via sweating. Cramping of voluntary muscles, usually in association with vigorous exercise, is the hallmark. Importantly, however, heat-dissipating mechanisms are sufficient to maintain a normal core body temperature.
- *Heat exhaustion*, probably the most common hyperthermic syndrome, is characterized by sudden onset prostration and collapse. It results from a failure of the cardiovascular system to compensate for hypovolemia caused by dehydration. After a period of collapse, which is usually brief, equilibrium is reestablished if the victim rehydrates.
- *Heat stroke* is associated with high ambient temperatures, high humidity, and exertion. Older adults, individuals undergoing intense physical stress (including young athletes and military recruits), and persons with cardiovascular disease are at particularly high risk for heat stroke. Thermoregulatory mechanisms fail, sweating ceases, and the core body temperature rises to more than 40°C, leading to multiorgan dysfunction that can be rapidly fatal. The hyperthermia is accompanied by marked generalized vasodilation, with peripheral pooling of blood and a decreased effective circulating blood volume. Hyperkalemia, tachycardia, arrhythmias, and other systemic effects are common. Particularly important, however, are sustained contractions of skeletal muscle that can exacerbate the hyperthermia and lead to muscle necrosis (rhabdomyolysis). These phenomena appear to stem from nitrosylation of *ryanodine receptor 1* (RYR1), which is located in the sarcoplasmic reticulum of skeletal muscle. RYR1 regulates the release of calcium from the sarcoplasm. Heat stroke deranges RYR1 function and allows calcium to leak into the cytoplasm, where it stimulates muscle contraction and additional heat production.
- *Malignant hyperthermia*, despite the name, is not caused by exposure to high temperatures. It is a genetic condition resulting from mutations in genes such as *RYR1* that control calcium levels in skeletal muscle cells. In affected individuals, exposure to certain anesthetics during surgery triggers a rapid rise in calcium levels in skeletal muscle, leading to muscle rigidity and increased heat production. The resulting hyperthermia has a mortality rate of approximately 80% if untreated, but this falls to less than 5% if the condition is recognized and muscle relaxants are administered promptly.

Hypothermia

Prolonged exposure to low ambient temperature leads to hypothermia, a dangerous condition that is of particular concern to people experiencing homelessness. High humidity and wet clothing, sometimes exacerbated by dilation of superficial blood vessels due to ingestion of alcohol, hasten the lowering of body temperature. At a body temperature of about 90°F, loss of consciousness occurs, followed by bradycardia and atrial fibrillation at lower core temperatures.

Hypothermia causes injury by two mechanisms:

- *Direct effects* are probably mediated by physical disruptions within cells by high salt concentrations caused by the crystallization of intracellular and extracellular water (frostbite).
- *Indirect effects* result from circulatory changes, which vary depending on the rate and duration of the temperature drop. Slow chilling may induce vasoconstriction and increase vascular permeability, leading to edema and hypoxia. Such changes are typical of trench foot. This condition developed in soldiers who spent long periods of time in waterlogged trenches during World War I (1914–18), frequently causing gangrene that necessitated amputation. With sudden, persistent chilling, the vasoconstriction and increased viscosity of the blood in the local area may cause ischemic injury and degenerative changes in peripheral nerves. In this situation, vascular injury and edema become evident only after the temperature begins to return to normal. However, if the period of ischemia is prolonged, hypoxic changes and infarction of the affected tissues (e.g., gangrene of toes or feet) may result.

Electrical Injury

Electrical injuries are often fatal and can arise from contact with low-voltage currents (i.e., in the home and workplace) or high-voltage currents carried by high-power lines or produced by lightning. Injuries are of two types: (1) burns and (2) ventricular fibrillation or cardiac and respiratory center failure, resulting from disruption of nerve impulse conduction. The type of injury and the severity and extent of burns depend on the strength (amperage), duration, and path of the electric current within the body.

Voltage in the household and workplace (120 or 220 V) is high enough that with low resistance at the site of contact (as when the skin is wet), sufficient current can pass through the body to cause serious injury, including ventricular fibrillation. If the current flow is sustained, it may generate enough heat to produce burns at the site of entry and exit as well as in internal organs. An important characteristic of alternating current, the type supplied to most homes, is that it induces tetanic muscle spasm, so that when a live wire or switch is grasped, irreversible clutching is likely to occur, prolonging the period of current flow. This results in a greater likelihood of developing extensive electrical burns and, in some cases, spasm of the chest wall muscles, producing death from asphyxia. Currents generated from high-voltage sources cause similar damage; however, because of the large current flows generated, these are more likely to produce paralysis of medullary centers and extensive burns. Lightning is a classic cause of high-voltage electrical injury.

Injury Produced by Ionizing Radiation

Radiation is energy that travels in the form of waves or high-speed particles. Radiation has a wide range of energies that span the electromagnetic spectrum; it can be divided into nonionizing and ionizing radiation. The energy of nonionizing radiation such as UV and infrared light, microwave, and sound waves, can move atoms in a molecule or cause them to vibrate but is not sufficient to displace bound electrons from atoms. By contrast, ionizing radiation has sufficient energy to remove tightly bound electrons.

Collision of free electrons with other molecules releases more electrons in a reaction cascade referred to as ionization. The main sources of ionizing radiation are (1) x-rays and gamma rays (electromagnetic waves of very high frequencies); (2) high-energy neutrons; (3) alpha particles (composed of two protons and two neutrons); and (4) beta particles, which may be either electrons or positrons. At equivalent amounts of energy, alpha particles induce heavy damage in a restricted area, whereas x-rays and gamma rays dissipate energy over a longer, deeper course and produce considerably less damage per unit of tissue. About 50% of the total dose of ionizing radiation received by the US population is human made, mostly originating from medical devices and radioisotopes, a proportion that has increased sharply over time, particularly with widespread use of computed tomography (CT) scans.

Ionizing radiation is a double-edged sword. It is indispensable in medical practice, being used in the treatment of cancer, in diagnostic imaging, and in therapeutic or diagnostic radioisotopes, but it also produces adverse short-term and long-term effects such as fibrosis, mutagenesis, carcinogenesis, and teratogenesis.

Radiation Units

Several somewhat confusing terms are used to describe radiation dose, which can be quantified according to the amount of radiation emitted by a source, the amount of radiation that is absorbed by a person, and the biologic effect of the radiation. Commonly used terms are:

- **Curie (Ci)** represents the disintegrations per second of a radionuclide (radioisotope); 1 Ci is equal to 3.7×10^{10} disintegrations per second. This is an expression of the amount of radiation emitted by a source.
- **Gray (Gy)** is a unit that expresses the energy absorbed by the target tissue per unit mass; 1 Gy corresponds to absorption of 10^4 erg/g of tissue. A centigray (cGy), which is the absorption of 100 erg/g of tissue, is equivalent to 100 Rad (radiation absorbed dose), abbreviated as R. The cGy terminology has now replaced the Rad in medical practice.
- **Sievert (Sv)** is a unit of equivalent dose that depends on the biologic rather than the physical effects of radiation (it replaced a unit called Rem). For the same absorbed dose, various types of radiation produce different amounts of damage. The equivalent dose controls for this variation and thereby provides a uniform measure of biologic dose. The equivalent dose (expressed in Sieverts) corresponds to the absorbed dose (expressed in Grays) multiplied by the relative biologic effectiveness of the radiation. The relative biologic effectiveness depends on the type of radiation, the type and volume of the exposed tissue, the duration of the exposure, and some other biologic factors (discussed next). The effective dose of x-rays in radiographs and CT is commonly expressed in milliSieverts (mSv). For x-radiation, $1 \text{ mSv} = 1 \text{ mGy}$.

Main Determinants of Biologic Effects of Ionizing Radiation

In addition to the physical properties of radiation, its biologic effects depend heavily on the following factors.

- **Rate of delivery.** The biologic effect of ionizing radiation is significantly modified by the rate of delivery. Although the effect of radiant energy is cumulative, divided doses may allow cells to repair some of the damage between exposures. Thus, fractionated doses of radiant energy have a cumulative effect only to the extent that repair during the “recovery” intervals is incomplete. Radiation therapy of tumors exploits the general capability of normal cells to repair themselves and recover more rapidly than tumor cells and thus not sustain as much cumulative radiation damage.

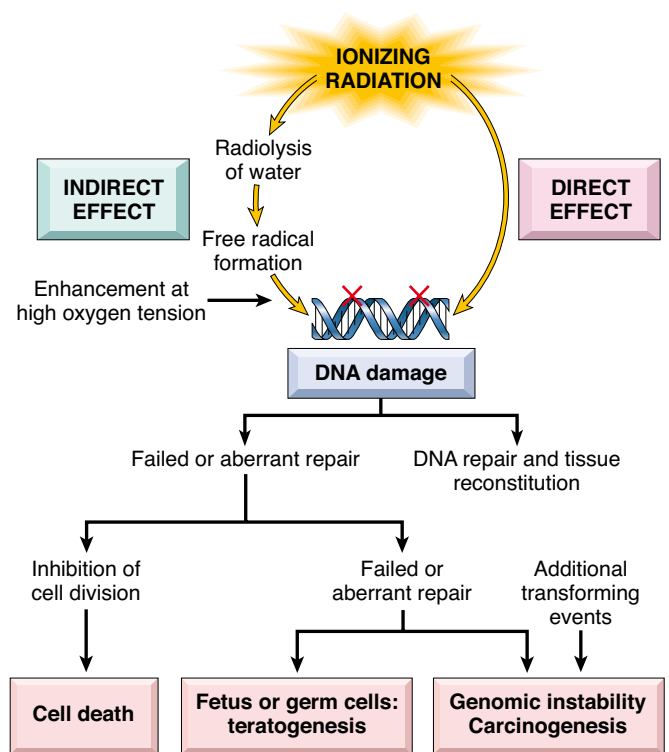


Fig. 9.15 Effects of ionizing radiation on DNA and downstream consequences. Ionizing radiation can damage DNA directly, or, more prominently, indirectly through free radical formation.

- **Field size** has a great influence on the consequences of irradiation. The body can sustain relatively high doses of radiation when delivered to small, carefully shielded fields, whereas smaller doses delivered to larger fields may be lethal.
- **Cell proliferation.** Because ionizing radiation damages DNA, rapidly dividing cells are more vulnerable to injury than quiescent cells (Fig. 9.15). Except at extremely high doses that impair DNA transcription, irradiation does not kill nondividing cells, such as neurons and muscle cells. Because radiation damages DNA and proliferating cells rely on DNA replication, tissues with a high rate of cell division, such as gonads, bone marrow, lymphoid tissue, and the mucosa of the gastrointestinal tract, are extremely vulnerable to radiation, and the injury is manifested early after exposure.
- **Oxygen effects and hypoxia.** The production of ROS from reactions with free radicals generated by radiolysis of water is the major mechanism by which DNA is damaged by ionizing radiation. Poorly vascularized tissues with low oxygenation, such as the center of rapidly growing tumors, are generally less sensitive to radiation therapy than nonhypoxic tissues.
- **Vascular damage.** Damage to endothelial cells, which are moderately sensitive to radiation, may cause narrowing or occlusion of blood vessels leading to impaired healing, fibrosis, and chronic ischemic atrophy. These changes may appear months or years after exposure.

Fig. 9.16 shows the overall consequences of radiation exposure. These consequences vary according to the dose of radiation and type of exposure. Table 9.7 lists the estimated threshold doses for acute effects of radiation aimed at specific organs; Table 9.8 lists the syndromes caused by exposure to various doses of total-body radiation.

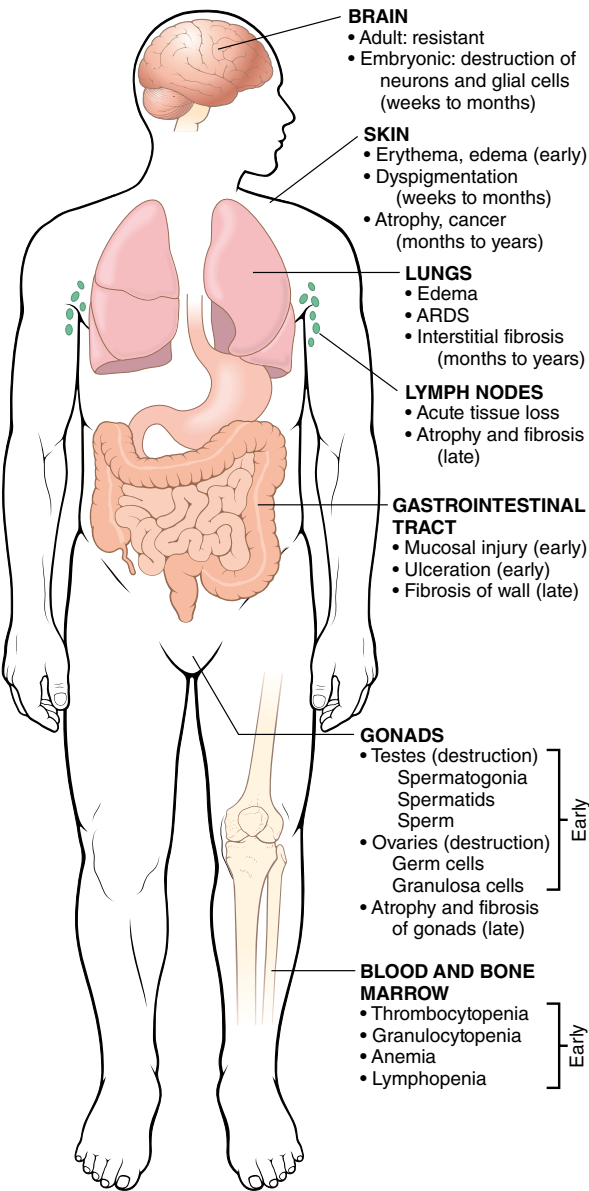


Fig. 9.16 Overview of the major morphologic consequences of radiation injury. Early changes occur in hours to weeks; late changes occur in months to years. ARDS, Acute respiratory distress syndrome.

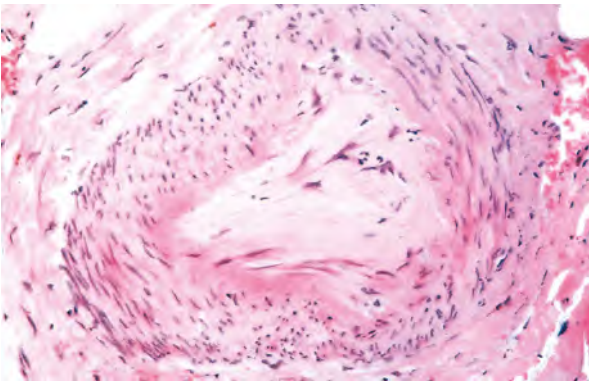


Fig. 9.17 Radiation-induced chronic vascular injury with subintimal fibrosis occluding the lumen. (From American Registry of Pathology, 1990.)

TABLE 9.7 Estimated Threshold Doses for Acute Radiation Effects on Specific Organs

Health Effect	Organ	Dose (Sv)
Temporary sterility	Testes	0.15
Depression of hematopoiesis	Bone marrow	0.50
Reversible skin effects (e.g., erythema)	Skin	1–2
Permanent sterility	Ovaries	2.5–6
Temporary hair loss	Skin	3–5
Permanent sterility	Testes	3.5
Cataract	Lens of eye	5

MORPHOLOGY

Cells surviving radiant energy damage show a wide range of structural **changes in chromosomes** that are related to double-stranded DNA breaks including deletions, translocations, and fragmentation. The mitotic spindle often becomes disorderly, and polyploidy and aneuploidy may be encountered. **Nuclear swelling** and condensation and clumping of chromatin may appear; disruption of the nuclear membrane may also be noted. Apoptosis may occur. Several **abnormal nuclear morphologies** may be seen. Giant cells with pleomorphic nuclei or more than one nucleus may appear and persist for years after exposure. At extremely high doses of radiant energy, indicators of cell death, such as nuclear pyknosis and lysis, appear quickly.

In addition to affecting DNA and nuclei, radiant energy may induce a variety of **cytoplasmic changes** including swelling, mitochondrial distortion, and degeneration of the endoplasmic reticulum. Plasma membrane breaks and focal defects may be seen. The histologic constellation of cellular pleomorphism, giant-cell formation, conformational changes in nuclei, and abnormal mitotic figures creates a more than passing similarity between radiation-injured cells and cancer cells, a problem that plagues the pathologist when evaluating irradiated tissues for the possible persistence of tumor cells.

Vascular changes (Fig. 9.17) and interstitial fibrosis (Fig. 9.18) are also prominent in irradiated tissues. During the immediate postirradiation period, vessels may show only dilation. With time or with higher doses, a variety of degenerative changes appear, including endothelial cell swelling and vacuolation, or even necrosis and dissolution of the walls of small vessels such as capillaries and venules. Affected vessels may rupture or thrombose. Still later, endothelial cell proliferation and collagenous hyalinization and thickening of the intima are seen in irradiated vessels, resulting in marked narrowing or even obliteration of the vascular lumens. By this time an increase in interstitial collagen in the irradiated field usually becomes evident, leading to scarring and contractions.

Total-Body Irradiation

Exposure of large areas of the body to even very small doses of radiation may have devastating effects. Doses below 1 Sv produce minimal symptoms, if any. However, higher levels of exposure cause health effects known as acute radiation syndromes, which at progressively higher doses involve the hematopoietic, gastrointestinal, and central nervous systems. The syndromes associated with total-body exposure to ionizing radiation are presented in Table 9.8.

Acute Effects on Hematopoietic and Lymphoid Systems

The hematopoietic and lymphoid systems are extremely susceptible to radiation injury and deserve special mention. With high dose levels and large exposure fields, severe lymphopenia appears within hours along with shrinkage of the lymph nodes and spleen. Radiation kills lymphocytes directly, both in the circulation and in tissues (nodes,

TABLE 9.8 Effects of Total-Body Ionizing Radiation

	0–1 Sv	1–2 Sv	2–10 Sv	10–20 Sv	>50 Sv
Main site of injury	None	Lymphocytes	Bone marrow	Small bowel	Brain
Main signs and symptoms	None	Moderate granulocytopenia, lymphopenia	Leukopenia, hemorrhage, hair loss, vomiting	Diarrhea, fever, electrolyte imbalance, vomiting	Ataxia, coma, convulsions, vomiting
Time of development	—	1 day to 1 week	2 to 6 weeks	5 to 14 days	1 to 4 hours
Lethality	None	None	Variable (0%–80%)	100%	100%

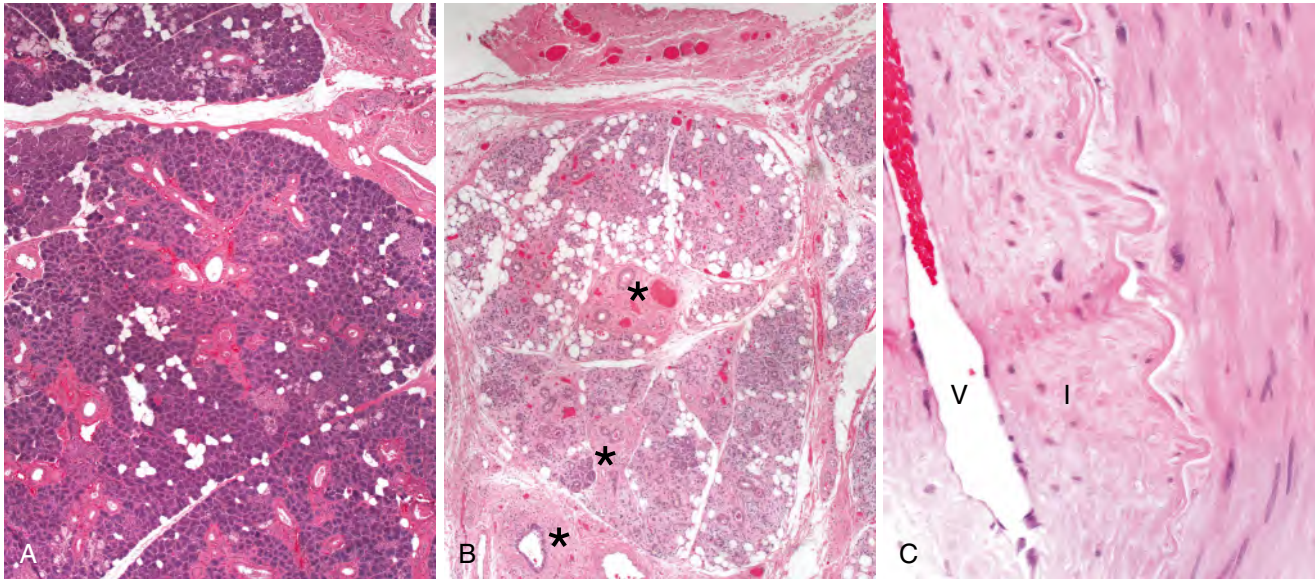


Fig. 9.18 Vascular changes and fibrosis of salivary glands produced by radiation therapy of the neck region. (A) Healthy salivary gland. (B) Fibrosis (asterisks) caused by radiation. (C) Vascular changes consisting of fibrointimal thickening and arteriolar sclerosis. *I*, Thickened intima; *V*, vessel lumen. (Courtesy Dr. Melissa Upton, Department of Pathology, University of Washington, Seattle.)

spleen, thymus, gut). With sublethal doses of radiation, regeneration from viable precursors is prompt, leading to restoration of a normal blood lymphocyte count within weeks to months. Hematopoietic precursors in the bone marrow are also quite sensitive to radiant energy, which produces a dose-dependent marrow aplasia. The acute effects of marrow irradiation on peripheral blood counts reflect the kinetics of turnover of the formed elements, the granulocytes, platelets, and red cells, which have half-lives of less than a day (granulocytes), 10 days (platelets), and 120 days (red cells). After a brief rise in the circulating neutrophil count, neutropenia appears within several days. Neutrophil counts reach their nadir, often at counts near zero, during the second week. If the patient survives, recovery of a normal granulocyte count may require 2 to 3 months. Thrombocytopenia appears by the end of the first week, with the platelet count nadir occurring somewhat later than that of granulocytes; recovery is similarly delayed. Anemia appears after 2 to 3 weeks and may persist for months. Understandably, higher doses of radiation produce more severe cytopenias and more prolonged periods of recovery. Very high doses kill hematopoietic stem cells and induce permanent aplasia (aplastic anemia) marked by a failure of blood count recovery, whereas with lower doses the aplasia is transient.

Fibrosis

A common consequence of radiation therapy for cancer is the development of fibrosis in the tissues included in the irradiated field (see

Fig. 9.18). Fibrosis may occur weeks or months after irradiation due to replacement of dead parenchymal cells by connective tissue, leading to the formation of scars and adhesions (**Fig. 9.19**). Vascular damage causing ischemic injury, the death of tissue stem cells, and the release of cytokines and chemokines that promote inflammation and fibroblast activation are the main contributors to the development of radiation-induced fibrosis. Common sites of fibrosis after radiation treatment are the lungs, the salivary glands after radiation therapy for head and neck cancers, and colorectal and pelvic areas after treatment for cancer of the prostate, rectum, or cervix.

DNA Damage and Carcinogenesis

Ionizing radiation can cause several types of DNA damage including single-base damage, single- and double-stranded breaks (DSBs), and DNA-protein crosslinks. In surviving cells, simple defects may be repaired by various enzyme systems present in most mammalian cells. The most serious damage to DNA consists of DSBs. Two types of mechanisms can repair DSBs in mammalian cells: homologous recombination and nonhomologous end joining (NHEJ), with NHEJ being the most common repair pathway. DNA repair through NHEJ often produces mutations including short deletions or duplications or gross chromosomal aberrations such as translocations and inversions. If the replication of cells containing DSBs is not stopped by cell cycle checkpoint controls (Chapters 1 and 7), cells with chromosomal damage persist and may initiate carcinogenesis many years later.

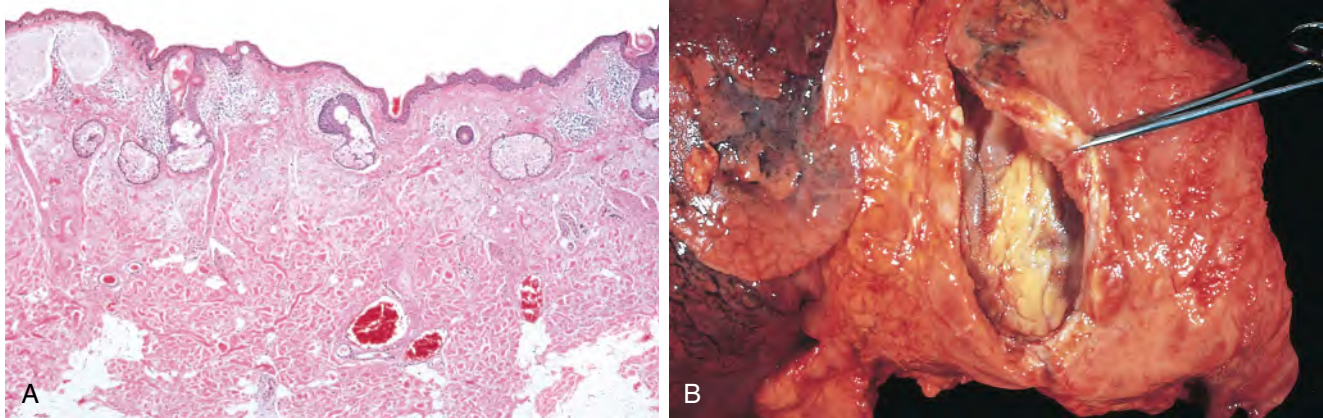


Fig. 9.19 Radiation-induced fibrosis. (A) Chronic radiation dermatitis with atrophy of epidermis, dermal fibrosis, and telangiectasia of the subcutaneous blood vessels. (B) Extensive mediastinal fibrosis after radiotherapy for carcinoma of the lung. Note the markedly thickened pericardium. (A, From American Registry of Pathology, 1990. B, From the Collection of the Department Pathology, Southwestern Medical School, Dallas.)

Cancer Risks From Exposures to Radiation

Any cell capable of division that has sustained a mutation has the potential to become cancerous. Thus, an increased incidence of neoplasms may occur in any organ after exposure to ionizing radiation. The level of radiation required to increase the risk of cancer development is difficult to determine, but there is little doubt that acute or prolonged exposures that result in doses of greater than 100 mSv cause serious consequences, including cancer. Proof of this risk is found in the increased incidence of leukemias and solid tumors in several organs (e.g., thyroid, breast, and lungs) in survivors of the atomic bombings of Hiroshima and Nagasaki; the high number of thyroid cancers in survivors of the Chernobyl accident; the high incidence of thyroid tumors and the elevated frequency of leukemias and birth defects in inhabitants of the Marshall Islands exposed to nuclear fallout; and the development of second cancers, such as acute myeloid leukemia, myelodysplastic syndrome, and solid tumors, in individuals who received radiation therapy for cancers such as Hodgkin lymphoma. The long-term cancer risks caused by radiation exposures in the range of 5 to 100 mSv are much more difficult to establish because accurate measurements of risks require large population groups ranging from 50,000 to 5 million people. Nevertheless, for x-rays and gamma rays there is good evidence for a statistically significant increase in the risk of cancer at acute doses of greater than 50 mSv and reasonable evidence for acute doses of greater than 5 mSv; as a point of reference, a single posteroanterior chest radiograph, a lateral chest film chest radiograph, and a CT scan of the chest deliver effective doses to the lungs of 0.01 mSv, 0.15 mSv, and 10 mSv. It is believed that the risk of secondary cancers following irradiation is greatest in children. This is based in part on a recent large-scale epidemiologic study showing that children who receive at least two CT scans have very small but measurable increased risks for leukemia and malignant brain tumors and on older studies showing that radiation therapy to the chest is particularly likely to produce breast cancers when administered to adolescent females.

Increased risk of cancer development may also be associated with occupational exposures. Radon gas is a ubiquitous product of the spontaneous decay of uranium. Its carcinogenic effects are largely attributable to two decay products, polonium 214 and polonium 218 (or “radon daughters”), which emit alpha particles. Polonium 214 and polonium 218 produced from inhaled radon tend to deposit in the lung, and chronic exposure in uranium miners may give rise to lung carcinomas. Risks are also present in homes in which the levels of

radon are very high, comparable to those found in mines. It is suspected that lower levels of household radon may also contribute to lung cancer development, particularly in those who also smoke tobacco. The CDC estimates that nearly 3000 deaths occur annually in the United States due to radon exposure in nonsmokers.

KEY CONCEPTS

Radiation Injury

- Ionizing radiation may injure cells directly or indirectly by generating free radicals from water or molecular oxygen.
- Ionizing radiation damages DNA; therefore, rapidly dividing cells such as germ cells and those in the bone marrow and gastrointestinal tract are very sensitive to radiation injury.
- Inadequately repaired DNA damage may result in mutations that predispose affected cells to neoplastic transformation.
- Ionizing radiation may cause vascular damage and sclerosis, resulting in ischemic necrosis of parenchymal cells and their replacement by fibrous tissue.

NUTRITIONAL DISEASES

Malnutrition is a consequence of inadequate intake of proteins and calories or deficiencies in the digestion or absorption of proteins resulting in the loss of fat and muscle mass, weight loss, and generalized weakness. Millions of people in low-resource nations are malnourished or living on the cruel edge of starvation. In the high-resource parts of the world and, more recently, in some low-resource countries, obesity has become a major public health problem due to its association with diseases such as diabetes, atherosclerosis, and cancer.

The sections that follow barely skim the surface of nutritional disorders. Particular attention is devoted to childhood malnutrition, anorexia nervosa and bulimia, deficiencies of vitamins and trace minerals, obesity, and the relationships of diet to cancer and atherosclerosis. Other nutrients and nutritional issues are discussed in the context of specific diseases.

Dietary Insufficiency

An appropriate diet should provide (1) sufficient energy, in the form of carbohydrates, fats, and proteins, for the body's daily metabolic needs;

(2) amino acids and fatty acids to be used as building blocks for synthesis of proteins and lipids; and (3) vitamins and minerals, which function as coenzymes or hormones in vital metabolic pathways or, as in the case of calcium and phosphate, as important structural components. In *primary malnutrition*, one or more of these components are missing from the diet. By contrast, *secondary malnutrition* results from malabsorption, impaired utilization or storage, excess loss, or increased need for nutrients.

There are several conditions that may lead to primary or secondary malnutrition.

- **Poverty.** Homeless persons, aged individuals, and children of the poor often suffer from severe malnutrition as well as trace nutrient deficiencies. In low-resource countries, poverty, crop failures, livestock deaths, and drought, often in times of war and political upheaval, create the setting for the malnourishment of children and adults.
- **Acute and chronic illnesses.** The basal metabolic rate rises in many illnesses, resulting in increased daily requirements for all nutrients. Failure to recognize these nutritional needs may delay recovery. Malnutrition is often present in patients with wasting diseases, such as advanced cancers and AIDS that are complicated by cachexia (Chapter 7).
- **Chronic excessive alcohol use.** People with alcohol use disorder may sometimes suffer from malnutrition but more frequently have deficiencies of vitamins—especially thiamine, pyridoxine, folate, and vitamin A—caused by poor diet, defective gastrointestinal absorption, abnormal nutrient utilization and storage, increased metabolic needs, and an increased rate of loss. A failure to recognize the likelihood of thiamine deficiency in persons with alcohol use disorder may result in irreversible brain damage (e.g., *Wernicke encephalopathy* and *Korsakoff psychosis*, discussed in Chapter 28).
- **Ignorance and failure of diet supplementation.** Even the affluent may fail to recognize that infants, adolescents, and pregnant females have increased nutritional needs. Ignorance about the nutritional content of foods is also a contributing factor. Some examples are iron deficiency in infants fed exclusively artificial milk diets; thiamine deficiency in diets in which polished rice is the mainstay; and lack of iodine due to consumption of locally produced food and water in regions removed from the oceans.
- **Self-imposed dietary restriction.** Anorexia nervosa, bulimia, and less overt eating disorders affect individuals who are concerned about body image and are obsessed with body weight (anorexia and bulimia are discussed later).
- **Other causes.** Additional causes of malnutrition include gastrointestinal diseases and malabsorption syndromes, genetic diseases, certain drug therapies (which block uptake or utilization of specific nutrients), and inadequate total parenteral nutrition.

Severe Acute Malnutrition

UNICEF/WHO/World Bank Group define malnutrition as a life-threatening wasting disorder of children caused by poor nutrient intake and/or recurrent illnesses. Children suffering from wasting have weakened immunity, are susceptible to long-term developmental delays, and face an increased risk of death, particularly when wasting is severe. In 2022, about 45 million children under the age of 5 years (6.8% of children in this age group) were affected by wasting, of which 13.6 million (2.1%) suffered from severe wasting (also known as severe acute malnutrition). Severe acute malnutrition is defined by several diagnostic criteria, including a mid-upper arm circumference of less than 115 mm, a weight-for-height ratio that is three standard deviations below normal, or the presence of bilateral pitting edema of the lower limbs. More than three-quarters of all children with severe

wasting live in Asia and another 22% live in Africa. Over 20% of children under 5 years (approximately 148 million children) suffer from stunting, which results from malnutrition in utero and during early childhood.

Severe acute malnutrition manifests as a spectrum of clinical syndromes, all resulting from a dietary intake of protein and calories that is inadequate to meet the body's needs. At two ends of severe acute malnutrition are *marasmus* and *kwashiorkor*. It should be noted that from a functional standpoint, there are two protein compartments in the body: the somatic compartment, represented by proteins in skeletal muscles, and the visceral compartment, represented by protein stores in the visceral organs, primarily the liver. These two compartments are regulated differently, as detailed subsequently. The somatic compartment is affected more severely in marasmus, and the visceral compartment is depleted more severely in kwashiorkor. Clinical assessment of undernutrition is discussed next, followed by descriptions of marasmus and kwashiorkor.

The diagnosis of severe acute malnutrition is obvious in its most severe forms. In mild to moderate forms, the usual approach is to compare the body weight for a given height against standard tables; other helpful parameters are fat stores, muscle mass, and levels of certain serum proteins. With a loss of fat, measured skinfold thickness (which includes skin and subcutaneous tissue) is reduced. If the somatic protein compartment (proteins in skeletal muscle) is catabolized, the resultant reduction in muscle mass is reflected by reduced circumference of the midarm. Measurement of serum proteins (albumin, transferrin, and others) provides an estimate of the adequacy of the visceral protein compartment. There may also be a role for the gut microbiome in the pathogenesis of severe acute malnutrition. There is a substantial difference in the microbial flora of children with severe acute malnutrition as compared to normal children, and some studies suggest that correcting these differences may improve recovery in children fed a nutrition-replete diet.

Marasmus

Marasmus develops when the diet is severely lacking in calories (Fig. 9.20A). A child is considered to have marasmus when weight falls to 60% of normal for sex, height, and age. A marasmic child suffers growth retardation and loss of muscle mass because of catabolism and depletion of the somatic protein compartment. This seems to be an adaptive response that provides the body with amino acids as a source of energy. Of interest, the visceral protein compartment, which is more critical for survival, is depleted only marginally, so serum albumin levels are either normal or only slightly reduced. In addition to muscle proteins, subcutaneous fat is also mobilized and used as fuel. Leptin (discussed later under “Obesity”) production is low, which may stimulate the hypothalamic-pituitary-adrenal axis to produce high levels of cortisol that contribute to lipolysis. Due to losses of muscle and subcutaneous fat, the extremities are emaciated; by comparison, the head appears too large for the body. Anemia and manifestations of multivitamin deficiencies are present, and there is evidence of immune deficiency, particularly of T-cell-mediated immunity. Hence, concurrent infections are usually present, which impose additional stress on a weakened body.

Kwashiorkor

Kwashiorkor occurs when protein deprivation is relatively greater than the reduction in total calories (Fig. 9.20B). This is the most common form of severe acute malnutrition seen in African children, who have been weaned too early and subsequently fed an almost exclusively carbohydrate diet (the name kwashiorkor, from the Ga language in Ghana, describes the illness in a young child that appears after the



Fig. 9.20 Childhood malnutrition. (A) Marasmus. Note the loss of muscle mass and subcutaneous fat; the head appears to be too large for the emaciated body. (B) Kwashiorkor. The infant shows generalized edema, seen as ascites and puffiness of the face, hands, and legs. (A, Courtesy Clinic Barak, Reisebericht, Kenya.)

arrival of another baby). The prevalence of kwashiorkor is also high in low-resource countries of Southeast Asia. Less severe forms occur worldwide in persons with chronic diarrheal states, in which protein is not absorbed, or in persons with chronic protein loss (e.g., protein-losing enteropathies, the nephrotic syndrome, or the aftermath of extensive burns). Rare cases of kwashiorkor resulting from fad diets or replacement of milk by rice-based beverages have been reported in the United States.

In kwashiorkor, unlike in marasmus, marked protein deprivation is associated with severe loss of the visceral protein compartment, and the resultant hypoalbuminemia gives rise to generalized or dependent edema (see Fig. 9.20B). The weight of children with severe kwashiorkor typically is 60% to 80% of normal, but the true loss of tissue mass is masked by increased fluid retention (edema). In further contrast with marasmus, there is relative sparing of subcutaneous fat and muscle mass. The modest loss of these compartments may also be masked by edema.

Children with kwashiorkor have characteristic skin lesions with alternating zones of hyperpigmentation, desquamation, and hypopigmentation, giving a “flaky paint” appearance. Hair changes include loss of color or alternating bands of pale and darker color, straightening, fine texture, and loss of firm attachment to the scalp. Other features that distinguish kwashiorkor from marasmus include an enlarged, fatty liver (resulting from reduced synthesis of carrier protein components of lipoproteins) and the development of apathy, listlessness, and loss of appetite. As in those with marasmus, vitamin deficiencies are likely to be present, as are defects in immunity. Secondary infections produce inflammation and a catabolic state that aggravates the malnutrition. As already mentioned, marasmus and kwashiorkor represent two ends of a spectrum, and considerable overlap exists.

Malnutrition in High-Resource Countries

In the United States, secondary malnutrition often develops in chronically ill, older, and bedridden patients. It is estimated that more than 50% of older residents in nursing homes in the United States are malnourished and that one-quarter of older adults in the general population are malnourished or at risk of malnutrition. The most obvious signs of secondary malnutrition include (1) depletion of subcutaneous fat in the arms, chest wall, shoulders, or metacarpal regions; (2) wasting of the quadriceps and deltoid muscles; and (3) ankle or sacral edema. Bedridden or hospitalized

malnourished patients have an increased risk of infection, sepsis, impaired wound healing, and death after surgery.

In older adults, it is important to assess the etiology of malnutrition. In this patient population, there are two main pathophysiological pathways of malnutrition: (1) energy and nutrient deficiency; and (2) inflammation-related malnutrition. Energy and nutrient deficiency occurs when there is inadequate nutrient intake or absorption, as may result from dysphagia following a stroke, or short bowel syndrome. Far more common in adults is inflammation-driven malnutrition, which is seen in the setting of chronic disorders such as cancer, chronic obstructive pulmonary disease, inflammatory bowel disease, and renal failure. It results from release of inflammatory cytokines and prostaglandins, which cause an increase in resting energy expenditure. This stimulates the catabolism of proteins in skeletal muscle, which provides amino acids for production of glucose at the expense of skeletal muscle mass, even in the setting of adequate protein intake. This mechanism may share features with that believed to be responsible for cancer-related cachexia (Chapter 7).

MORPHOLOGY

The main anatomic changes in severe acute malnutrition are (1) growth failure; (2) peripheral edema in kwashiorkor; and (3) loss of body fat and atrophy of muscle, more marked in marasmus.

The **liver** in kwashiorkor, but not in marasmus, is enlarged and fatty; superimposed cirrhosis is rare. In kwashiorkor (but rarely in marasmus) the **small bowel** shows a decrease in mitotic cells in the crypts of the glands, which is associated with mucosal atrophy and loss of villi and microvilli. In such cases concurrent loss of small intestinal enzymes occurs, most often manifested as disaccharidase deficiency. Hence, infants with kwashiorkor initially may not respond well to full-strength, milk-based diets. With treatment, the mucosal changes are reversible.

The **bone marrow** in both kwashiorkor and marasmus may be hypoplastic, mainly due to decreased numbers of red cell precursors. The peripheral blood commonly reveals mild to moderate anemia, which is often multifactorial in origin; nutritional deficiencies of iron, folate, and protein, as well as the suppressive effects of infection (anemia of chronic inflammation) may all contribute. Depending on the predominant factor, the red cells may be microcytic, normocytic, or macrocytic.

The **brain** in infants who are born to malnourished mothers and who suffer severe acute malnutrition during the first 1 or 2 years of life has been reported by some to show cerebral atrophy, a reduced number of neurons, and impaired myelination of white matter.

Many other changes may be present, including (1) thymic and lymphoid atrophy (more marked in kwashiorkor than in marasmus); (2) anatomic alterations induced by intercurrent infections, particularly with all manner of endemic worms and other parasites; and (3) deficiencies of other required nutrients such as iodine and vitamins.

Cachexia

Secondary malnutrition is a common complication in patients with AIDS or advanced cancers; in these settings it is known as *cachexia*. Because of its frequent association with advanced cancer, cachexia is discussed in Chapter 7.

Anorexia Nervosa and Bulimia

People with anorexia nervosa severely restrict their food intake, leading to significant weight loss, whereas people with bulimia experience episodes of uncontrollable eating (bingeing) followed by purging behaviors such as vomiting or laxative misuse to prevent weight gain. Anorexia nervosa has the highest death rate of any psychiatric disorder. Bulimia is more common than anorexia nervosa and generally has a better prognosis; it is estimated to occur in 1% to 2% of females and 0.1% of males, with an average onset at 20 years of age. These eating disorders occur primarily in previously healthy young females who have become preoccupied with body image and thinness. The neurobiologic underpinnings of these diseases are unknown, but altered serotonin metabolism may play a role.

The clinical findings in anorexia nervosa are generally like those in severe acute malnutrition. In addition, effects on the endocrine system are prominent. *Amenorrhea*, resulting from decreased secretion of gonadotropin-releasing hormone and, hence, reduced levels of luteinizing hormone and follicle-stimulating hormone, is so common that its presence is considered a diagnostic feature. Other common findings related to *decreased thyroid hormone release* include cold intolerance, bradycardia, constipation, and changes in the hair and skin, which becomes dry and scaly. In addition, dehydration and electrolyte abnormalities are frequently present. The bone density is decreased, most likely because of low estrogen levels, mimicking the postmenopausal acceleration of osteoporosis. Anemia, lymphopenia, and hypoalbuminemia may be present. Decreased marrow hematopoiesis is associated with a peculiar deposition of mucinous matrix material that is referred to as *gelatinous transformation*, a finding that is virtually pathognomonic for anorexia nervosa (Fig. 9.21). A major complication of anorexia nervosa (and bulimia) is an increased susceptibility to *cardiac arrhythmia* and *sudden death*, resulting from hypokalemia.

In bulimia, *binge eating* is the norm. Large amounts of food, principally carbohydrates, are ingested, followed by induced vomiting. Although menstrual irregularities are common, amenorrhea occurs in less than 50% of bulimic patients because weight and gonadotropin levels remain near normal. The major medical complications relate to frequent vomiting and the chronic use of laxatives and diuretics. They include (1) electrolyte imbalances (hypokalemia), which predispose the patient to cardiac arrhythmias; (2) pulmonary aspiration of gastric contents; and (3) esophageal and gastric rupture. Nevertheless, since there are no specific signs or symptoms, the diagnosis of bulimia relies on a comprehensive psychologic assessment.

Vitamin Deficiencies

Thirteen vitamins are necessary for health. Vitamins A, D, E, and K are fat-soluble, and all others are water-soluble, a distinction that is important. Fat-soluble vitamins are more readily stored in the body, but they may be

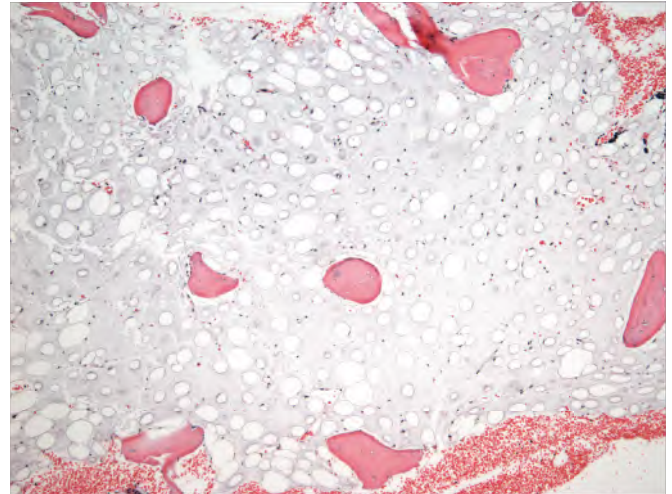


Fig. 9.21 Bone marrow changes in anorexia nervosa. Decreased cellularity associated with deposition of mucinous matrix material (gelatinous transformation) is pathognomonic of this disease.

poorly absorbed in fat malabsorption disorders, caused by disturbances of digestive functions (Chapter 17). Certain vitamins can be synthesized endogenously—vitamin D from precursor steroids; vitamin K and biotin by the intestinal microflora; and niacin from tryptophan, an essential amino acid. Notwithstanding this endogenous synthesis, a dietary supply of all vitamins is essential for health.

A deficiency of vitamins may be primary (dietary in origin) or secondary to disturbances in intestinal absorption, transport in the blood, tissue storage, or metabolic conversion. In the following sections, vitamins A, D, and C are discussed in further detail because of their wide-ranging activities and the morphologic changes of deficiency states. In addition, Tables 9.9 and 9.10 provide overviews of the main consequences of deficiencies of several vitamins and essential minerals. However, it should be emphasized that deficiency of a single vitamin is uncommon, and that single or multiple vitamin deficiencies may be associated with severe acute malnutrition.

Vitamin A

The major functions of vitamin A are maintenance of normal vision, regulation of cell growth and differentiation, and regulation of lipid metabolism. Vitamin A is the name given to a group of related compounds that include *retinol*, *retinal*, and *retinoic acid*, which have similar biologic activities. The generic term *retinoid* encompasses vitamin A in its various forms and natural and synthetic chemicals that are structurally related to vitamin A but may not necessarily have vitamin A–like biologic activity. Animal-derived foods such as liver, fish, eggs, milk, and butter are important dietary sources of preformed vitamin A. Yellow and leafy green vegetables such as carrots, squash, and spinach supply large amounts of carotenoids, provitamins that can be metabolized to vitamin A in the body. Carotenoids contribute approximately 30% of the vitamin A in human diets; the most important of these is β -carotene, which is efficiently converted to vitamin A. The recommended dietary allowance for vitamin A is expressed in retinol equivalents to account for intake of both preformed vitamin A and β -carotene.

Vitamin A is a fat-soluble vitamin, and its absorption requires bile, pancreatic enzymes, and some level of antioxidant activity in food. Retinol (generally ingested as retinol ester) and β -carotene are absorbed in the intestine, where β -carotene is converted to retinol (Fig. 9.22). Retinol is then transported in chylomicrons to the liver, where it is taken up through the apolipoprotein E receptor, esterified, and stored. More than

TABLE 9.9 Vitamins: Major Functions and Deficiency Syndromes

Vitamin	Functions	Deficiency Syndromes
Fat Soluble		
Vitamin A	A component of visual pigment Maintenance of specialized epithelia Maintenance of resistance to infection	Night blindness, xerophthalmia, blindness Squamous metaplasia Vulnerability to infection, particularly measles
Vitamin D	Facilitates intestinal absorption of calcium and phosphorus and mineralization of bone	Rickets in children Osteomalacia in adults
Vitamin E	Major antioxidant; scavenges free radicals	Spinocerebellar degeneration, hemolytic anemia
Vitamin K	Cofactor in hepatic carboxylation of procoagulants—factors II (prothrombin), VII, IX, and X and protein C and protein S	Abnormal bleeding (Chapter 14)
Water Soluble		
Vitamin B ₁ (thiamine)	As pyrophosphate, is coenzyme in decarboxylation reactions	Dry and wet beriberi, Wernicke-Korsakoff syndrome (Chapter 28)
Vitamin B ₂ (riboflavin)	Converted to coenzymes flavin mononucleotide and flavin adenine dinucleotide, cofactors for many enzymes in intermediary metabolism	Ariboflavinosis, cheilosis, stomatitis, glossitis, dermatitis, corneal vascularization
Niacin	Incorporated into nicotinamide adenine dinucleotide (NAD) and NAD phosphate, involved in a variety of redox reactions	Pellagra—"three Ds": dementia, dermatitis, diarrhea
Vitamin B ₆ (pyridoxine)	Derivatives serve as coenzymes in many intermediary reactions	Cheilosis, glossitis, dermatitis, peripheral neuropathy (Chapter 28) Maintenance of myelination of spinal cord tracts
Vitamin B ₁₂	Required for normal folate metabolism and DNA synthesis	Megaloblastic pernicious anemia and degeneration of posterolateral spinal cord tracts (Chapter 14)
Vitamin C	Serves in many oxidation-reduction (redox) reactions and hydroxylation of collagen	Scurvy
Folate	Essential for transfer and use of one-carbon units in DNA synthesis	Megaloblastic anemia, neural tube defects (Chapter 14)
Pantothenic acid	Incorporated in coenzyme A	No nonexperimental syndrome recognized
Biotin	Cofactor in carboxylation reactions	No clearly defined clinical syndrome

TABLE 9.10 Selected Trace Elements and Deficiency Syndromes

Element	Function	Basis of Deficiency	Clinical Features
Zinc	Component of enzymes, principally oxidases	Inadequate supplementation in artificial diets Interference with absorption by other dietary constituents Inborn error of metabolism	Rash around eyes, mouth, nose, and anus called acrodermatitis enteropathica Anorexia and diarrhea Growth retardation in children Depressed mental function Depressed wound healing and immune response Impaired night vision Infertility
Iron	Essential component of hemoglobin as well as several iron-containing metalloenzymes	Inadequate diet Chronic blood loss	Hypochromic microcytic anemia (Chapter 14)
Iodine	Component of thyroid hormone	Inadequate supply in food and water	Goiter and hypothyroidism (Chapter 24)
Copper	Component of cytochrome c oxidase, dopamine β -hydroxylase, tyrosinase, lysyl oxidase, and unknown enzymes involved in cross-linking collagen	Inadequate supplementation in artificial diet Interference with absorption	Muscle weakness Neurologic defects Abnormal collagen cross-linking Anemia and ring sideroblasts
Fluoride	Mechanism unknown	Inadequate supply in soil and water Inadequate supplementation	Dental caries (Chapter 16)
Selenium	Component of glutathione peroxidase Antioxidant with vitamin E	Inadequate amounts in soil and water	Myopathy Cardiomyopathy (Keshan disease)

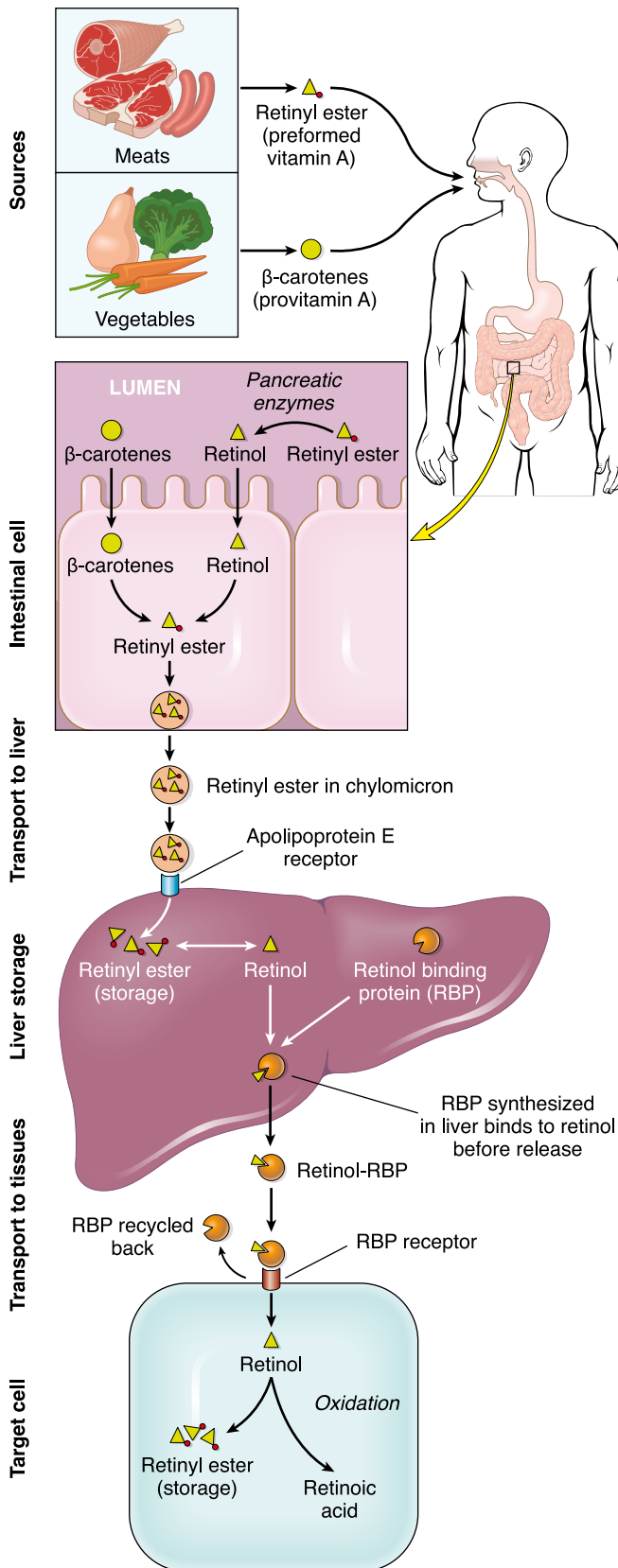


Fig. 9.22 Vitamin A metabolism.

90% of the body's vitamin A reserves are found in the liver, predominantly in the perisinusoidal stellate (Ito) cells. In healthy persons who consume an adequate diet, these reserves are sufficient to meet the body's demands for at least 6 months. Retinol esters stored in the liver can be mobilized; before release, retinol binds to a specific retinol-binding protein (RBP) that is synthesized in the liver. The uptake of retinol/RBP in peripheral tissues is dependent on cell surface receptors specific for RBP. Once taken up into peripheral tissues, retinol may also be stored as retinol ester or may be oxidized to form retinoic acid, which has important effects on epithelial differentiation and growth.

Function. In humans, the main functions of vitamin A are:

- **Maintenance of normal vision.** The visual process involves four forms of vitamin A-containing pigments: rhodopsin in the rods, the most light-sensitive pigment and therefore important in reduced light; and three iodopsins in cone cells, each responsive to specific colors in bright light. The synthesis of rhodopsin from retinol involves (1) oxidation to all-*trans*-retinal, (2) isomerization to 11-*cis*-retinal, and (3) covalent association with the 7-transmembrane rod protein opsin to form rhodopsin. A photon of light causes the isomerization of 11-*cis*-retinal to all-*trans*-retinal, which dissociates from rhodopsin. This induces a conformational change in opsin, triggering a series of downstream events that generate a nerve impulse, which is transmitted via neurons from the retina to the brain. During dark adaptation, some of the all-*trans*-retinal is reconverted to 11-*cis*-retinal, but most is reduced to retinol and lost from the retina, resulting in the need for continuous supply.
- **Cell growth and differentiation.** Vitamin A and retinoids play an important role in the orderly differentiation of mucus-secreting epithelium; when a deficiency state exists, the epithelium undergoes squamous metaplasia, differentiating into a keratinizing epithelium. Activation of retinoic acid receptors (RARs) by their ligands causes the release of corepressors and the obligatory formation of heterodimers with another retinoid receptor, RXR. Both RAR and RXR have three isoforms, α , β , and γ . The RAR/RXR heterodimers bind to retinoic acid response elements located in the regulatory regions of genes that encode receptors for growth factors, tumor suppressor genes, and secreted proteins. Through these effects, retinoids regulate cell growth and differentiation, cell cycle control, and other biologic responses. All-*trans*-retinoic acid, a potent acid derivative of vitamin A, has the highest affinity for RARs of any retinoid.
- **Metabolic effects of retinoids.** RXR, believed to be activated by 9-*cis*-retinoic acid, can also form heterodimers with other nuclear receptors, such as receptors involved in drug metabolism, peroxisome proliferator-activated receptors (PPARs), and vitamin D receptors. PPARs are key regulators of lipid metabolism including fatty acid oxidation in fat cells and muscle, adipogenesis, and lipoprotein metabolism. The cofunction of RXR/PPAR γ heterodimers underlies the metabolic effects of retinoids on adipogenesis.
- **Enhancing immunity to infections.** Vitamin A supplementation can reduce morbidity and mortality rates from diarrhea by 15% and 28%, respectively. The effects of vitamin A on infections probably derive in part from its ability to stimulate the immune system through unclear mechanisms. Infections may reduce the bioavailability of vitamin A, possibly by inducing the acute phase response, which appears to inhibit RBP synthesis in the liver. The drop in hepatic RBP causes a decrease in circulating retinol, which reduces the tissue availability of vitamin A.
- **Other activities.** Retinoids, β -carotene, and some related carotenoids function as photoprotective and antioxidant agents.

Retinoids are used clinically in the treatment of skin disorders such as severe acne and certain forms of psoriasis, as well as in the treatment of acute promyelocytic leukemia. As discussed in Chapter 7, all-*trans*-retinoic acid induces the differentiation and subsequent apoptosis of acute promyelocytic leukemia cells through its ability to bind to a PML-RAR α fusion protein that characterizes this form of cancer. A different isomer, 13-*cis*-retinoic acid, is used as part of a therapeutic regimen designed to prevent the recurrence of childhood neuroblastoma.

Vitamin A Deficiency. Vitamin A deficiency occurs worldwide due to primary malnutrition and in conditions that cause malabsorption of fats. In children, stores of vitamin A are depleted by infections, and the absorption of the vitamin is poor in newborn infants. Adult patients with malabsorption syndromes such as celiac disease, Crohn disease, and colitis may develop vitamin A deficiency in conjunction with depletion of other fat-soluble vitamins. Bariatric surgery and, in older persons, continuous use of mineral oil as a laxative may lead to deficiency. The pathologic effects of vitamin A deficiency are summarized in Fig. 9.23.

As already discussed, vitamin A is a component of rhodopsin and other visual pigments. Not surprisingly, one of the earliest manifestations of vitamin A deficiency is impaired vision, particularly in reduced light (night blindness). Other effects of deficiency are related to the role of vitamin A in regulating the differentiation of epithelial cells. Persistent deficiency gives rise to epithelial metaplasia and keratinization. The most devastating changes occur in the eyes and are referred to as xerophthalmia (dry eye). First, there is dryness of the conjunctiva (xerosis conjunctivae) as the normal lacrimal and mucus-secreting epithelium is replaced by keratinized epithelium. This is followed by a buildup of keratin debris in small opaque plaques (*Bitot spots*) that progresses to erosion of the roughened corneal surface, softening and destruction of the cornea (keratomalacia), and blindness.

In addition to the ocular epithelium, the epithelia lining the upper respiratory passage and urinary tract also undergo squamous metaplasia. Loss of the mucociliary epithelium of the airways predisposes to secondary pulmonary infections, and desquamation of keratin debris in the urinary tract predisposes to renal and urinary bladder stones. Hyperplasia and hyperkeratinization of the epidermis and plugging of the ducts of the adnexal glands may produce follicular or papular dermatosis. Another serious consequence is immune deficiency, which is

responsible for higher mortality rates from common infections such as measles, pneumonia, and infectious diarrhea. In parts of the world where deficiency of vitamin A is prevalent, providing dietary vitamin A supplements reduces mortality from these infections by 20% to 30%.

Vitamin A Toxicity. Both short-term and long-term excesses of vitamin A may produce toxic manifestations. The consequences of acute hypervitaminosis A were first described by Gerrit de Veer in 1597, a ship's carpenter stranded in the Arctic, who recounted in his diary the serious symptoms that he and other members of the crew developed after eating polar bear liver. With this cautionary tale in mind, the adventurous eater should be aware that acute vitamin A toxicity has also been described in individuals who ingested the livers of whales, sharks, and tuna.

The symptoms of acute vitamin A toxicity include headache, dizziness, vomiting, stupor, and blurred vision, symptoms that may be confused with those of a brain tumor (*pseudotumor cerebri*). Chronic toxicity is associated with weight loss, anorexia, nausea, vomiting, and bone and joint pain. Retinoic acid stimulates osteoclast production and activity, leading to increased bone resorption and high risk of fractures. Although synthetic retinoids used for the treatment of acne are not associated with these types of conditions, their use in pregnancy should be avoided because of the well-established teratogenic effects of retinoids.

Vitamin D

The major function of vitamin D is the maintenance of adequate plasma levels of calcium and phosphorus to support metabolic functions, bone mineralization, and neuromuscular transmission. Vitamin D is a fat-soluble vitamin required for the prevention of bone diseases known as *rickets* (in children whose epiphyses have not already closed) and *osteomalacia* (in adults), as well as *hypocalcemic tetany*. With respect to tetany, vitamin D maintains the correct concentration of ionized calcium in the extracellular fluid compartment. When deficiency develops, the drop in ionized calcium in the extracellular fluid results in continuous excitation of muscle (tetany). It should be noted, however, that any reduction in the level of serum calcium is usually corrected by increased secretion of parathyroid hormone followed by bone resorption; hence, tetany is quite uncommon. Our attention here is focused on the function of vitamin D in the regulation of serum calcium levels.

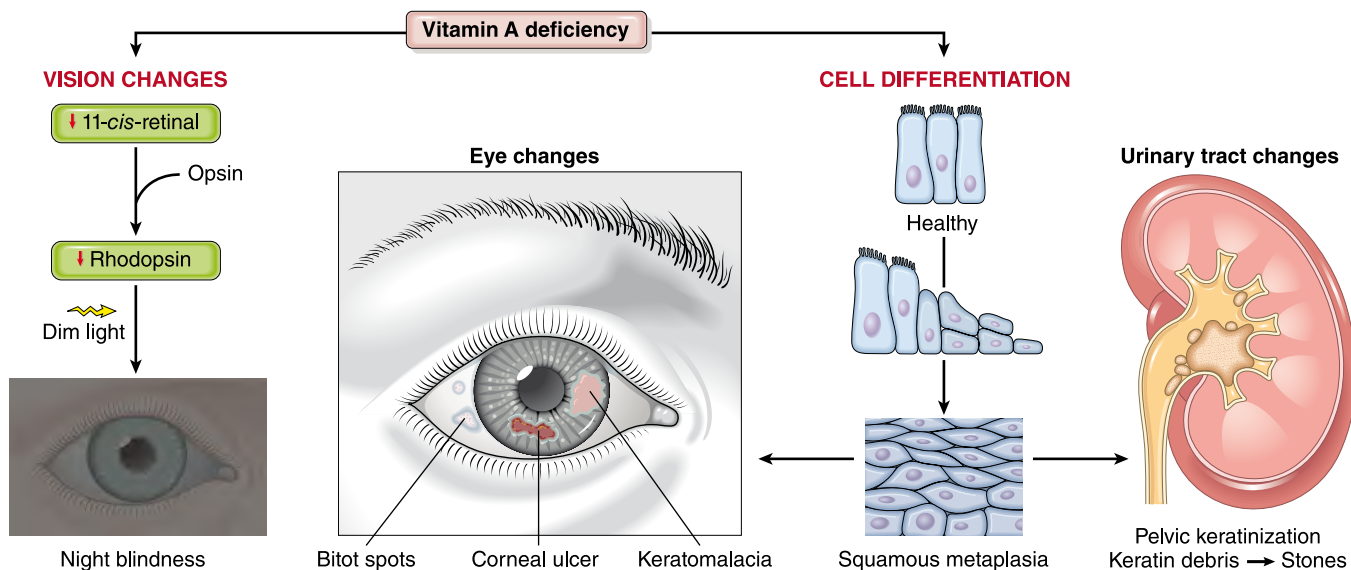


Fig. 9.23 Effects of vitamin A deficiency. Major effects include eye disease and complications due to squamous metaplasia. Not depicted is immune deficiency.

Metabolism of Vitamin D. The major source of vitamin D for humans is its endogenous synthesis from a precursor, 7-dehydrocholesterol, in a photochemical reaction that requires solar or artificial UV light in the range of 290 to 315 nm (UVB radiation). This reaction results in the synthesis of cholecalciferol, known as vitamin D₃. Herein, the term vitamin D is used to refer to this compound. Under usual conditions of sun exposure, about 90% of the required vitamin D is endogenously synthesized by the skin. However, individuals with dark skin generally have a lower level of vitamin D production because of absorption of UV light by melanin. Dietary sources, such as deep-sea fish, plants, and grains, contribute the remaining required vitamin D and depend on adequate intestinal fat absorption. In plants, vitamin D is present in a precursor form (ergosterol), which is converted to vitamin D in the body.

The main steps of vitamin D metabolism are summarized:

1. Photochemical synthesis of vitamin D from 7-dehydrocholesterol in the skin and absorption of vitamin D from foods and supplements in the gut
2. Binding of vitamin D from both these sources to plasma α_1 -globulin (D-binding protein [DBP]) and transport to the liver
3. Conversion of vitamin D into 25-hydroxycholecalciferol (25-OH-D) in the liver, through the action of 25-hydroxylases, including CYP27A1 and other CYPs
4. Conversion of 25-OH-D into 1,25-dihydroxyvitamin D [1,25(OH)₂D₃], the most active form of vitamin D, by the enzyme 1 α -hydroxylase in the kidney

The activity of renal 1 α -hydroxylase, the final step in production of 1,25-dihydroxyvitamin D, is in turn regulated by four factors (Fig. 9.24).

- *Hypocalcemia stimulates secretion of parathyroid hormone (PTH)*, which upregulates 1 α -hydroxylase expression.
- *Hypophosphatemia*, which also upregulates 1 α -hydroxylase expression.
- *1,25-dihydroxyvitamin D itself*, which inhibits 1 α -hydroxylase expression.
- *FGF23*, a factor produced by osteoblasts and osteocytes that inhibits 1 α -hydroxylase expression.

Functions. Like retinoids and steroid hormones, 1,25-dihydroxyvitamin D acts by binding to a high-affinity nuclear receptor (vitamin D receptor), which associates with the already mentioned RXR. This heterodimeric complex binds to vitamin D response elements located in the regulatory sequences of vitamin D target genes. The receptors for 1,25-dihydroxyvitamin D are present in most cells of the body. In the small intestine, bones, and kidneys, signals transduced via these receptors regulate plasma levels of calcium and phosphate. Beyond its role in skeletal homeostasis, vitamin D has immunomodulatory and antiproliferative effects. 1,25-Dihydroxyvitamin D also appears to act through mechanisms that do not require the transcription of target genes. These alternative mechanisms involve the binding of 1,25-dihydroxyvitamin D to a membrane-associated vitamin D receptor, leading to the activation of protein kinase C and opening of calcium channels.

Effects of Vitamin D on Calcium and Phosphate Homeostasis. The main effects of 1,25-dihydroxyvitamin D on calcium and phosphate homeostasis are:

- *Stimulation of intestinal calcium absorption.* 1,25-Dihydroxyvitamin D stimulates intestinal absorption of calcium in the duodenum through the interaction of 1,25-dihydroxyvitamin D with nuclear vitamin D receptor and the formation of a complex with RXR. The complex binds to vitamin D response elements and activates the transcription of TRPV6 (a member of the transient receptor potential vanilloid family), which encodes a critical calcium transport channel.
- *Stimulation of calcium reabsorption in the kidney.* 1,25-Dihydroxyvitamin D increases calcium influx in distal tubules of

the kidney through the increased expression of TRPV5, another member of the transient receptor potential vanilloid family. TRPV5 expression is also regulated by PTH in response to hypocalcemia.

- *Interaction with PTH in the regulation of blood calcium.* Vitamin D maintains calcium and phosphate at supersaturated levels in the plasma. The parathyroid glands have a key role in the regulation of extracellular calcium concentrations. These glands have a calcium receptor that senses even small changes in blood calcium concentrations. In addition to their effects on calcium absorption in the intestine and kidneys, both 1,25-dihydroxyvitamin D and PTH enhance the expression of RANKL (receptor activator of NF- κ B ligand) on osteoblasts. RANKL binds to its receptor (RANK) located in preosteoclasts, thereby inducing the differentiation of these cells into mature osteoclasts (Chapter 26). Through the secretion of hydrochloric acid and activation of proteases such as cathepsin K, osteoclasts dissolve bone and release calcium and phosphate into the circulation.
- *Mineralization of bone.* Vitamin D contributes to the mineralization of osteoid matrix and epiphyseal cartilage in both flat and long bones. It stimulates osteoblasts to synthesize the calcium-binding protein osteocalcin, which is involved in the deposition of calcium during bone development. Flat bones develop by intramembranous bone formation, in which mesenchymal cells differentiate directly into osteoblasts, which synthesize the collagenous osteoid matrix on which calcium is deposited. Long bones develop by endochondral ossification, through which growing cartilage at the epiphyseal plates is provisionally mineralized and then progressively resorbed and replaced by osteoid matrix that is mineralized to create bone (Fig. 9.25A).

When *hypocalcemia* occurs due to vitamin D deficiency (Fig. 9.26), PTH production is elevated, causing (1) activation of renal 1 α -hydroxylase, increasing the level of vitamin D and enhancing calcium absorption; (2) increased resorption of calcium from bone by osteoclasts; (3) decreased renal calcium excretion; and (4) increased renal excretion of phosphate. Although a normal serum level of calcium may be restored, hypophosphatemia persists, impairing the mineralization of bone. Inappropriately high production of FGF-23, which is seen in rare inherited disorders and as a paraneoplastic syndrome with certain tumors, leads to rare forms of hypophosphatemic rickets and osteomalacia.

Deficiency States. The normal reference range for circulating 25-(OH)-D is 20 to 100 ng/mL; concentrations of less than 20 ng/mL constitute vitamin D deficiency.

Rickets in growing children and osteomalacia in adults are skeletal diseases with worldwide distributions. They may result from diets deficient in calcium and vitamin D, but an equally important cause of vitamin D deficiency is limited exposure to sunlight. This most often affects inhabitants of northern latitudes, but can even be a problem in tropical countries, in heavily veiled females, and in children born to mothers who have frequent pregnancies followed by lactation. In all these situations, vitamin D deficiency can be prevented by a diet high in fish oils. Other, less common causes of rickets and osteomalacia include renal disorders, which lead to decreased synthesis of 1,25-dihydroxyvitamin D, phosphate depletion, malabsorption disorders, and rare inherited disorders. Although rickets and osteomalacia rarely occur outside high-risk groups, milder forms of vitamin D deficiency resulting in an increased risk of bone loss and hip fractures are quite common in older adults in the United States and Europe. Some genetically determined variants of the vitamin D receptors are also associated with accelerated loss of bone minerals during aging and certain familial forms of osteoporosis (Chapter 26).

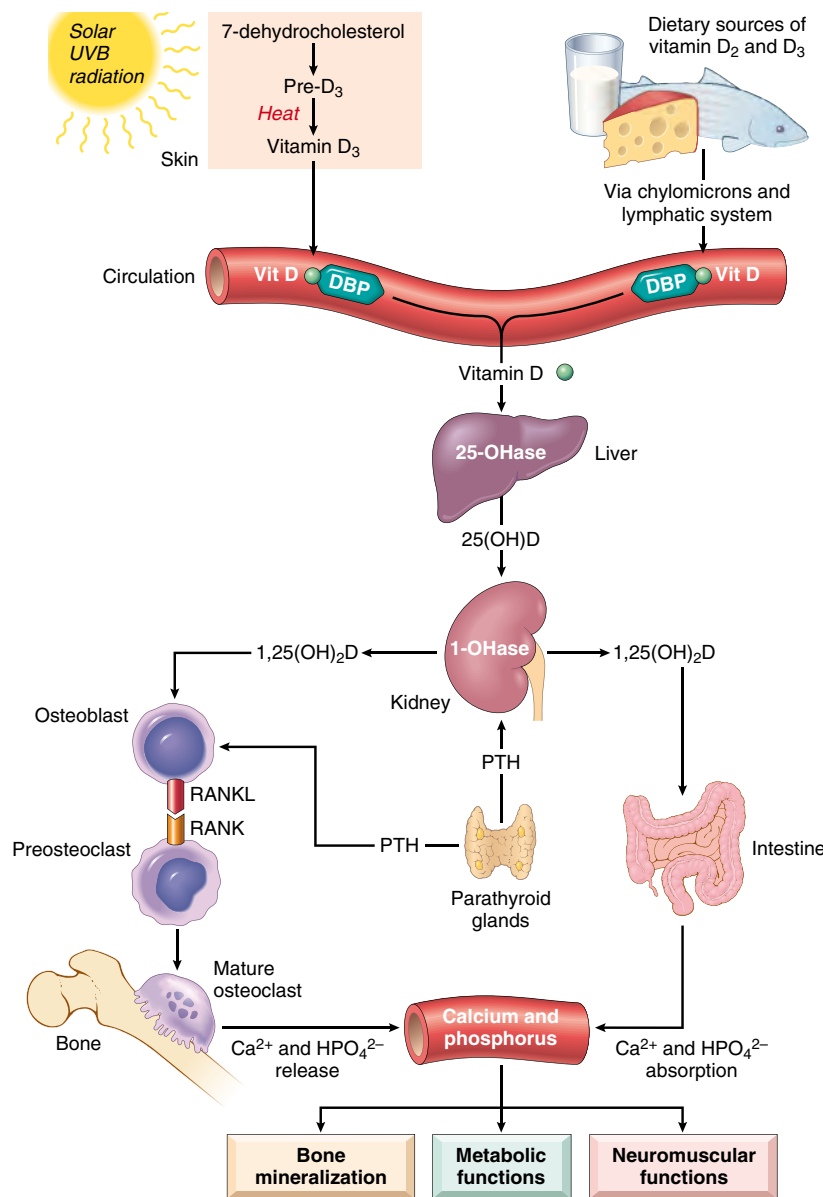


Fig. 9.24 Vitamin D metabolism. Vitamin D is produced from 7-dehydrocholesterol in the skin or is ingested in the diet. It is converted in the liver into 25(OH)D and in the kidney into 1,25-dihydroxyvitamin D ($1,25(\text{OH})_2\text{D}$), the active form of the vitamin. $1,25(\text{OH})_2\text{D}$ stimulates the expression of RANKL, an important regulator of osteoclast maturation and function, on osteoblasts and enhances the intestinal absorption of calcium and phosphate. In the setting of hypocalcemia, increased release of PTH from the parathyroid glands increases the level of 1- α -hydroxylase in the kidney and expression of RANKL on osteoblasts. *1-OHase*, 1-Hydroxylase; *25-OHase*, 25-hydroxylase; *DBP*, vitamin D-binding protein (α_1 -globulin); *PTH*, parathyroid hormone.

Nonskeletal Effects of Vitamin D. As mentioned earlier, the vitamin D receptor is present in various cells and tissues that do not participate in calcium and phosphate homeostasis. In addition, macrophages, keratinocytes, and tissues such as breast, prostate, and colon can produce 1,25-dihydroxyvitamin D. Within macrophages, synthesis of 1,25-dihydroxyvitamin D occurs through the activity of CYP27B located in the mitochondria. It appears that pathogen-induced activation of toll-like receptors in macrophages causes increased expression of vitamin D receptor and CYP27B, leading to local synthesis of 1,25-dihydroxyvitamin D and activation of vitamin D-dependent gene expression in macrophages and other neighboring immune cells. As a result, granulomatous disease is occasionally associated with hypercal-

cemia. The net effect of this altered gene expression on the immune response remains to be determined, however, and clinical trials have failed to demonstrate beneficial effects of vitamin D supplements on the course of respiratory infections, including tuberculosis.

Vitamin D Toxicity. Prolonged exposure to normal sunlight does not produce an excess of vitamin D, but megadoses of orally administered vitamin can lead to hypervitaminosis. In children, hypervitaminosis D may take the form of metastatic calcifications of soft tissues such as the kidney; in adults, it causes bone pain and hypercalcemia. The toxic potential of this vitamin is so great that in sufficiently large doses it is a potent rodenticide.

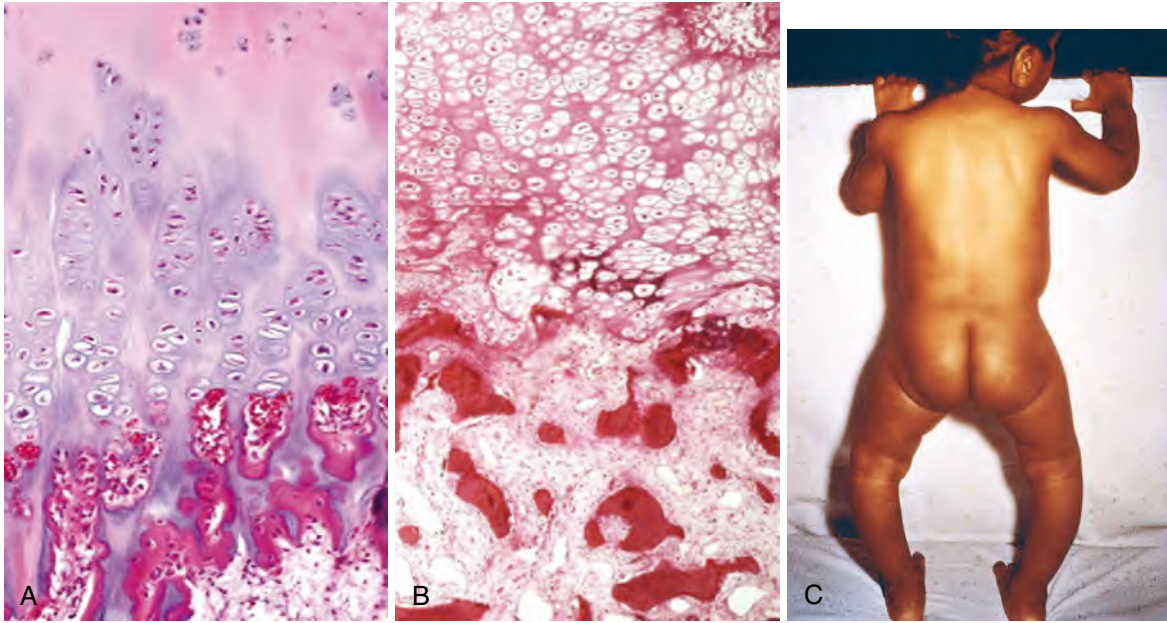


Fig. 9.25 Rickets. (A) Normal costochondral junction of a young child illustrating formation of cartilage palisades and orderly transition from cartilage to new bone. (B) Detail of a rachitic costochondral junction in which the palisades of cartilage are lost. Darker trabeculae are well-formed bone; paler trabeculae consist of uncalcified osteoid. (C) Bowing of legs due to poor bone mineralization. (B, Courtesy Dr. Andrew E. Rosenberg, Massachusetts General Hospital, Boston.)

MORPHOLOGY

Vitamin D deficiency in both rickets and osteomalacia results in an **excess of unmineralized matrix**. The following sequence ensues in rickets:

- Overgrowth of epiphyseal cartilage due to inadequate provisional calcification and failure of the cartilage cells to mature and disintegrate
- Persistence of distorted, irregular masses of cartilage that project into the marrow cavity
- Deposition of osteoid matrix on inadequately mineralized cartilaginous remnants
- Disruption of the orderly replacement of cartilage by osteoid matrix, with enlargement and lateral expansion of the osteochondral junction (see Fig. 9.25B)
- Abnormal overgrowth of capillaries and fibroblasts in the disorganized zone resulting from microfractures and stresses on the inadequately mineralized, weak, poorly formed bone
- Deformation of the skeleton due to the loss of structural rigidity of the developing bones

The gross skeletal changes in rickets depend on the severity and duration of the process and the stresses to which individual bones are subjected. During the nonambulatory stage of infancy, the head and chest sustain the greatest stresses. The softened occipital bones may become flattened, and the parietal

bones can be buckled inward by pressure; with the release of the pressure, elastic recoil snaps the bones back into their original positions (**craniotabes**). An excess of osteoid produces **frontal bossing** and a **squared appearance to the head**. Deformation of the chest results from overgrowth of cartilage or osteoid tissue at the costochondral junction, producing the **rachitic rosary**. The weakened metaphyseal areas of the ribs are subject to the pull of the respiratory muscles and thus bend inward, creating anterior protrusion of the sternum (**pigeon breast deformity**). When an ambulating child develops rickets, deformities are likely to affect the spine, pelvis, and tibia, causing **lumbar lordosis** and **bowing of the legs** (see Fig. 9.25C).

In adults with **osteomalacia**, the lack of vitamin D deranges the normal bone remodeling that occurs throughout life. The newly formed osteoid matrix laid down by osteoblasts is inadequately mineralized, thus producing an excess of persistent osteoid that is characteristic of osteomalacia. Although the contours of the bone are not affected, the bone is weak and vulnerable to gross fractures or microfractures, which are most likely to affect vertebral bodies and femoral necks. The unmineralized osteoid appears as a thickened layer of matrix (which stains pink in hematoxylin and eosin preparations) arranged about the more basophilic, normally mineralized trabeculae.

Vitamin C (Ascorbic Acid)

A deficiency of water-soluble vitamin C leads to the development of **scurvy**, characterized principally by bone disease in growing children and by hemorrhages and healing defects in children and adults. Sailors of the British Royal Navy were nicknamed “limeys” because at the end of the 18th century the Navy began to provide lime and lemon juice (rich sources of vitamin C) to sailors to prevent scurvy during their long sojourn at sea. It was not until 1932 that ascorbic acid was identified and synthesized. Ascorbic acid is not synthesized endogenously in humans; therefore, we are entirely dependent on diet for this nutrient. Vitamin C is present in milk and some animal products

(liver, fish) and is abundant in a variety of fruits and vegetables. All but the most restricted diets provide adequate amounts of vitamin C.

Function. Ascorbic acid has several functions:

- **Collagen synthesis.** The best established function of vitamin C is the activation of prolyl and lysyl hydroxylases, which catalyze hydroxylation of procollagen. Inadequately hydroxylated procollagen cannot acquire a stable helical configuration and is inefficiently secreted from fibroblasts. Those molecules that are secreted are inadequately cross-linked, lack tensile strength, and are more soluble and vulnerable to enzymatic degradation. Collagen, which normally has

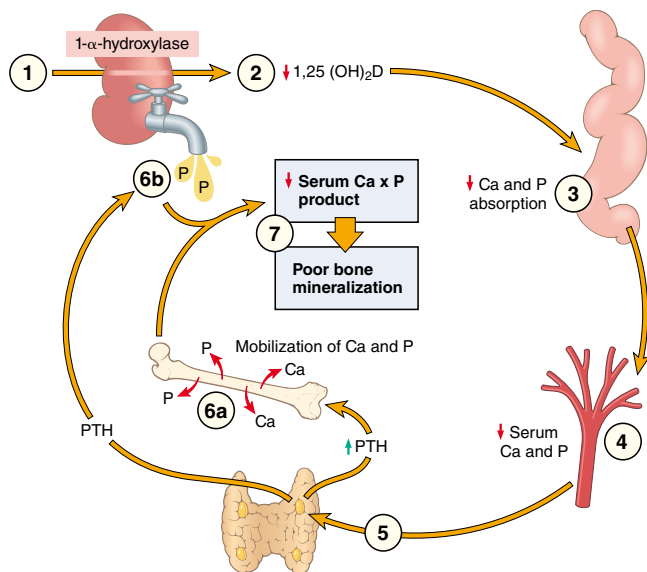


Fig. 9.26 Vitamin D deficiency. There is inadequate substrate for renal 1α -hydroxylase (1), yielding a deficiency of $1,25(\text{OH})_2\text{D}$ (2), and deficient absorption of calcium (Ca) and phosphorus (P) from the gut (3), with consequently depressed serum levels of both (4). The hypocalcemia activates the parathyroid glands (5), causing mobilization of calcium and phosphorus from bone (6a). Simultaneously, the parathyroid hormone (PTH) induces wasting of phosphate in the urine (6b) and calcium retention. As a result, the serum levels of calcium are normal or nearly normal, but phosphate levels are low; hence mineralization is impaired (7).

the highest content of hydroxyproline of any polypeptide, is most affected, particularly in blood vessels, accounting for the predisposition to hemorrhages in scurvy.

- **Neurotransmitter synthesis.** Synthesis of norepinephrine involves hydroxylation of dopamine, a step that requires vitamin C.

Vitamin C has long been touted as an antioxidant (possibly protecting against ROS and other oxidants) and an immune modulator, but clinical trials of vitamin C based on these proposed functions have generally been disappointing (discussed later).

Deficiency States. Consequences of vitamin C deficiency (scurvy) are illustrated in Fig. 9.27. Because of the abundance of ascorbic acid in many foods, scurvy has ceased to be a global problem. It is sometimes encountered even in affluent populations as a secondary deficiency, particularly among older individuals, persons who live alone, and those with alcohol use disorder, all groups that often have erratic and inadequate eating patterns. Occasionally, scurvy occurs in patients undergoing peritoneal dialysis and hemodialysis and in food faddists. The condition also sometimes appears in infants who are maintained on formulas of evaporated milk without supplementation of vitamin C.

Vitamin C Excess. The popular notion that megadoses of vitamin C protect against the common cold, or at least allay the symptoms, has not been borne out by controlled clinical studies. Such slight relief as may be experienced is probably due to the mild antihistamine action of ascorbic acid. Similarly, randomized clinical trials of large doses of vitamin C have failed to identify any protective effect against cancer development, cardiovascular disease, or cataracts and macular degeneration. The physiologic availability of excess vitamin C is limited due to its inherent instability, poor intestinal absorption, and rapid urinary excretion. Fortunately, toxicities related to high doses of vitamin C are rare, consisting

of possible iron overload (due to increased absorption), hemolytic anemia in individuals with glucose-6-phosphate dehydrogenase (G6PD) deficiency (Chapter 14), and calcium oxalate kidney stones.

Other vitamins and some essential minerals are listed and briefly described in Tables 9.9 and 9.10; many are discussed in other chapters.

KEY CONCEPTS

Nutritional Diseases

- Severe acute malnutrition is a common cause of childhood deaths in low-resource countries. The two main primary forms of severe acute malnutrition syndromes are marasmus and kwashiorkor. Secondary malnutrition occurs in the chronically ill and in patients with advanced cancer (due to cachexia).
- Kwashiorkor is characterized by hypoalbuminemia, generalized edema, fatty liver, skin changes, and defects in immunity. It is caused by diets low in protein but normal in calories.
- Marasmus is characterized by emaciation resulting from loss of muscle mass and fat with relative preservation of serum albumin. It is caused by diets severely lacking in both protein and nonprotein calories.
- Anorexia nervosa is self-induced starvation; it is associated with amenorrhea and multiple manifestations of low thyroid hormone levels. Bulimia is a condition in which food binges alternate with induced vomiting.
- Vitamins A and D are fat-soluble vitamins with a wide range of activities. Vitamin A is required for vision, epithelial differentiation, and immune function. Vitamin D is a key regulator of calcium and phosphate homeostasis.
- Vitamin C and members of the vitamin B family are water-soluble. Vitamin C is needed for collagen synthesis and collagen cross-linking and tensile strength. B vitamins have diverse roles in cellular metabolism.

Obesity

Obesity is a condition where excess body fat accumulates to a level that can impair health. How does one measure fat accumulation? Several high-tech methods have been devised, but for practical purposes the *body mass index* (BMI) is used most commonly. BMI is calculated as (weight in kilograms)/(height in meters)², or kg/m². The normal BMI range is 18.5 to 25 kg/m², although the range differs across countries due to variation in ethnicity and genetic backgrounds. Individuals with BMI greater than 30 kg/m² are classified as obese; individuals with BMI between 25 kg/m² and 30 kg/m² are considered overweight. Unless otherwise noted, the term obesity herein applies to both overweight and obese individuals.

Obesity is associated with several of the most important diseases of humans, including type 2 diabetes, dyslipidemia, cardiovascular disease, hypertension, and cancer. The strength of this association is affected not only by the quantity of excess fat but also by its distribution. Central, or visceral, obesity, in which excess fat accumulates preferentially in the trunk and in the abdominal cavity (in the mesentery and around viscera), is associated with a much higher risk for several diseases than is excess accumulation of subcutaneous fat.

Obesity is a major public health problem in high-resource countries and an emerging health problem in some lower resource nations, such as India. In the United States, obesity has reached epidemic proportions. According to the CDC, in 2020 the overall prevalence of obesity was 41.9% in adults and approximately 20% in children and adolescents. The increase in obesity in the United States has been associated with the higher caloric content of the diet, mostly caused by increased consumption of refined sugars, sweetened beverages, and vegetable oils. Unfortunately, much of the rest of the world appears to be following the path taken by the United States. The World Obesity Federation estimates that 3.12 billion people (39% of the global population) met the criteria for obesity in 2023 (an increase from 23.9% in 2008), and that by the year

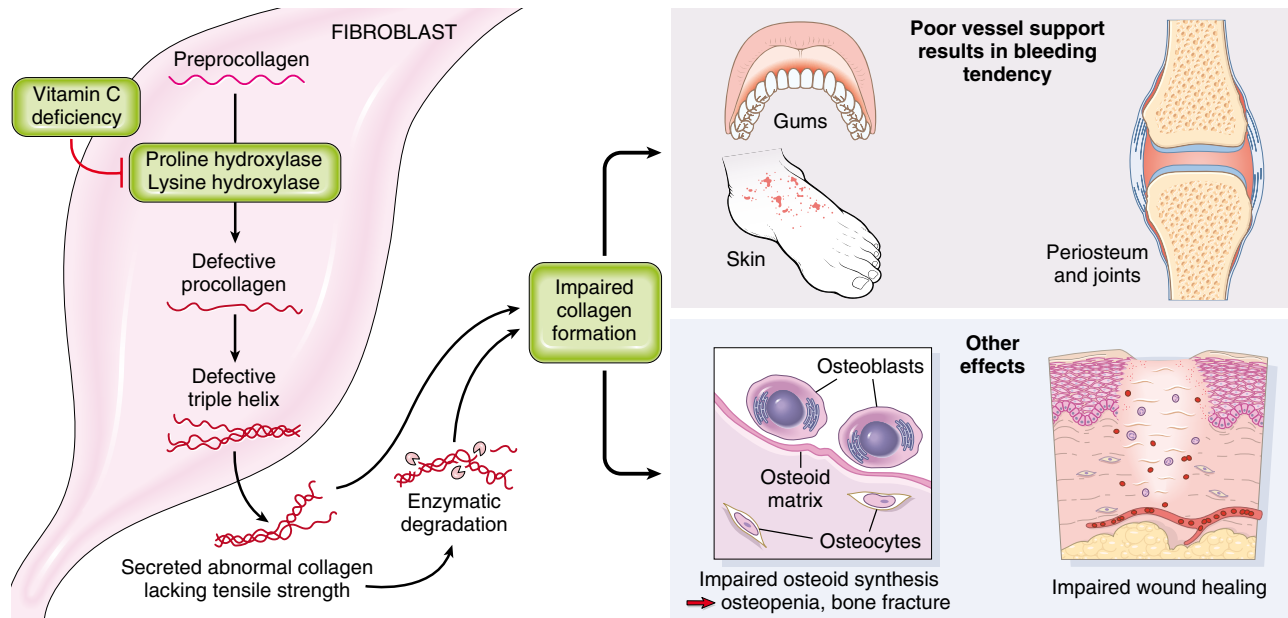


Fig. 9.27 Causes and consequences of impaired collagen formation due to vitamin C deficiency.

2035, 4 billion people (51% of the global population) will be obese. On a global basis, the economic impact of this level of obesity is estimated to be a staggering \$4.3 trillion annually.

The etiology of obesity is complex and incompletely understood. Genetic, environmental, and psychologic factors are involved. However, simply put, obesity is a disorder of energy homeostasis. **The two sides of the energy equation, intake and expenditure, are finely regulated by neural and hormonal mechanisms so that body weight is maintained within a narrow range for many years.** Apparently, this fine balance is controlled by an internal set-point, or “lipostat,” that senses the quantity of energy stores (adipose tissue) and appropriately regulates food intake as well as energy expenditure. The hypothalamus is the master regulator of energy homeostasis. It receives inputs from the periphery about the state of energy stores; if they are inadequate, it triggers anabolic circuits, and if they are adequate, catabolic circuits are activated. The effect of the anabolic circuits is to increase intake of food and reduce energy expenditure, whereas catabolic circuits reduce food intake and increase energy expenditure. The neurohumoral mechanisms that regulate energy balance can be subdivided into three components (Fig. 9.28).

- The *peripheral or afferent system* generates signals from various sites. Its main components are leptin produced by fat cells; ghrelin from the stomach; peptide YY (PYY); the incretins glucagon-like peptide 1 (GLP-1) and glucose-dependent insulinotropic polypeptide (GIP, also called gastric inhibitory polypeptide), produced by enteroendocrine cells in the small intestine and colon; and insulin from the pancreas. The afferent systems provide signals to the central processing system in the brain.
- The *central processing system* resides in the arcuate nucleus of the hypothalamus where neurohumoral peripheral signals are integrated to generate efferent signals. Two sets of neurons participate in central processing (see Fig. 9.28).
 - A pair of first-order neurons: (1) pro-opiomelanocortin (POMC) and cocaine and amphetamine-regulated transcript (CART) neurons and (2) neurons containing neuropeptide Y (NPY) and agouti-related peptide (AgRP). These first-order neurons communicate with second-order neurons.
 - A pair of second-order neurons: (1) neurons that bear melanocortin receptors 3 and 4 (MC3/4R) and receive signals from

first-order POMC/CART neurons and (2) neurons that bear Y1 and Y5 receptors and receive signals from first-order NPY/AgRP neurons.

- The *efferent system* consists of signals generated by second-order neurons and is organized along two pathways, catabolic (downstream of MC3/4R) and anabolic (downstream of Y1 and Y5 receptors) that control food intake and energy expenditure. In addition to these circuits (within the hypothalamus), the hypothalamic nuclei also communicate with forebrain and midbrain centers that control the autonomic nervous system.

With this background on the organization of the hypothalamic centers that regulate energy balance, we can now discuss how they function. Upon nutrient intake, POMC is cleaved from POMC/CART neurons and gives rise to α -melanocyte-stimulating hormone (MSH), which activates MC3/4R receptors in second-order neurons. These second-order neurons are responsible for reducing food intake and increasing energy expenditure by producing brain-derived neurotrophic factor (BDNF), thyroid-releasing hormone (TRH), and corticotropin-releasing hormone (CRH). By contrast, fasting activates the NPY/AgRP neurons to release NPY, which activates Y1 and Y5 receptors in second-order neurons. These second-order neurons are responsible for increasing food intake by producing melanin-concentrating hormone (MCH) and orexin, and for reducing energy expenditure by downregulation of sympathetic output. NPY/AgRP neurons also directly inhibit POMC/CART neurons, thus blunting their anorexigenic effect.

To summarize, NPY/AgRP neurons may be thought of as the gas pedals for appetite, whereas POMC/CART neurons represent the brake pedal. The orderly functioning of these two pedals maintains energy homeostasis.

Discussed next are two important components of the afferent system that regulates appetite and satiety, leptin and gut hormones, as well as adiponectin, which regulates energy consumption.

Leptin

Leptin is secreted by fat cells, and its output is regulated by the adequacy of fat stores. BMI and body fat stores are directly related to

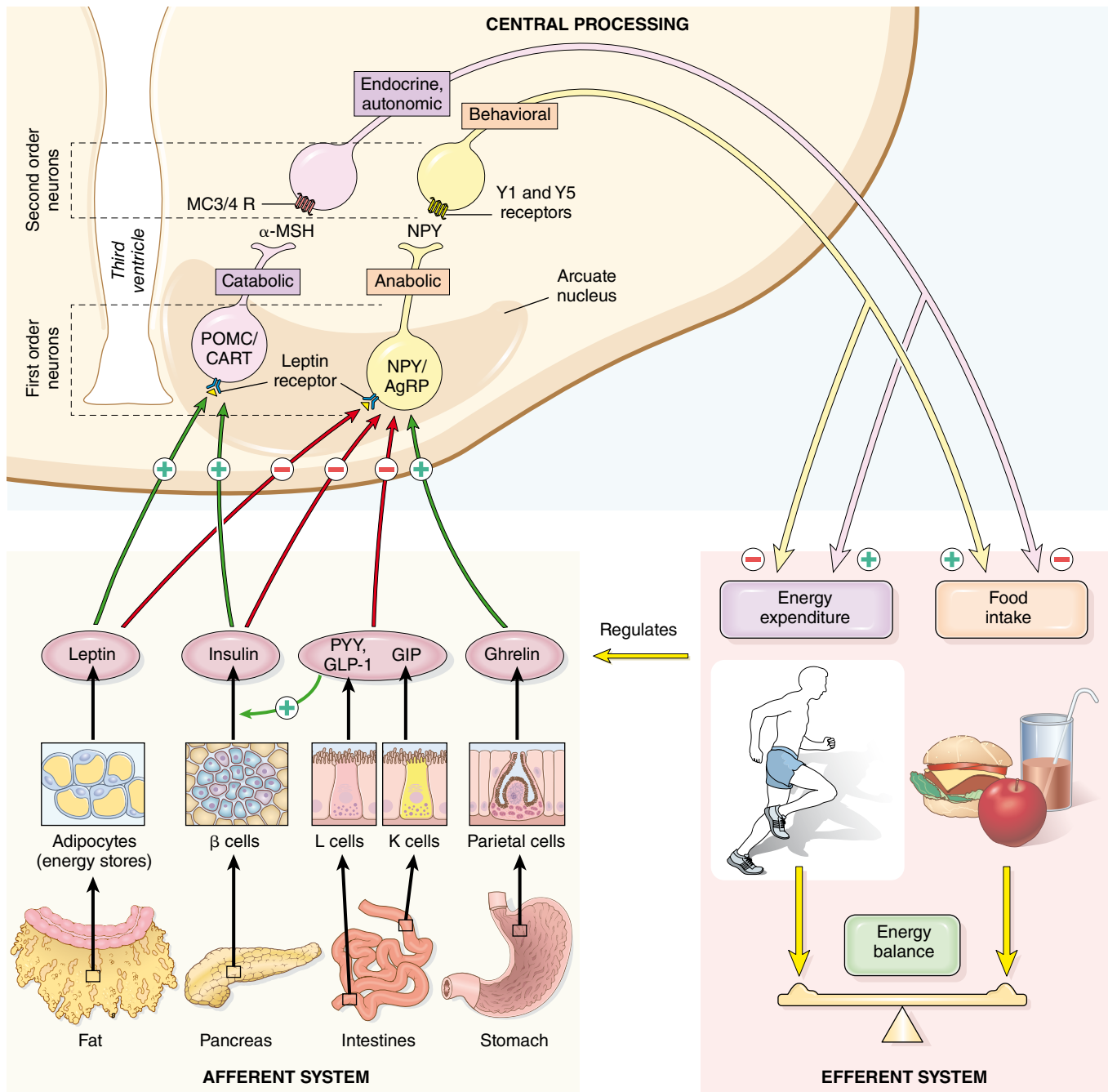


Fig. 9.28 Energy balance regulatory circuitry. When sufficient energy is stored in adipose tissue and the individual is well fed, afferent adiposity signals (insulin, leptin, ghrelin, peptide YY) are delivered to the central neuronal processing units in the hypothalamus, where they inhibit anabolic circuits and activate catabolic circuits. The effector arms of these central circuits then influence energy balance by inhibiting food intake and promoting energy expenditure. This in turn reduces energy stores and blunts proadiposity signals. Conversely, when energy stores are low, anabolic circuits are stimulated that generate energy stores in the form of adipose tissue.

leptin secretion. With abundant adipose tissue, leptin secretion is stimulated. After release into the blood, it crosses the blood-brain barrier and travels to the hypothalamus, where it reduces food intake by stimulating POMC/CART neurons and inhibiting NPY/AgRP neurons. The opposite sequence of events occurs when there are inadequate stores of body fat: leptin secretion is diminished, and food intake is increased. In people of stable weight, the activities of these pathways are balanced.

Leptin regulates not only food intake but also energy expenditure through a distinct set of pathways. Thus, an abundance of leptin stimulates physical activity, heat production, and energy expenditure. The neurohumoral mediators of leptin-induced energy expenditure are less well defined. *Thermogenesis*, an important catabolic effect mediated by leptin, is controlled in part by hypothalamic signals that increase the release of norepinephrine from sympathetic nerve endings in adipose tissue.

The central role of leptin in energy homeostasis is underscored by the study of mutations affecting components of the leptin pathway in mice and in humans. Mice with mutations that disable the genes for leptin or its receptor fail to sense the adequacy of fat stores, so they behave as if they are undernourished, eating ravenously and becoming massively obese. As in mice, disabling mutations of the leptin gene or receptor in humans, although rare, cause massive obesity. More common are loss-of-function mutations in the melanocortin receptor 4 gene (*MC4R*), found in 4% to 5% of patients with massive obesity. These monogenic traits confirm the importance of leptin signaling in the control of body weight, and it is possible that other genetic or acquired defects in the leptin pathway play a role in more common forms of obesity. In keeping with this notion, the anorexigenic response of leptin is blunted in states of obesity despite high levels of circulating leptin (leptin resistance). This phenomenon of leptin resistance explains why injection of leptin in obese humans fails to affect food intake and energy expenditure, dashing initial enthusiasm of leptin therapy for obesity.

In closing it should be noted that although our discussion has centered on the actions of leptin, there is increasing evidence that insulin, like leptin, exerts anorexigenic responses. Both POMC/CART and NPY/AgRP neurons express insulin receptors. However, while insulin can mimic the actions of leptin, most of the evidence suggests the primacy of leptin in the regulation of energy homeostasis.

Gut Hormones

Gut peptides act as short-term meal initiators and terminators. They include ghrelin, PYY, GLP-1 (glucagon-like peptide-1), and glucose-dependent insulintropic polypeptide (GIP), among others.

Ghrelin is produced in the stomach and is the only known gut hormone that increases food intake (orexigenic effect). Its injection in rodents elicits voracious feeding, even after repeated administration. Long-term injections cause weight gain by increasing caloric intake and reducing energy utilization. Ghrelin acts centrally by activating orexigenic NPY/AgRP neurons. Ghrelin levels normally rise before meals and fall 1 to 2 hours afterward, but this drop is attenuated in obese people. Ghrelin levels are lower in obese individuals compared with normal-weight individuals, and levels increase with a reduction in obesity. Interestingly, the rise in ghrelin levels is much reduced in individuals in whom gastric bypass surgery is performed for the treatment of obesity, suggesting that the beneficial effects of such surgery may be in part due to a reduced surface of gastric mucosa that is exposed to food.

PYY and GLP-1 are secreted from L cells, enteroendocrine cells in the lower small intestine (ileum) and colon, while GIP is produced by enteroendocrine K cells in the upper small intestine (duodenum and jejunum). Plasma levels of PYY, GLP-1 and GIP are low during fasting and increase shortly after food intake. GLP-1 and GIP are *incretins*, a class of hormone defined by the ability to stimulate insulin release following ingestion of glucose. Glucose in the gut stimulates the release of incretins, which then bind to specific G-protein coupled receptors in pancreatic β cells. Binding activates adenylate cyclase, thereby increasing levels of intracellular cyclic adenosine monophosphate and enhancing insulin secretion. GLP-1 and GIP as well as PYY also act centrally on neurons in the hypothalamus to cause a decrease in food intake.

These insights have led to the development of incretin agonists, which have been shown to markedly improve glucose tolerance in patients with type 2 diabetes. Moreover, these agents also produce substantial, sustained reductions in body weight, offering great promise for treating individuals with obesity. As a result of these findings, GLP-1 agonists and GLP-1-GIP co-agonists have been approved for treatment of type 2 diabetes and obesity. Remarkably, other recent

studies indicate that GLP-1 agonists are beneficial in the treatment of congestive heart failure and renal insufficiency, pointing to a broad role for metabolic dysregulation in multiple common diseases of adults. Based on these successes, many other agents mimicking the effects of nutrient-stimulated hormones are under development.

Adiponectin

Adiponectin, produced in the adipose tissue, has been called a “fat-burning molecule,” as it stimulates fatty acid oxidation in skeletal muscle, thereby reducing fatty acid levels. Evidence suggests that excess fatty acids can cross the blood-brain barrier and enter the hypothalamus, where they are sensed by microglial cells. These cells respond by releasing inflammatory factors that appear to act on hypothalamic neurons to cause leptin resistance, thereby blunting its antiadiposity signals. Because of its actions on reducing fatty acids by promoting their oxidation, adiponectin is called the “guardian angel against obesity.” Adiponectin also decreases glucose production in the liver and increases insulin sensitivity, protecting against the metabolic syndrome. In addition to its metabolic effects, adiponectin has antiinflammatory, antiatherogenic, antiproliferative, and cardioprotective effects. Its serum levels are lower in obese than in lean individuals. These effects contribute to obesity-associated insulin resistance, type 2 diabetes, and nonalcoholic fatty liver disease (Chapter 18).

Adiponectin binds to two receptors, AdipoR1 and AdipoR2. These receptors are found in many tissues, including the brain, but AdipoR1 and AdipoR2 are most highly expressed in skeletal muscle and liver, respectively. Binding of adiponectin to its receptors triggers signals that activate cyclic adenosine monophosphate (cAMP)-dependent protein kinase (protein kinase A), which in turn phosphorylates and inactivates acetyl coenzyme A carboxylase, a key enzyme required for fatty acid synthesis.

Adipose Tissue

It has been known for some time that there are two types of adipose tissues: white adipose tissue and brown adipose tissue. Brown adipose tissue has the unique property of expending energy by nonshivering thermogenesis. Brown adipose tissue functions by uncoupling energy production from energy storage and converting the energy produced into heat. Brown adipose tissue is abundant in newborns and is located primarily in interscapular and supraclavicular areas. Lesser amounts are present around kidneys, aorta, heart, pancreas, and trachea. Until recently, it was thought that brown adipose tissue is lost in adults, but imaging studies have revealed that some brown adipose tissue is preserved in adolescents and adults. Much effort is now focused on determining how brown adipose tissue can be preserved in adulthood and activated to burn energy.

In addition to leptin and adiponectin, white adipose tissue produces cytokines such as TNF, IL-6, IL-1, and IL-18; chemokines; and steroid hormones. The increased production of cytokines and chemokines by adipose tissue in obese patients creates a chronic proinflammatory state marked by high levels of circulating C-reactive protein (CRP). This relationship may be more than a one-way street, as emerging evidence suggests that immune cells, particularly tissue macrophages and some T cells, have important roles in regulating adipocyte function. It is now evident that through this panoply of mediators, adipose tissue participates in the control of energy balance and energy metabolism, functioning as a link between lipid metabolism, nutrition, and inflammatory responses.

Role of the Gut Microbiome

An interesting series of observations in mice suggest that the gut microbiome may be involved in the development of obesity. In support of this notion is the finding that the profiles of gut microbiota differ between

genetically obese mice and their lean littermates. The microbiome of genetically obese mice can harvest much more energy from food compared with that of lean mice. Colonization of the gut of germ-free mice by microbiota from obese mice (but not microbiota from lean mice) is associated with increased body weight. The relevance of the mouse models to human obesity is tantalizing but remains to be proven. Differences between the gut microbiome of obese and lean humans have also been reported. Whether this difference is causal or just an association is unclear.

Clinical Consequences of Obesity

Obesity, particularly central obesity, is associated with an increase in mortality from most diseases and is a known risk factor for several common conditions including type 2 diabetes, cardiovascular disease, and cancer (Fig. 9.29). Central obesity also stands at the center of a cluster of alterations known as the *metabolic syndrome*, characterized by abnormalities of glucose and lipid metabolism coupled with hypertension and evidence of a systemic proinflammatory state. This seems to be caused by activation of the inflammasome by free fatty

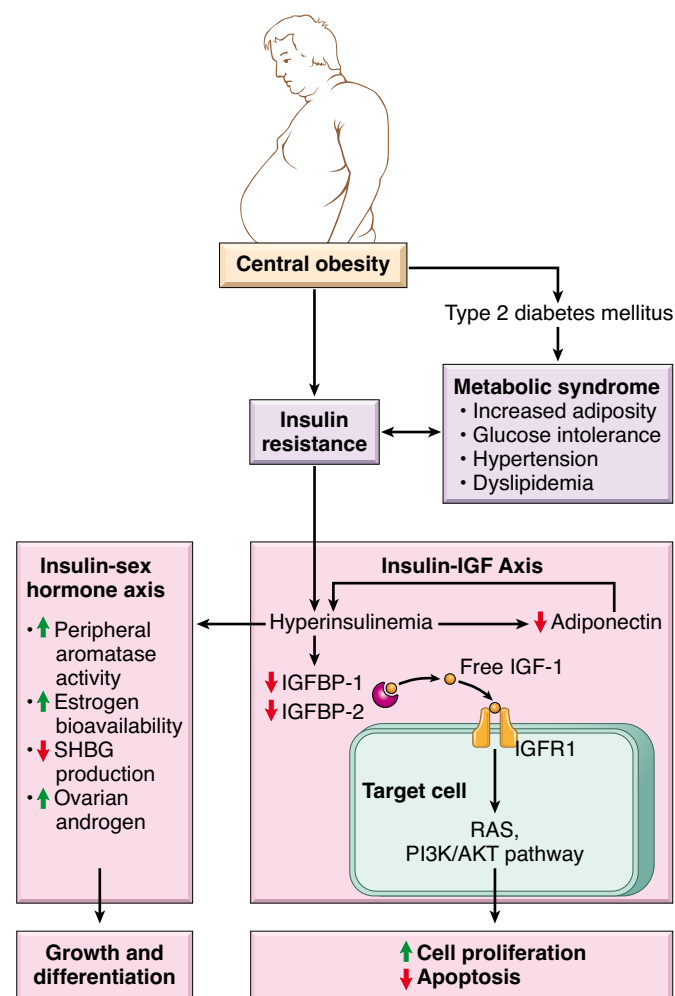


Fig. 9.29 Obesity, metabolic syndrome, and cancer. Excessive weight and obesity are precursors of the metabolic syndrome, which is associated with insulin resistance, type 2 diabetes, and hormonal changes. Increases in insulin and insulin-like growth factor-1 stimulate cell proliferation and inhibit apoptosis and may contribute to tumor development. *IGF*, Insulin-like growth factor; *IGFBP*, insulin-like growth factor-binding protein; *SHBG*, sex hormone-binding globulin. (Modified from Renehan AG, Frystyk J, Flyvbjerg A. Obesity and cancer risk: the role of the insulin-IGF axis. *Trends Endocrinol Metab.* 2006;17:328-336.)

acids and excess levels of lipids in cells and tissue. This in turn stimulates secretion of IL-1, which induces systemic inflammation. The following associations are worthy of note.

- *Obesity is associated with insulin resistance and hyperinsulinemia*, important features of type 2 diabetes (Chapter 24). Inflammation induced by IL-1 and other factors likely contributes to insulin resistance. Excess insulin, in turn, may play a role in the retention of sodium, expansion of blood volume, production of excess norepinephrine, and smooth muscle proliferation that are the hallmarks of hypertension. Whatever the mechanism, the risk of developing hypertension among previously normotensive persons increases proportionately with weight.
- *Obese persons generally have hypertriglyceridemia and low HDL cholesterol levels*, factors that increase the risk of coronary artery disease. The risk of heart disease is compounded by the existence of comorbid conditions including diabetes, hypertension, and dyslipidemia.
- *Nonalcoholic fatty liver disease* is commonly associated with obesity and type 2 diabetes. It can progress to fibrosis and cirrhosis (Chapter 18).
- *Cholelithiasis (gallstones)* is six times more common in obese than in lean subjects. The obesity-associated increase in total-body cholesterol and cholesterol turnover leads to elevated biliary excretion of cholesterol in the bile, which in turn predisposes affected persons to the formation of cholesterol-rich gallstones (Chapter 18).
- *Obstructive sleep apnea* and consequent right-sided heart failure is strongly associated with obesity. Hypoventilation syndrome is a constellation of respiratory abnormalities in very obese persons. It has been called the pickwickian syndrome, after the fat lad who was constantly falling asleep in Charles Dickens' novel *The Pickwick Papers*.
- *Marked adiposity is a predisposing factor for the development of degenerative joint disease (osteoarthritis)*. This form of arthritis, which typically appears in older persons, is attributed in large part to the cumulative effects of wear and tear on joints. The greater the body burdens of fat, the greater the trauma to joints with the passage of time.
- *Markers of inflammation, such as CRP and proinflammatory cytokines like TNF, are often elevated in obese persons, in particular people with central obesity.* The basis for the inflammation is uncertain; both a direct proinflammatory effect of excess circulating lipids and increased release of cytokines from fat-laden adipocytes have been proposed. Whatever the cause, it is thought that chronic inflammation may contribute to many of the complications of obesity including insulin resistance, metabolic abnormalities, thrombosis, cardiovascular disease, and cancer.

Obesity and Cancer

There is an increased incidence of certain cancers in overweight and obese people, including carcinomas of the esophagus and upper stomach, colon and rectum, gallbladder, liver, pancreas, kidney, endometrium, ovaries, and breast, meningioma, and multiple myeloma. Evidence linking obesity to other cancer types is growing as well. Although the "risk" associated with obesity is modest, because of the prevalence of obesity in the population, it is associated with at least 40% of all cancers in the United States, somewhat more in females than in males. In addition, it appears that the risk of developing cancer is elevated the longer a person is overweight or obese, the higher the BMI, and the younger a person's age at the onset of obesity. The underlying mechanisms are unknown and are likely to be multiple.

- *Elevated insulin levels.* Insulin resistance leads to hyperinsulinemia, which includes multiple effects that may directly or indirectly contribute to cancer (see Fig. 9.29). For example, hyperinsulinemia causes a rise in levels of free insulin-like growth factor 1 (IGF-1). IGF-1 is a mitogen, and its receptor, IGFR-1, is highly expressed in many human cancers. IGFR-1 activates the RAS and PI3K/AKT

pathways, which promote the growth of both normal and neoplastic cells (Chapter 7).

- Obesity has effects on *steroid hormones* that regulate cell growth and differentiation in the breast, uterus, and other tissues. Specifically, obesity increases the synthesis of estrogen from androgen precursors, increases androgen synthesis in ovaries and adrenals, and enhances estrogen availability in obese persons by inhibiting the production of sex hormone-binding globulin (SHBG) in the liver.
- As discussed earlier, *adiponectin* secretion from adipose tissue is reduced in obese individuals. Adiponectin suppresses cell proliferation and promotes apoptosis. It does so in part by promoting the actions of p53 and p21. In obese individuals these antineoplastic actions of adiponectin may be compromised.
- The *proinflammatory state* that is associated with obesity may itself be carcinogenic through mechanisms discussed in Chapter 7.

KEY CONCEPTS

Obesity

- Obesity is a disorder of energy regulation. It increases the risk for several important conditions such as insulin resistance, type 2 diabetes, hypertension, and hypertriglyceridemia, which are associated with coronary artery disease, certain cancers, fatty liver disease, and gallstones.
- The regulation of energy balance has three main components: (1) afferent signals provided mostly by insulin, leptin, ghrelin, GLP-1, GIP, and peptide YY; (2) the central hypothalamic system, which integrates afferent signals and triggers the efferent signals; and (3) efferent signals, which control energy balance.
- Leptin plays a key role in energy balance. Its output from adipose tissues is increased by the abundance of fat stores. Leptin binding to its receptors in the hypothalamus decreases appetite and increases energy consumption by stimulating POMC/CART neurons and inhibiting NPY/AgRP neurons.
- Incretins GLP-1 and GIP produced by enteroendocrine cells in the intestines stimulate insulin secretion from pancreatic β cells; agonists of their receptors are effective in treating type 2 diabetes and obesity and appear to be beneficial in other diseases such as heart failure and renal insufficiency.

Diet and Cancer

As you will recall from Chapter 7, the incidence of specific cancers varies as much as 100-fold in different geographic areas. It is well known that differences in incidence of various cancers are not fixed and can be modified by environmental factors including changes in diet. For instance, the incidence of colon cancer in Japanese males and females 55 to 60 years of age was negligible about 50 years ago, but it is now higher than that in males of the same age in the United Kingdom. Studies have also shown a progressive increase in colon cancers in Japanese populations as they moved from Japan to Hawaii and from there to the continental United States. Nevertheless, despite extensive experimental and epidemiologic research, mechanisms that link diet and specific types of cancer have not yet been established.

With respect to carcinogenesis, three aspects of the diet are of major concern: (1) the content of exogenous carcinogens, (2) the endogenous synthesis of carcinogens from dietary components, and (3) the lack of protective factors.

- Regarding exogenous substances, *aflatoxin* is involved in the development of hepatocellular carcinomas in parts of Asia and Africa, generally in cooperation with hepatitis B virus. Exposure to aflatoxin causes a specific mutation in codon 249 of the *TP53* gene; when found in hepatocellular carcinomas, this mutation serves as a molecular signature for aflatoxin exposure. Debate continues about

the carcinogenicity of food additives, artificial sweeteners, and contaminating pesticides.

- The concern about endogenous synthesis of carcinogens or enhancers of carcinogenicity from components of the diet relates principally to gastric carcinomas. *Nitrosamines* and *nitrosamides* are implicated in the generation of these tumors, as they induce gastric cancer in animals. These compounds can be formed in the body from nitrites and amines or amides derived from digested proteins. Sources of nitrites include sodium nitrite added to foods as a preservative, and nitrates, present in common vegetables, which are reduced in the gut by bacterial flora. There is then the potential for endogenous production of carcinogenic agents from dietary components, which might in turn contribute to gastric carcinogenesis.
- *High animal fat intake combined with low fiber intake has been implicated in the causation of colon cancer.* It has been estimated that doubling the average level of total fiber consumption to about 40 g/day per person in most populations decreases the risk of colon cancer by 50%. The most plausible explanation for this association is that high fat intake increases the level of bile acids in the gut, which in turn modifies intestinal flora, favoring the growth of microaerophilic bacteria. Bile acid metabolites produced by these bacteria may function as carcinogens. The *protective effect of a high-fiber diet* might relate to (1) increased stool bulk and decreased transit time, which decreases the exposure of mucosa to putative offenders, and (2) the capacity of certain fibers to bind carcinogens and thereby protect the mucosa. However, attempts to document these theories in clinical and experimental studies have not generated consistent results.
- *Vitamins C and E, β -carotene, and selenium* have been assumed to have anticarcinogenic effects because of their antioxidant properties. However, thus far there is no convincing evidence that these antioxidants act as chemopreventive agents. As discussed earlier in this chapter, retinoids are highly effective in treating acute promyelocytic leukemia, and associations between low levels of vitamin D and cancer of the colon, prostate, and breast have been reported.

On one hand, despite many tantalizing trends and proclamations by diet gurus, thus far there is no definitive proof that a particular diet can cause or prevent cancer. On the other hand, given the relationship between obesity and cancer development, prevention of obesity through the consumption of a healthy diet is a commonsense measure that goes a long way in preserving good health. It will also be of great interest to determine if new, effective weight loss drugs such as GLP-1 agonists decrease the incidence of obesity-associated cancers.

Diet and Systemic Diseases

The problems of undernutrition and overnutrition as well as specific nutrient deficiencies have been discussed; however, the composition of the diet, even in the absence of any of these problems, may make a significant contribution to the causation and progression of several diseases. A few examples suffice here.

Currently, one of the most important and controversial issues is the contribution of diet to atherogenesis. The central question is whether dietary modification—specifically, reduction in the consumption of foods high in cholesterol and saturated animal fats (e.g., eggs, butter, beef)—can reduce serum cholesterol levels and prevent or retard the development of atherosclerosis (of most importance, coronary artery disease) in people with no previous episode of cardiovascular disease. This is called primary prevention. We know some, but not all, of the answers. The average adult in the United States consumes a large amount of fat and cholesterol daily, with a ratio of saturated fatty acids to polyunsaturated fatty acids of about 3:1. Lowering the level of saturated fatty acids to the level of the polyunsaturated fatty acids causes a 10% to 15%

reduction in serum cholesterol within a few weeks. Vegetable oils (e.g., corn and safflower oils) and fish oils contain polyunsaturated fatty acids and are good sources of such cholesterol-lowering lipids. Fish oil fatty acids belonging to the omega-3, or *n*-3, family have more double bonds than the omega-6, or *n*-6, fatty acids found in vegetable oils. A corollary of this idea is that supplementation of diet with fish oils might protect against atherosclerosis. However, a large meta-analysis of 79 randomized controlled trials showed that dietary supplements of omega-3 fatty acids or consumption of oily fish had little or no effect on cardiovascular disease (ischemic heart disease, stroke).

Other specific effects of diet on disease have been recognized:

- Restricting sodium intake reduces hypertension.
- Dietary fiber, or roughage, resulting in increased fecal bulk is thought by some investigators to provide a preventive effect against diverticulosis of the colon.
- Caloric restriction has been convincingly demonstrated to increase lifespan in experimental animals, including monkeys. The basis for this striking observation is not clear (Chapter 2), and the degree of calorie restriction that is needed to prolong life is so great that one can question if such a life is worth living. Perhaps by understanding the underlying mechanism, we may someday be able to have our cake and eat it, too.
- Even lowly garlic has been touted to protect against heart disease (and also against devils, werewolves, vampires, and, alas, kisses), although research has yet to prove this effect unequivocally. Of these, the effect on kisses is the best established!

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